Digital Signal Processing 21/01/25

Applications of DSP:-

- Audio Processing → Noise cancellation, blind source separation
- Image Processing → 2D signal, 1D signal, enhancing images through SC, Insta cameras.
  Image restoration
- Speech Processing → speech recognition, youtube, SIRI

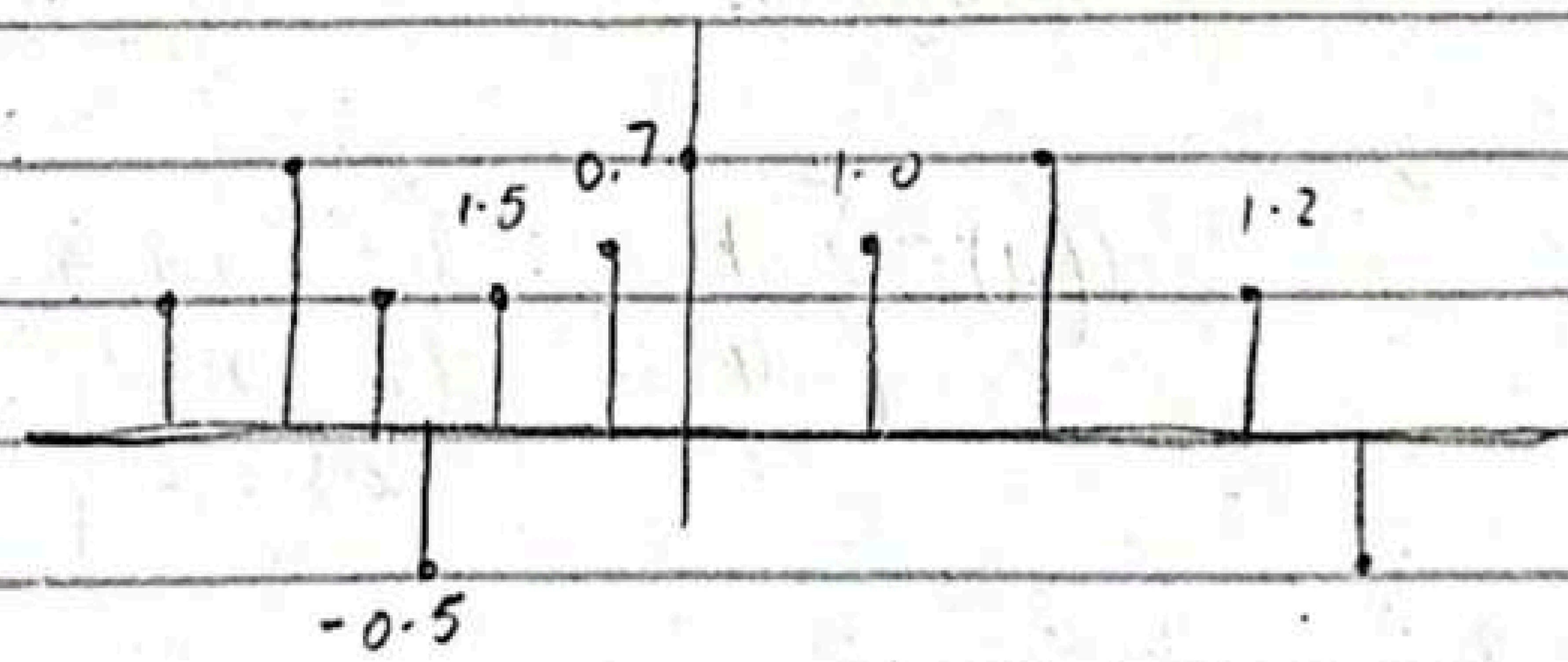
- communication:-
  sensor & control
  Modem, modulation, demodulation, "control signals, motor control", "signal detection and tracking of moving targets."
  signal detection

space, medical, etc.

Chapter #2

Discrete Time Signals And System

A discrete time signal $x(n)$ is a function of an independent variables that is an integer.



Discrete time signal is defined for every integer value n for $-\infty < n < \infty$.

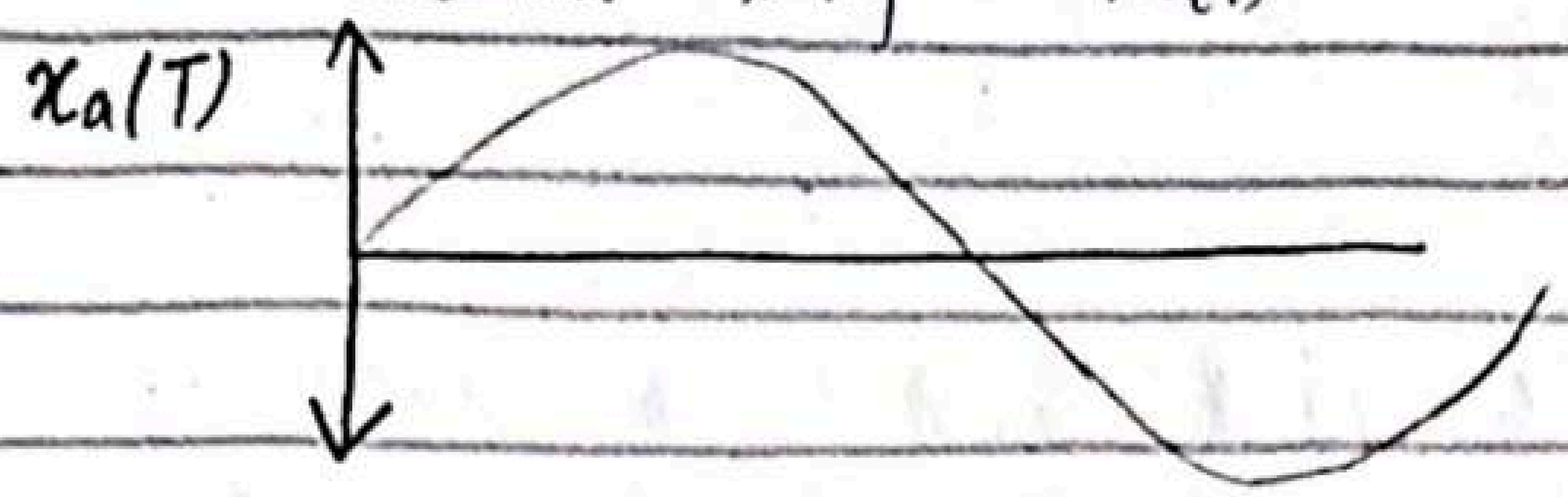
$x(n)$ is the n^{th} sample.

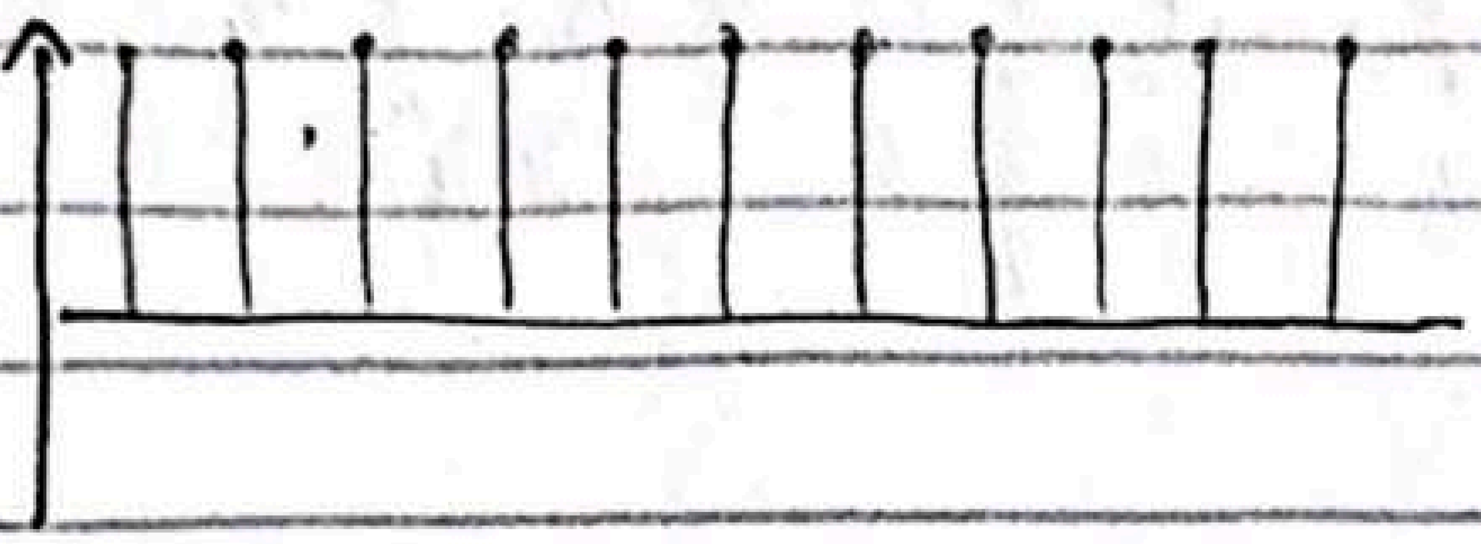
T is in b/w two values.

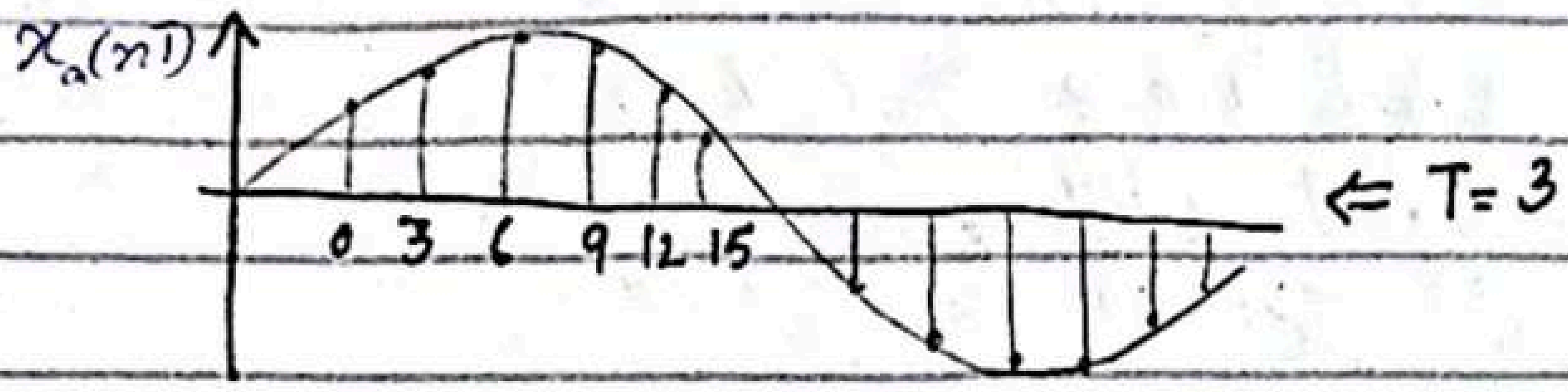
successive samples $x_a(t)$

Graphical representation

Analog signal







if we take less samples it is called under sampling, if we take maximum samples it is called over sampling.

Representation of signals :-

Functional Representation :-

$$x(n) = \begin{cases} 1 & \text{for } n = 1, 3 \\ 4 & \text{for } n = 2 \\ 0 & \text{elsewhere.} \end{cases}$$

Tabular Representation :-

n	......	-2	-1	0	1	2	3	4	...
$x(n)$	......	0	0	0	1	4	1	0	...

Sequence Representation:-

$$x(n) = \{ \ldots 0, \underset{\uparrow}{0}, 1, 4, 1, 0 \ldots \}$$

sequence $x(n)$ which is zero for $n<0$, can be represented as

$$x(n) = \{0, 1, 4, 1, 0, \ldots\}$$

It depends on arrow if arrow is at 3 then $x(0) = 3$, $x(1) = -1$

$$x(n) = \{\underset{\uparrow}{3}, -1, \frac{2}{3}, \ldots\}$$

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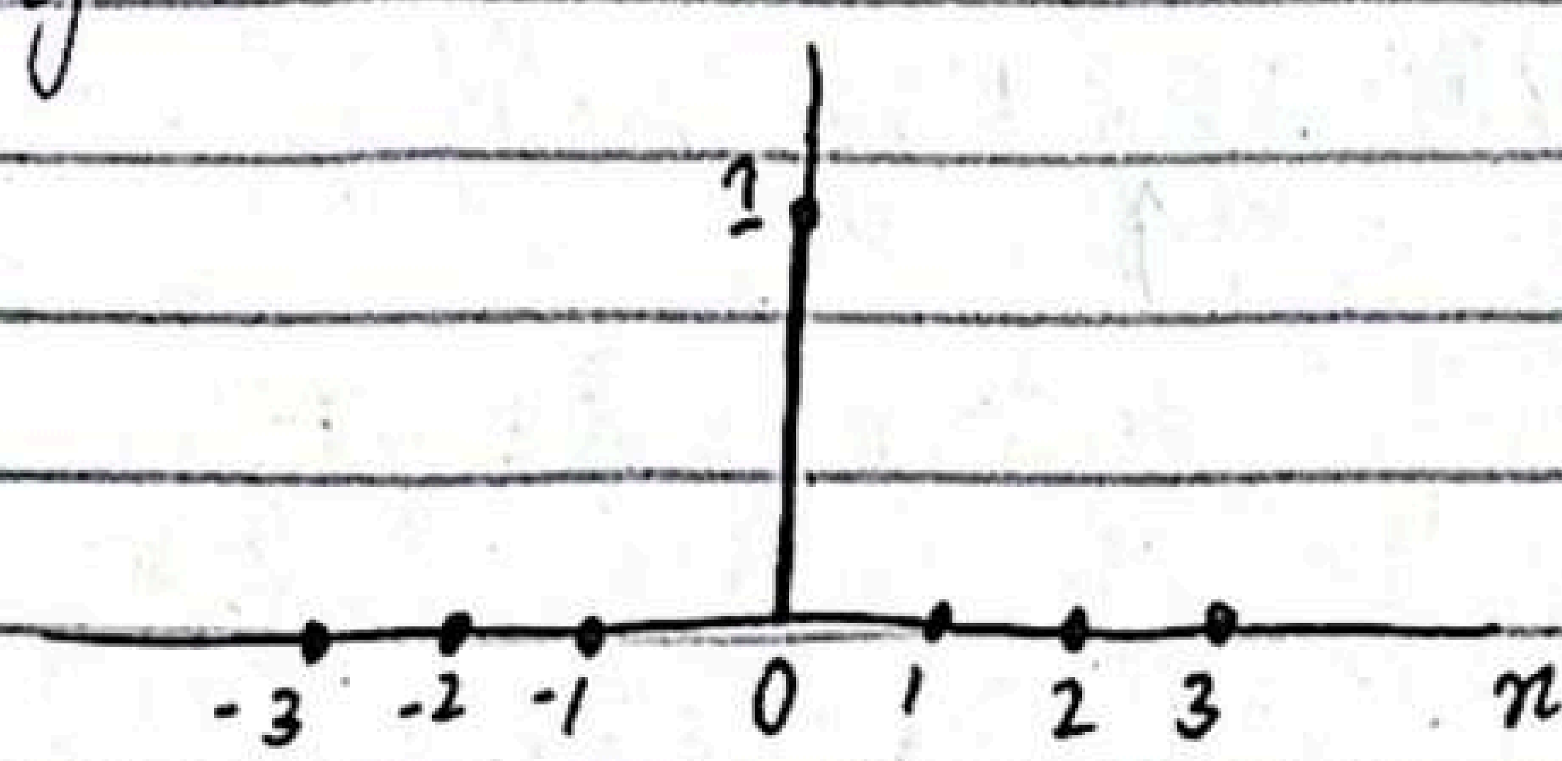
Some Elementary Discrete-Time Signals

Unit sample sequence:-

The unit sequence is denoted as $\delta(n)$ and is defined as

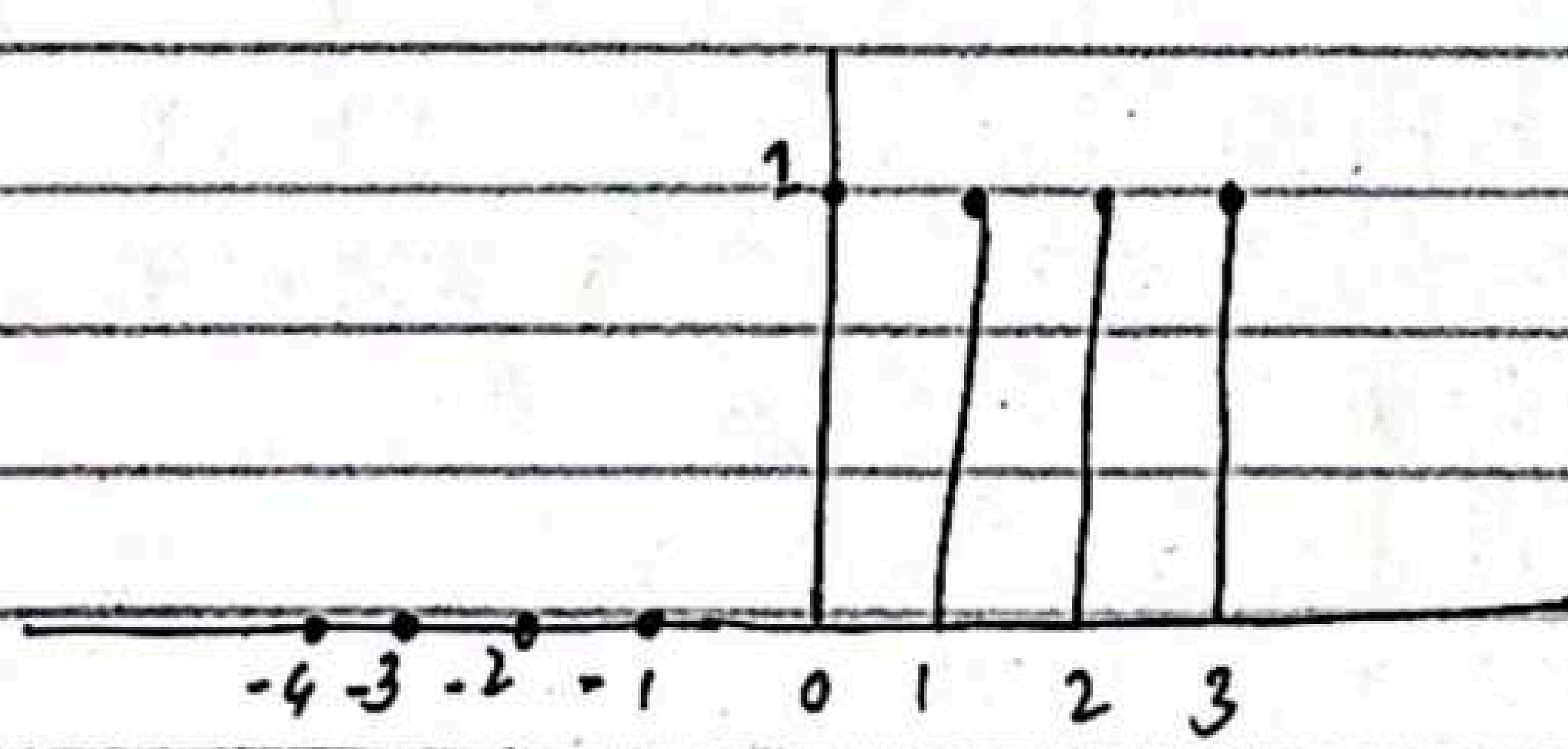
$$\delta(n) = \begin{cases} 1 & \text{for } n = 0 \\ 0 & \text{for } n \neq 0 \end{cases}$$

The unit sample sample sequence is a signal that is 0 everywhere exept at $n=0$ where it is unity.



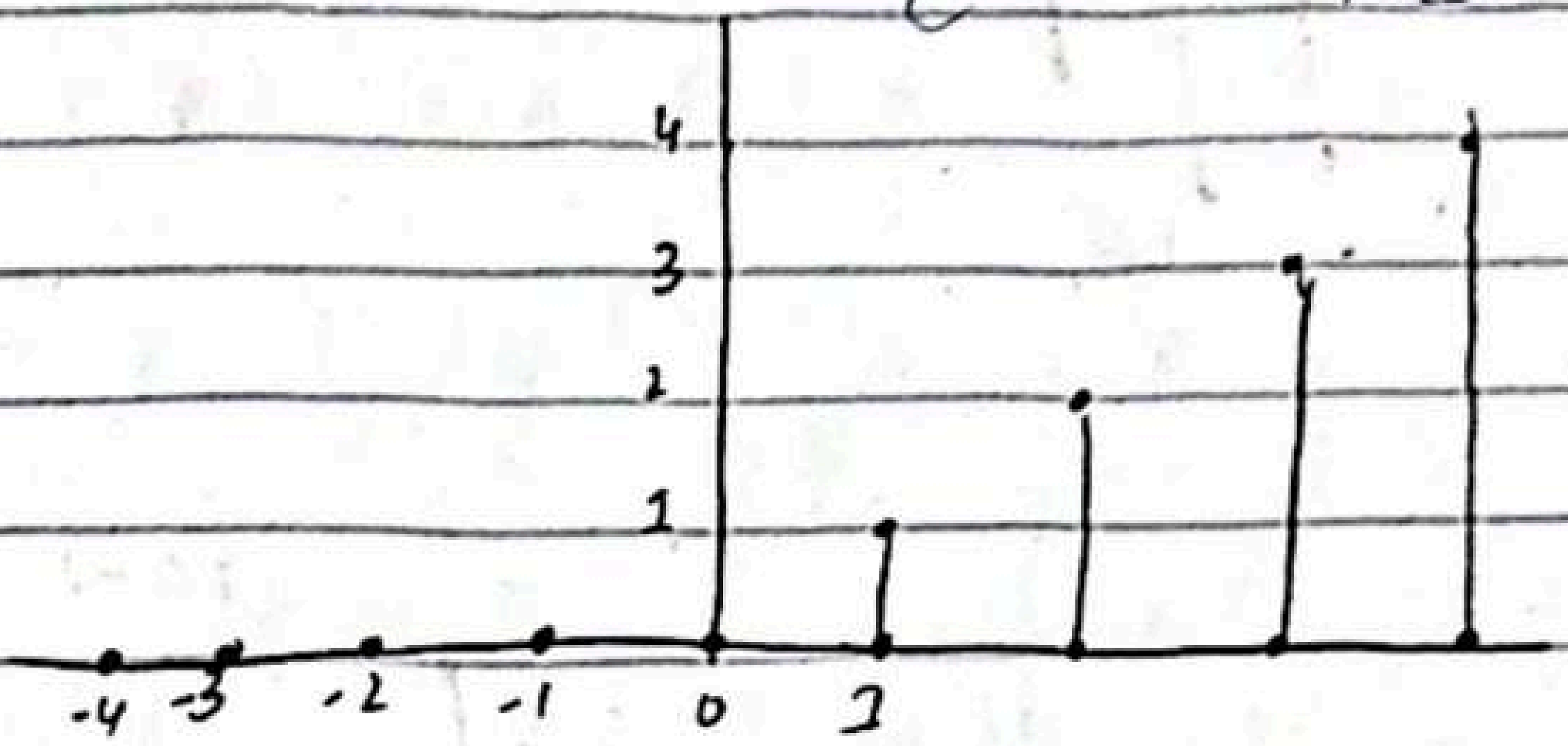
Unit step signal for

$$u(n) = \begin{cases} 1 & n \geq 0 \\ 0 & n < 0 \end{cases}$$



unit ramp signal is denoted as $U_r(n)$

$$U_r(n) = \begin{cases} n & n \geq 0 \\ 0 & n < 0 \end{cases}$$

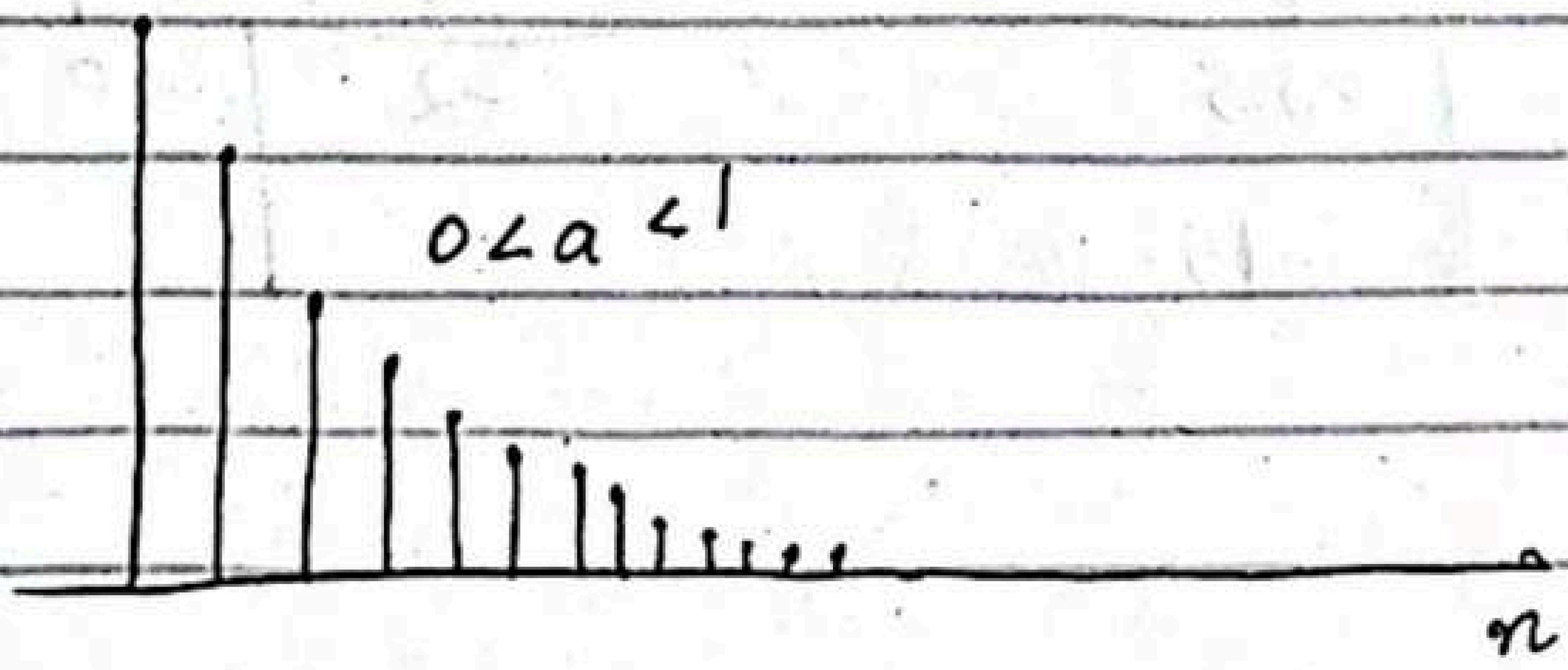


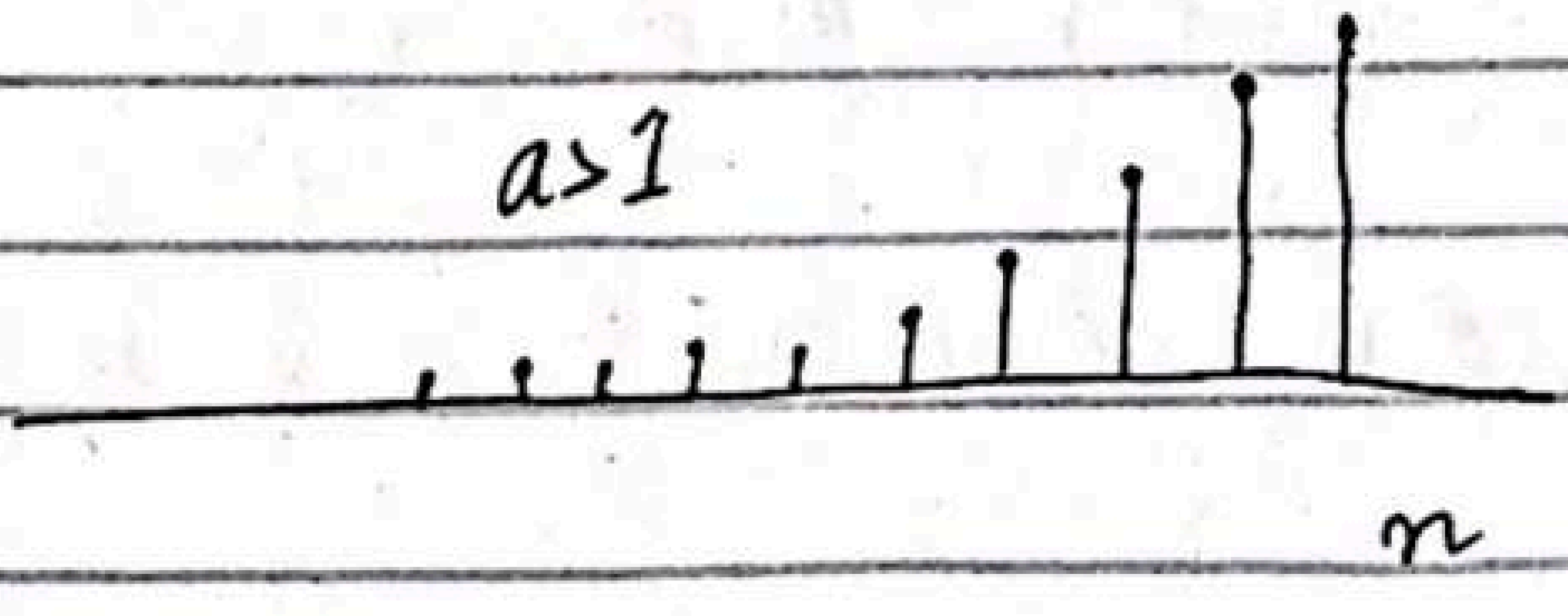
Exponential signal :-

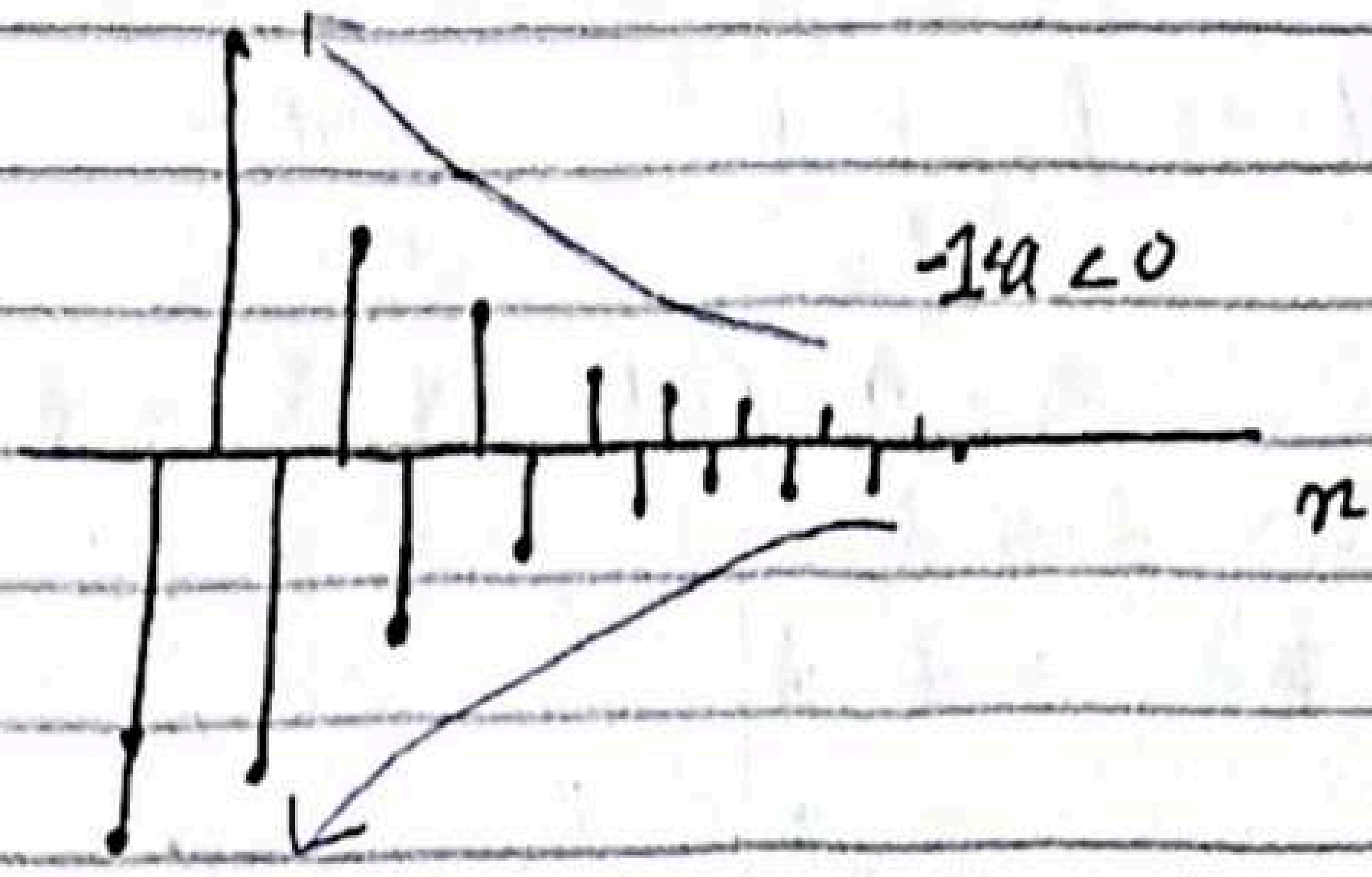
The exponential signal is a sequence of the form

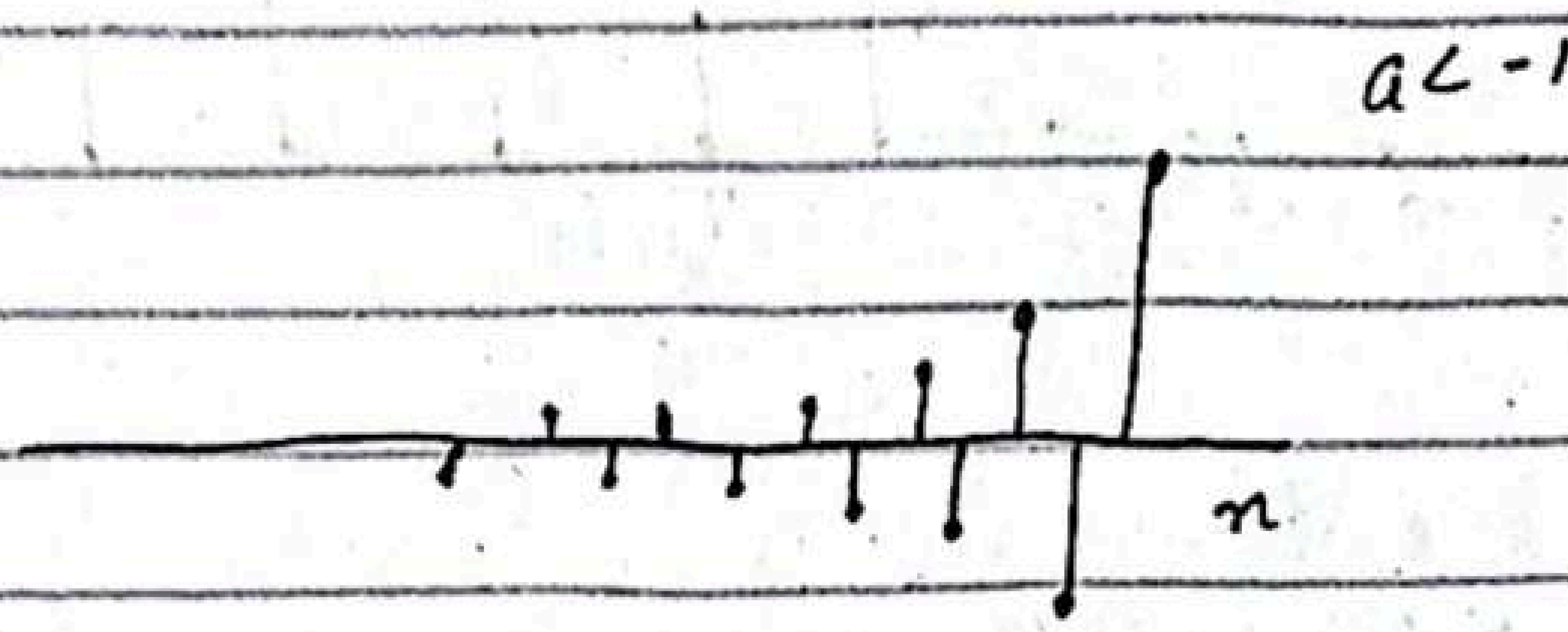
$$x(n) = a^n \quad \text{for all } 'n'$$

If the parameter a is real, then $x(n)$ is a real signal.









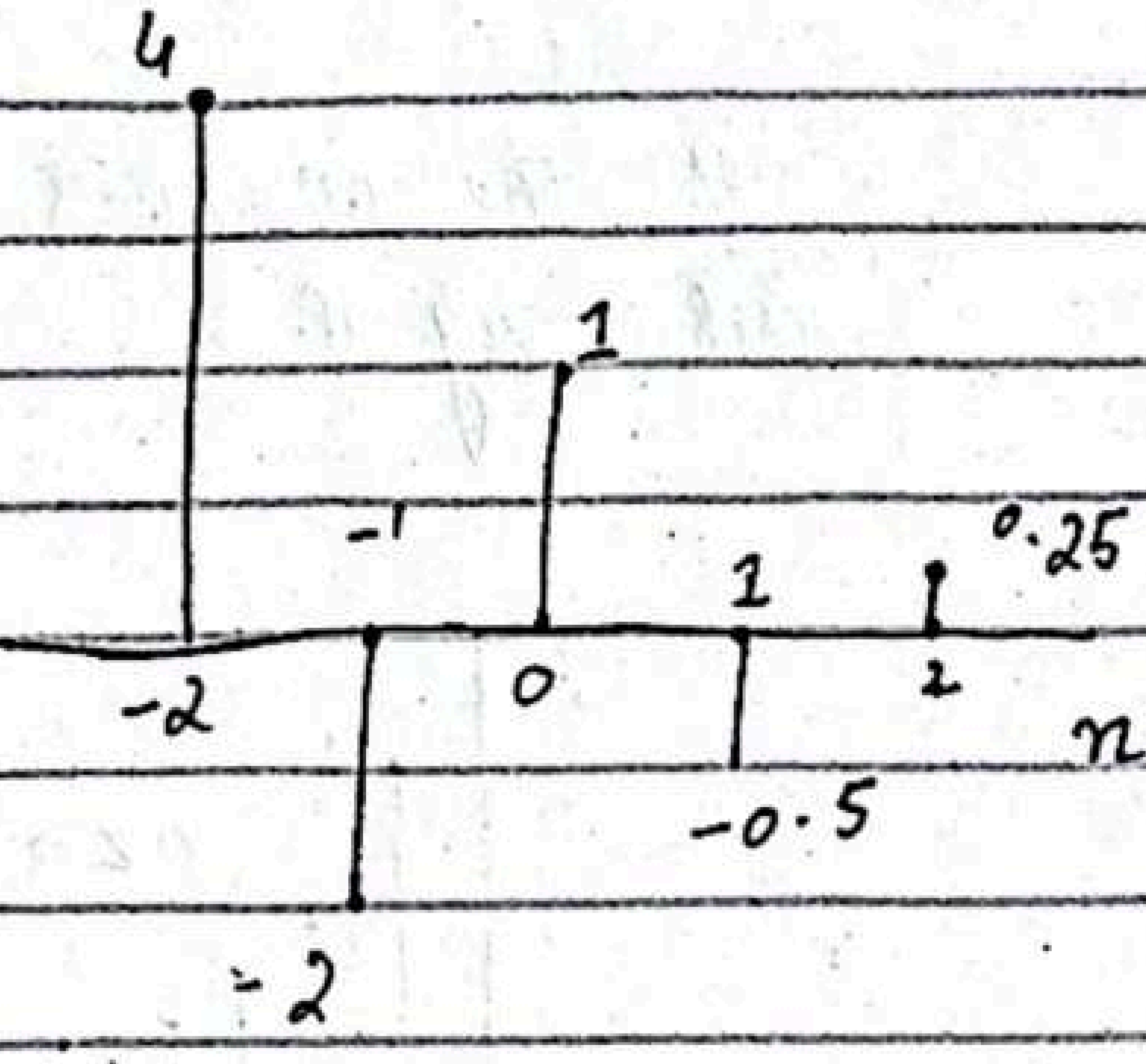
Example:-

$x(n) = a^n$ for all n

$n = -2 : 1 : 2$

$a = -0.5$

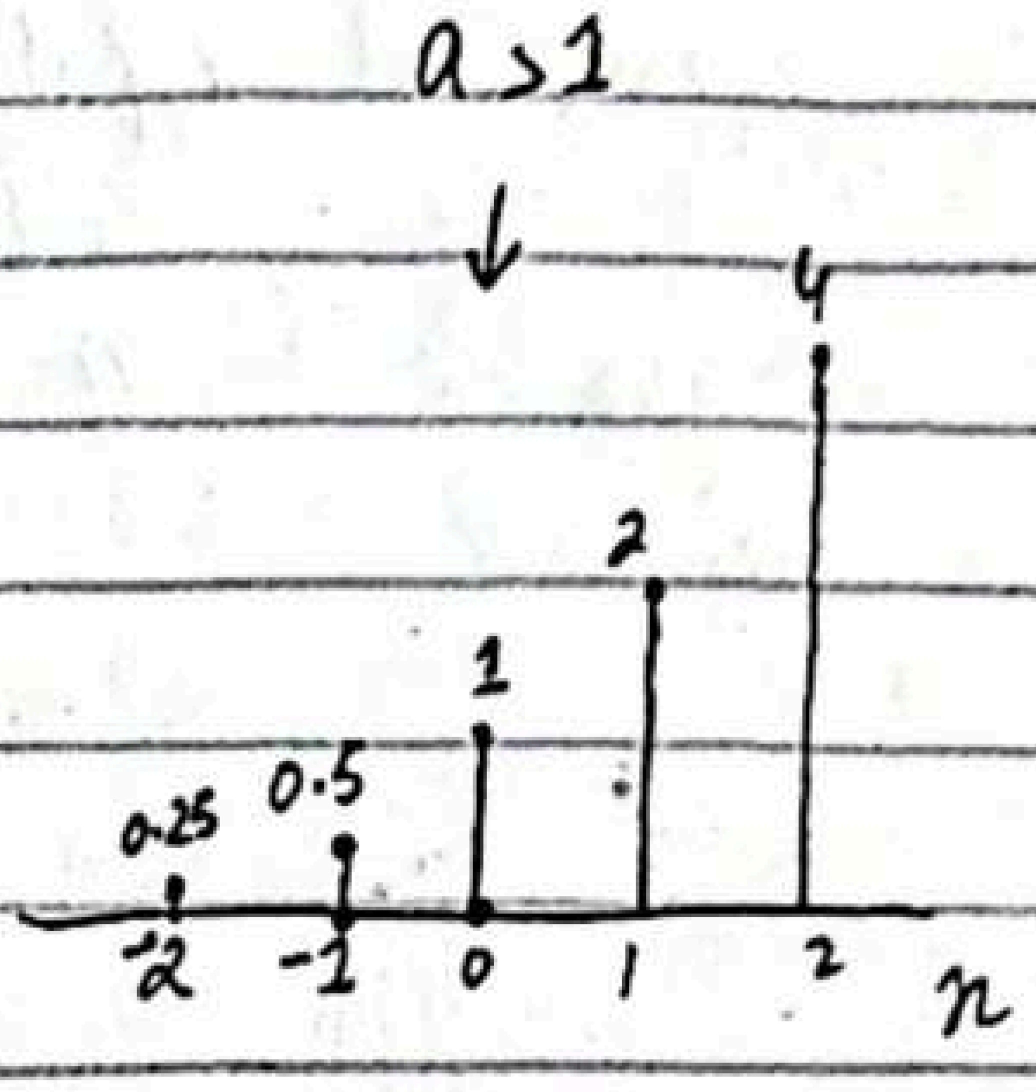
n	a^n
-2	4
-1	-2
0	1
1	-0.5
2	0.25

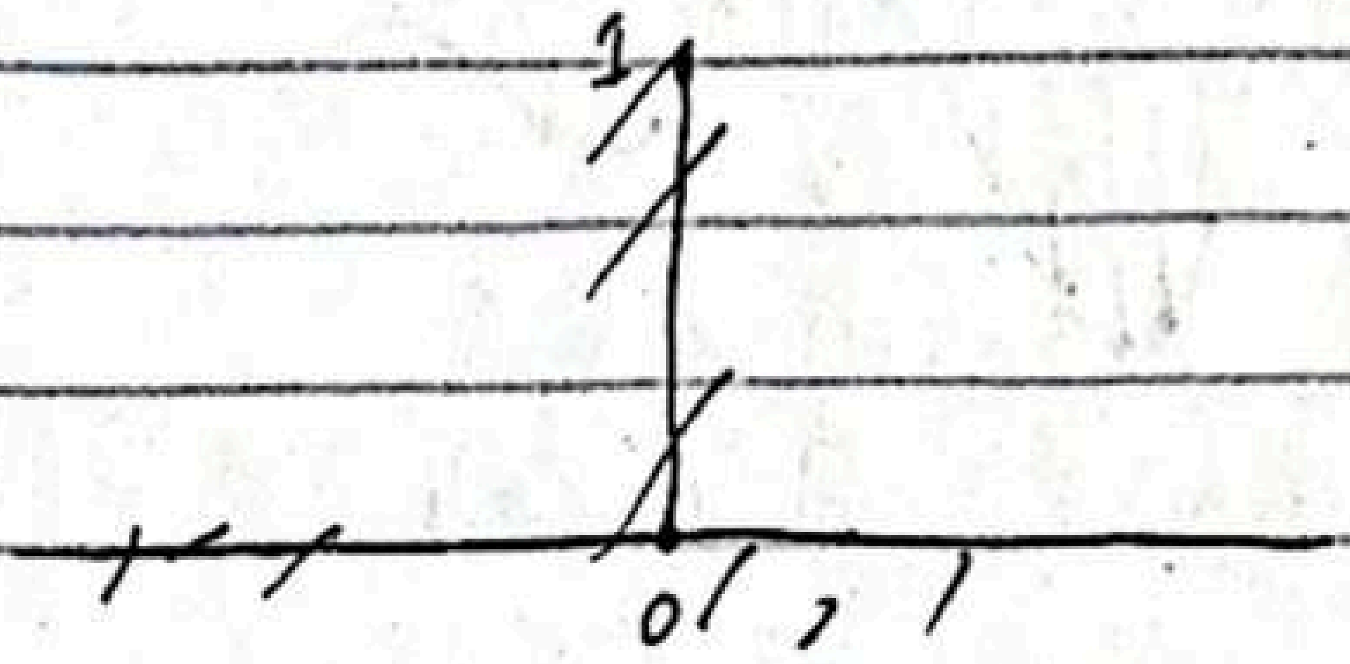


Example:

$$a = 2$$
$$n = -2:1:2$$
$$y = a^n$$

n	a
-2	0.25
-1	0.5
0	1
1	2
2	4





When the parameter a is a complex valued it can be expressed as

$$a = re^{j\theta}$$

r and θ are now parameters

$$x(n) = r^n e^{j\theta n}$$
$$= r^n(\cos\theta n + j\sin\theta n)$$

$x(n)$ is now complex valued, it can be represented graphically by plotting real part

$$x_R(n) = r^n \cos\theta n$$

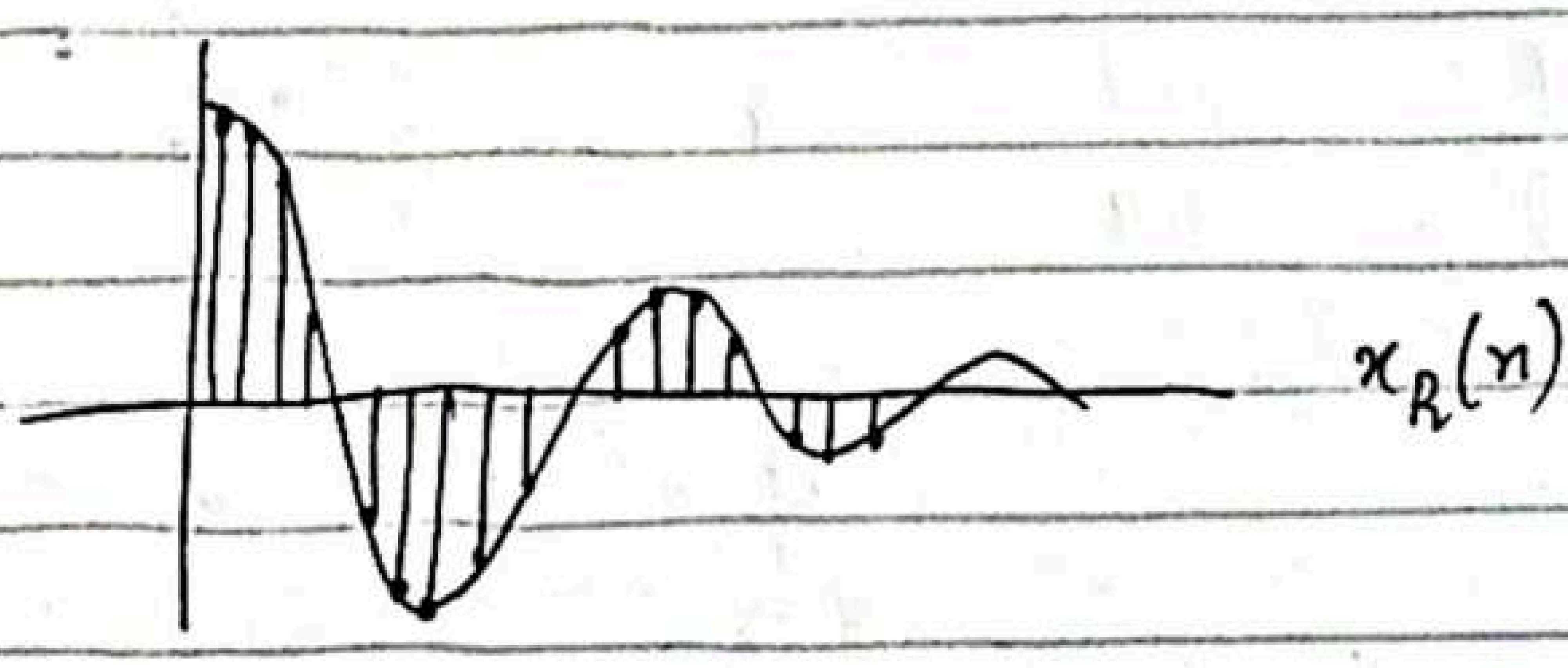
as a function of n and seperately plotting the imaginary part

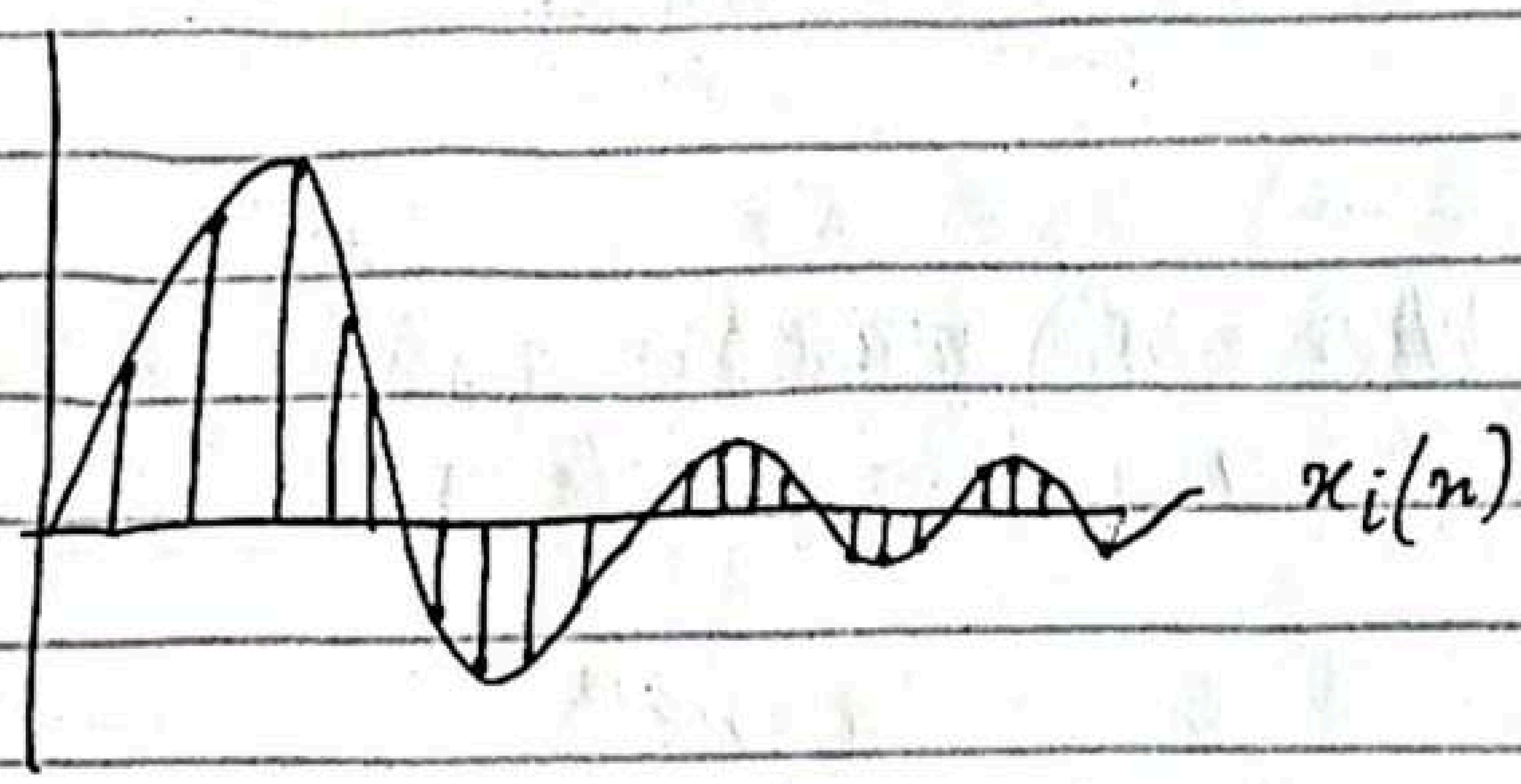
$x_i(n) = r^n \sin\theta n$

as a function of n.

graphs of $x_R(n)$ and $x_i(n)$, $r = 0.9$

$\theta = \pi/10$, we observe that signals $x_R(n)$ and $x_i(n)$ are damped (decaying) cosine function and damped sine function. The angle variable θ is simply the frequency of the sinusoid





amplitude :-

$$|x(n)| = A(n) = r^n$$

phase function :-

$$\angle x(n) = \phi(n) = \theta n$$

$A(n)$ and $\phi(n)$ for $r = 0.9$ and $\theta = \pi/10$

we observe the phase function is linear with n. However, the phase is defined only over interval $-\pi < \theta < \pi$ or $0 \le \theta \le 2\pi$,
In other words we subtract multiples of 2π form $\phi(n)$ ~~ff~~ before plotting

The subtraction of multiples of 2π from $\phi(n)$ is equivalent to interpreting of the function $\phi(n)$ as $\phi(n)$ modulo 2π.

$\Rightarrow$ Phase is defined only over the interval $-\pi < \theta < \pi$

$$\phi_{normalized} = \phi(n) \circ \cancel{2\pi} \text{ rad } 2\pi \, round\left(\frac{\phi(n)}{2\pi}\right)$$

suppose you have a phase signal $\phi(n)$

$$\phi(n) = \{-3\pi, \pi, \pi, \frac{3\pi}{2}, 3\pi\}$$

without normalized values
a wide range $(-3\pi$ to $3\pi\}$

~~$\pi/3\pi$~~

After normalization

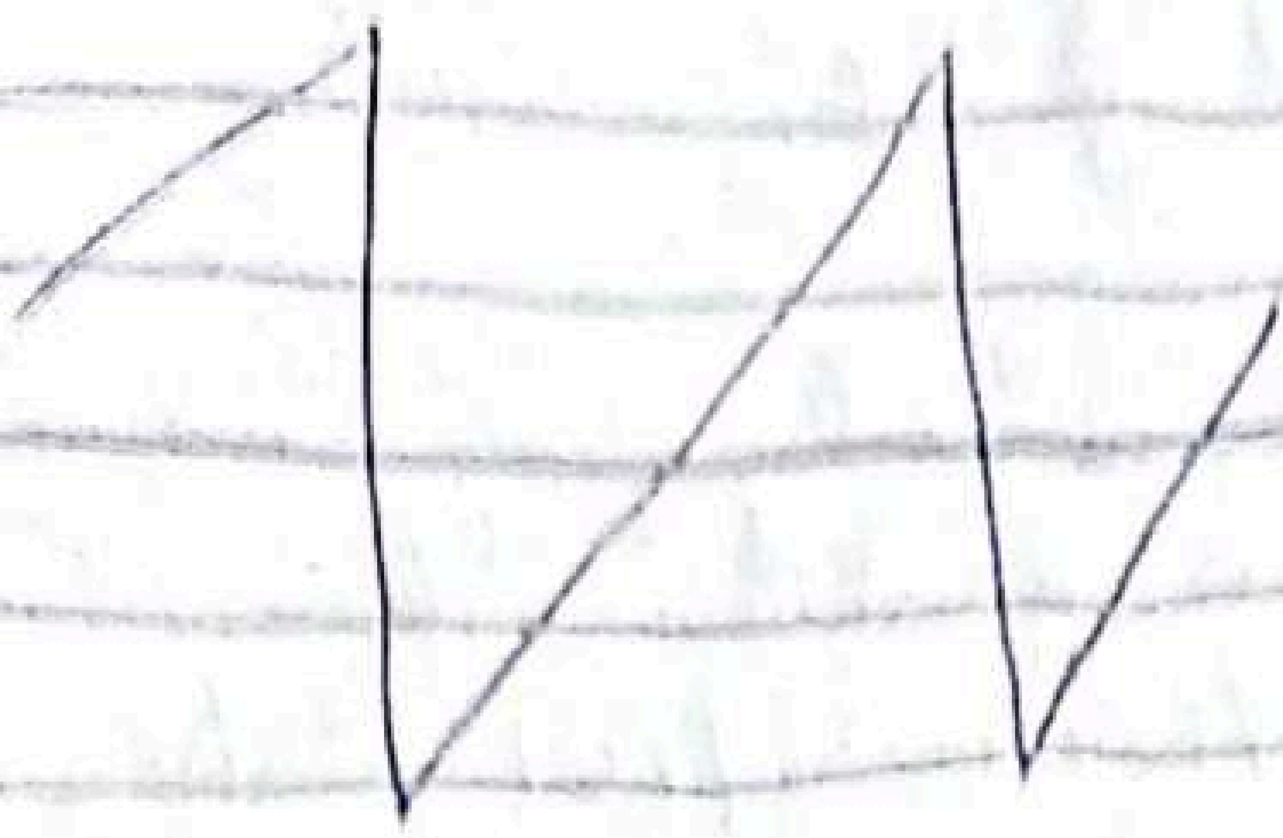
$$\{\pi, -\frac{\pi}{2}, \pi, \pi/2, \pi\}$$

$$\phi(n) = \frac{\pi}{10} n . \text{modulo } 2\pi$$

$n = -20 : 20$

y = (pi/10)* (n) - 2pi round (n/20)

— round $\left(\frac{2\pi/1}{10/2\pi}\right)$



Energy signals and power signals:- The energy E of a signal $x(n)$ is defined as

$$E = \sum_{n=-\infty}^{\infty} |x(n)|^2$$

The energy of a signal can be finite or infinite. If E is finite $(0 < E < \infty)$ then $x(n)$ is called energy signal.

Many signal that posses infinite energy, have finite average power

Energy of signal:- $\sum_{n=-\infty}^{\infty} |x(n)|^2$

Power of a signal:-

$$P = \lim_{N \to \infty} \frac{1}{2N+1} \sum_{n=-N}^{N} |x(n)|^2$$

If we define the signal energy of $x(n)$ over finite interval $-N \le n \le N$ as

$$E_N = \sum_{n=-N}^{N} |x(n)|^2$$

then we can express the signal energy E as

$$E = \lim_{N \to \infty} E_N$$

$$E = \lim_{N \to \infty} \sum_{n=-N}^{N} |x(n)|^2$$

And average power of signal $x(n)$ as

$$P = \lim_{N \to \infty} \frac{1}{2N+1} E_N$$

Clearly, if E is finite, $P = 0$

if E is infinite, the average power P may be finite or infinite.

if P is finite the signal is called power signal

Example:-

E = finite, $P = 0$

let

$$E_N = \sum_{n=-N}^{N} |x(n)|^2$$

$$P = \lim_{N \to \infty} \frac{1}{2N+1} \sum_{n=-N}^{N} |x(n)|^2$$ → finite energy

$$P = \frac{1}{\infty} = 0$$

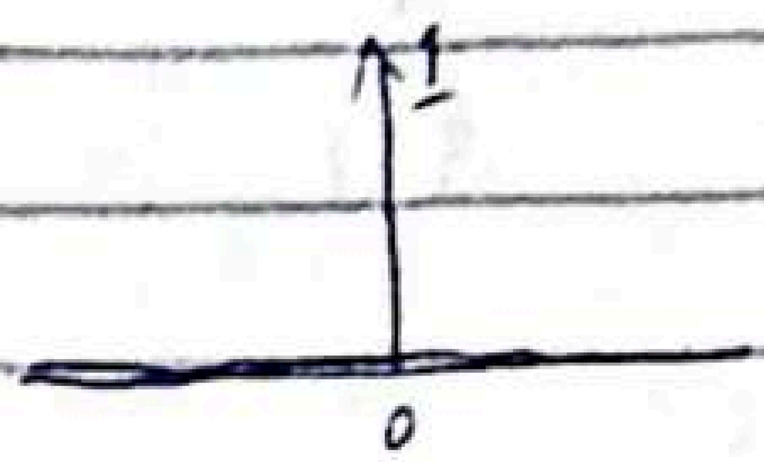
Example:-

Determine the power and energy of the unit step sequence. The average power of unit step signal is

$$P = \lim_{N \to \infty} \frac{1}{2N+1} \sum_{n=0}^{N} u^2(n)$$

$$E_N = \sum_{n=0}^{N} u^2(n)$$

$$= \sum_{n=0}^{N} (1)^2$$



$$E_N = 1$$

$$= \lim_{N \to \infty} \frac{1}{2N+1} (1)$$

$$= \frac{1}{\infty + 1} = 0$$

$$P = \lim_{N \to \infty} \frac{1}{2N+1} \sum_{n=0}^{N} u^2(n)$$

$$= \lim_{N \to \infty} \frac{N+1}{2N+1} = \lim_{N \to \infty} \frac{1 + 1/N}{2 + 1/N} = \frac{1}{2}$$

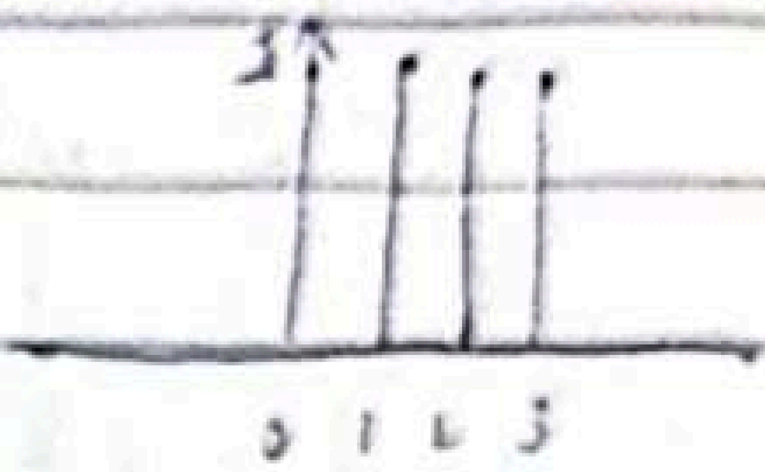
power is finite so it is a power signal and energy is infinite.

Let $N = 2$;

then

$$\sum_{n=0}^{2} u^2(n)$$

$u^2(0) + u^2(1) + u^2(2)$ → $u^2(2)$ is eq to 1 because



$= 1^2 + 1^2 + 1^2$

$= 3 \Rightarrow N+1$

Find Energy of a unit step function if limit is 0 to ∞

The energy a discrete-time signal $x(n)$ is given by

$$E = \sum_{n=\infty}^{\infty} |x(n)|^2$$

$$E_n = \sum_{n=\infty}^{\infty} |u(n)|^2$$

since $u(n) = 0$ for $n<0$ and $u(n) = 1$ → missing
we only

•

similarly, it can be shown that the complex exponential sequence $x(n) = Aj^{e\theta n}$

Power is A^2 it is power signal
ramp signal is neither energy nor power signal.

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Periodic signals and aperiodic signals:

signal $x(n)$ is periodic with period $N (N>0)$ if and only if $x(n+N) = x(n)$ for all n.

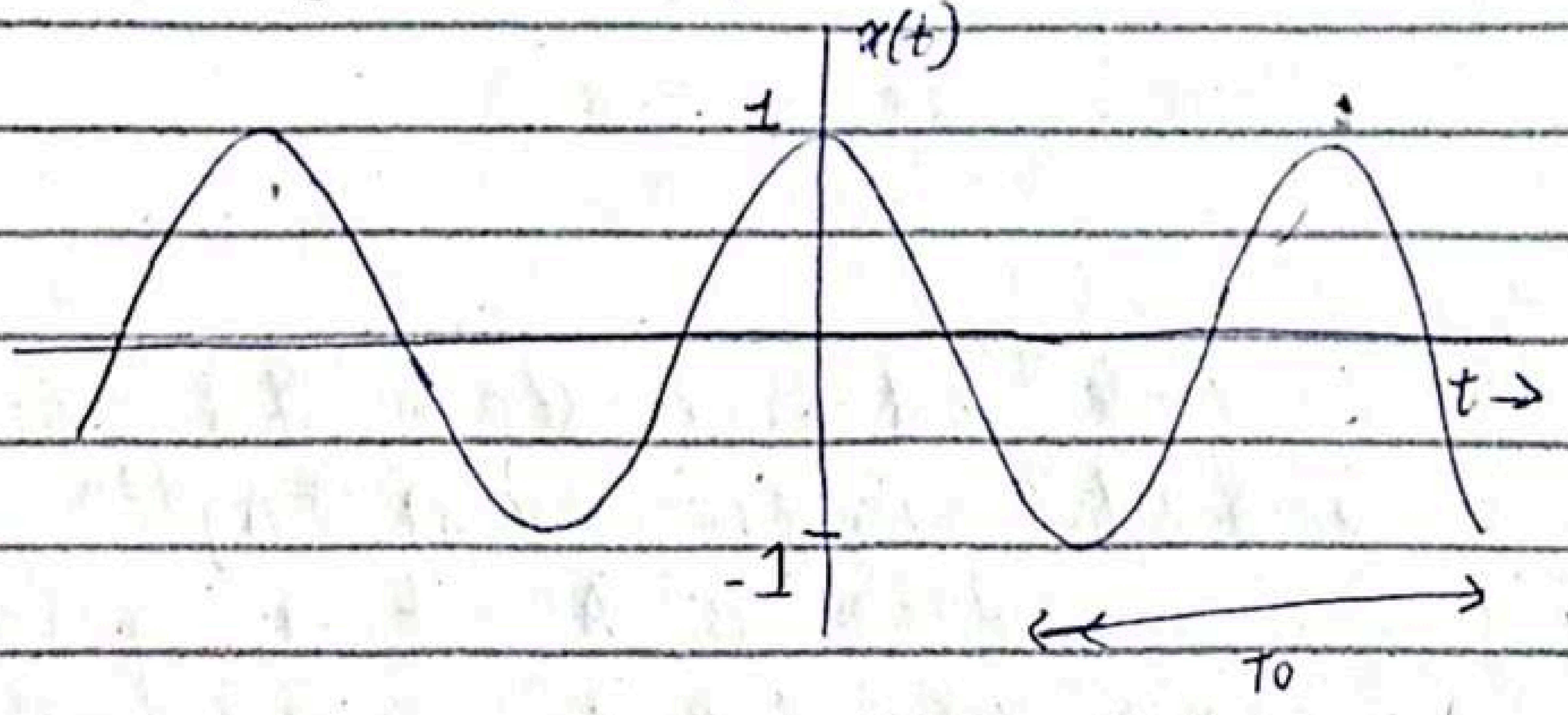
The smallest value of N for which above equation holds is called fundamental period.

If there is no value of N that satisfies the eq, signal is non periodic or a periodic

Example:

$$x(t) = \cos(2\pi f_0 t + \phi)$$

f_0 Hz $\rightarrow$ f_0 cycles in 1 sec
$\Rightarrow$ each cycle $T_0 = \frac{1}{f_0}$ sec.

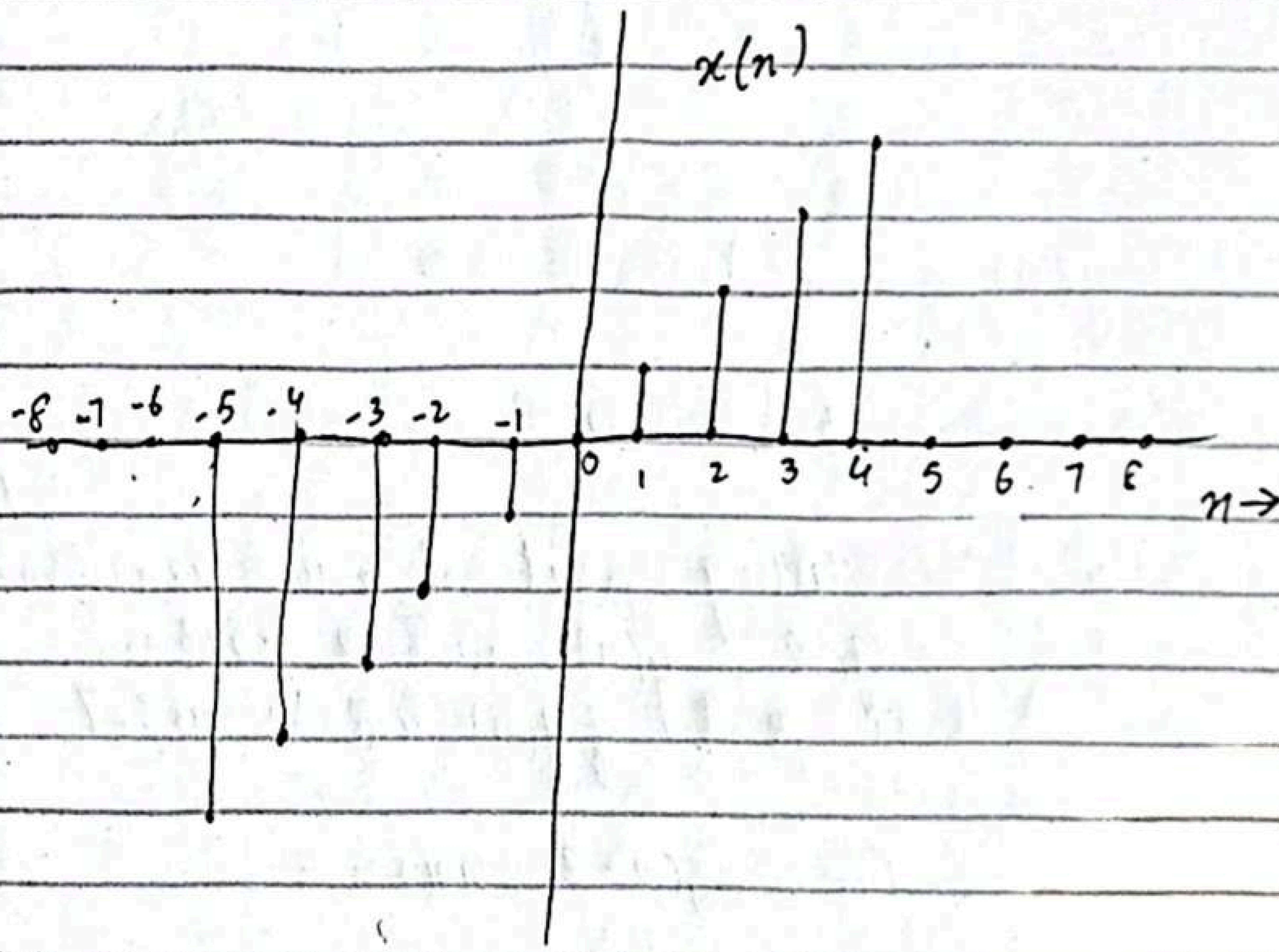


Math: $x(t+T_0) = \cos(2\pi f_0 (t+T_0) + \phi)$
$= \cos(2\pi f_0 t + 2\pi f_0 T_0 + \phi)$
$= \cos(2\pi f_0 t + \phi)$ as $T_0 = \frac{1}{f_0}$
$x(t)$ for all t

Fundamental Period: smallest T_0

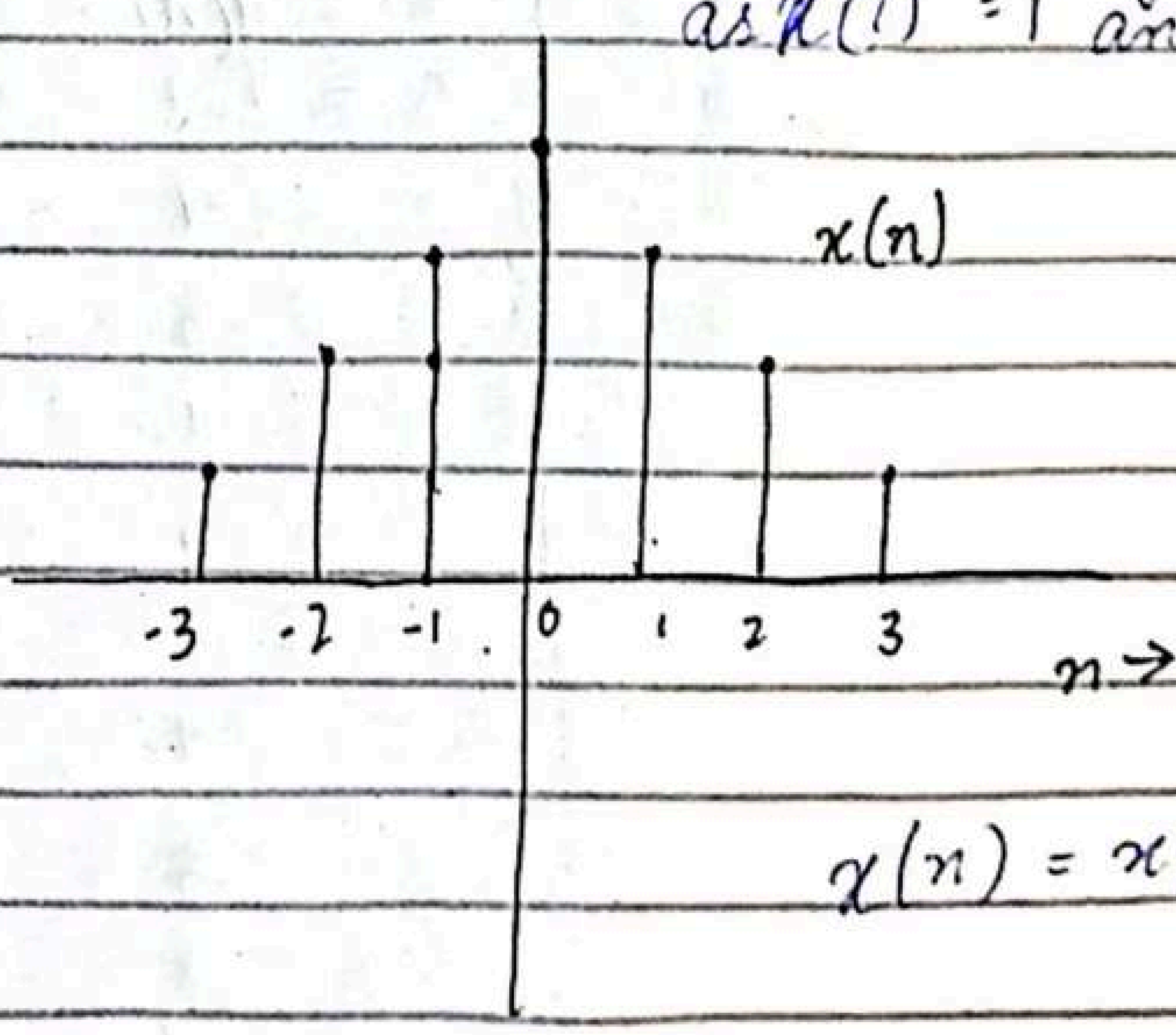
$x(n) = A \sin 2\pi f_0 n$ is periodic when f_0 is rational

$f_0 = \frac{K}{N}$, where K and N are integers



$x(-n) = -x(n) \Rightarrow$ odd signal

as $x(1) = 1$ and $x(-1) = -1$



$x(n) = x(-n) \Rightarrow$ even signal

Example 2.12

signal $x(n)$ is graphically shown show a graphical representation of signal $x(n-$

$$\boxed{y(n) = x(n-3)}$$

step 1: $n=3=0 \Rightarrow n=3$

step 2: Starting point of new transformed signal $y(n)$ will be $-5+3=-2$
End point of $y(n)$ will be $4+3=7$

step 3:- Find $x_N(n-3)$ now :-

$x(n-3)$	$y(n)$
$-2-3=-5$	-2
-4	-1
-3	0
-2	1
-1	2
0	3
1	4
2	5
3	6
4	7

$y(n) = x(n+2)$

step 1: $n+2=0$, $n=-2$

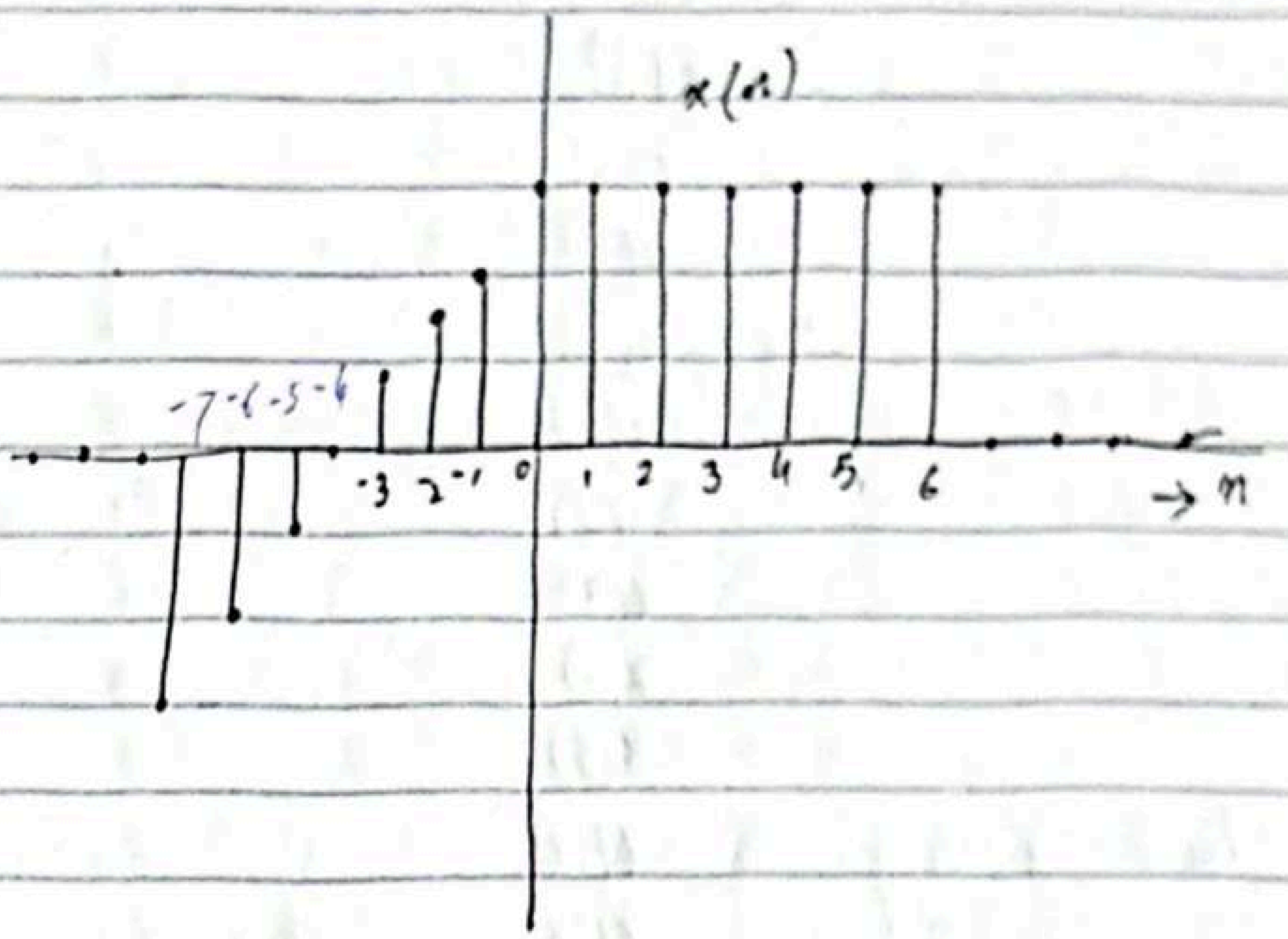
step 2: Range of $y(n)$ will be from -7 to 2

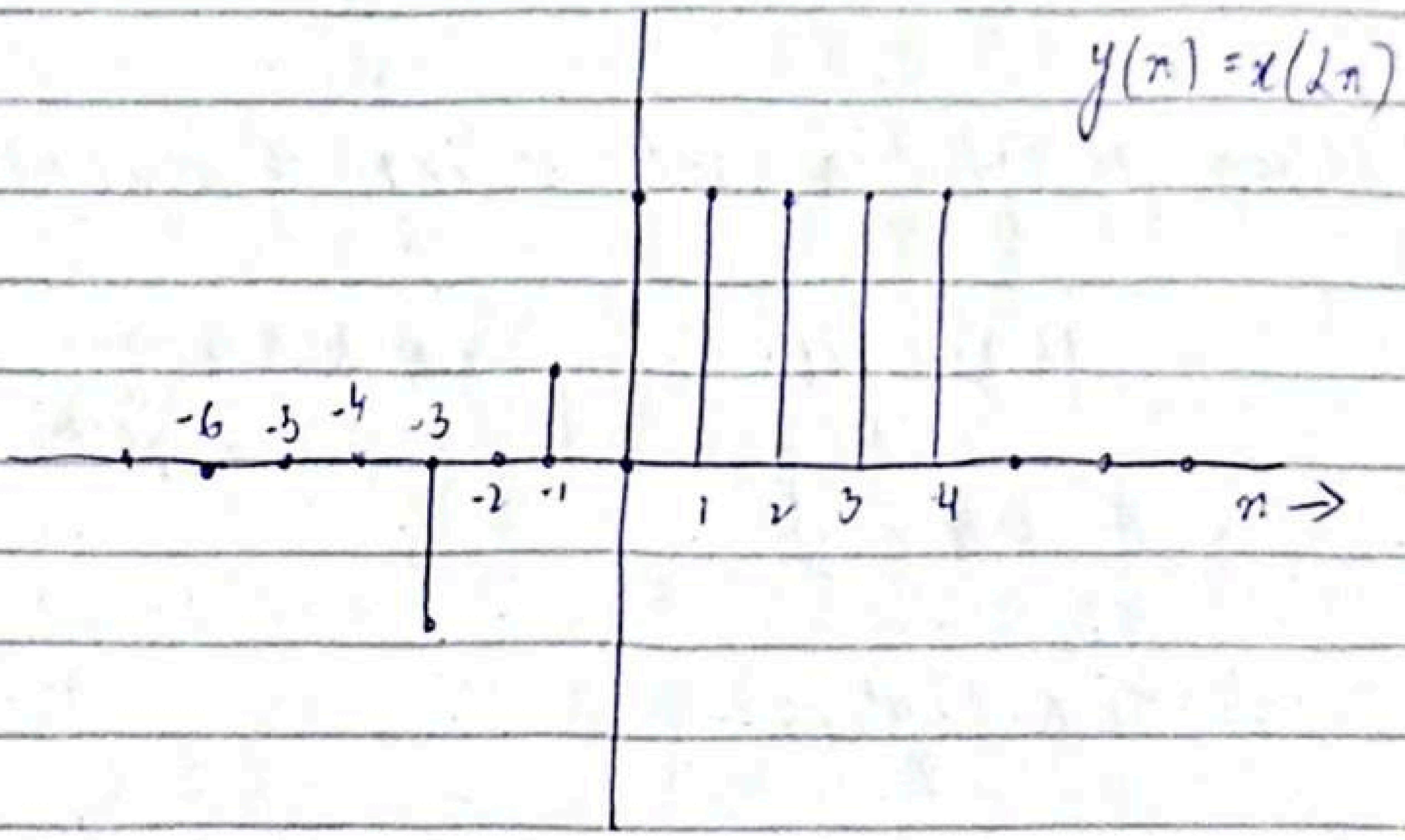
step 3: Find $x(n+2)$ now

$x(n+2)$	$y(n)$
-5	-7
-4	-6
-3	-5
-2	-4
-1	-3
0	-2
1	-1
2	0
3	1
4	2

Example 2.1.4

$y(n) = x(2n)$ where $x(n)$ is the signal





step 1 : $n = 0$

step 2: $-7 \le y(n) \le 6$

step 3: $x(2n)$

step 3:-

	$x(2n)$	$y(n)$
	$x(2(-7)) = -14$	-7
	$x(-12)$	-6
	$x(-10)$	-5
	$x(-8)$	-4
	$x(-6)$	-3
	$x(-4)$	-2
	$x(-2)$	-1
	$x(0)$	0
	$x(2)$	1
	$x(4)$	2
	$x(6)$	3
	$x(8)$	4
down sampling	$x(10)$	5
	$x(12)$	6

Addition, Multiplication, and scaling of sequences:-

$y(n) = Ax(n) \qquad -\infty < n < \infty$

slides

Discrete Time systems :-

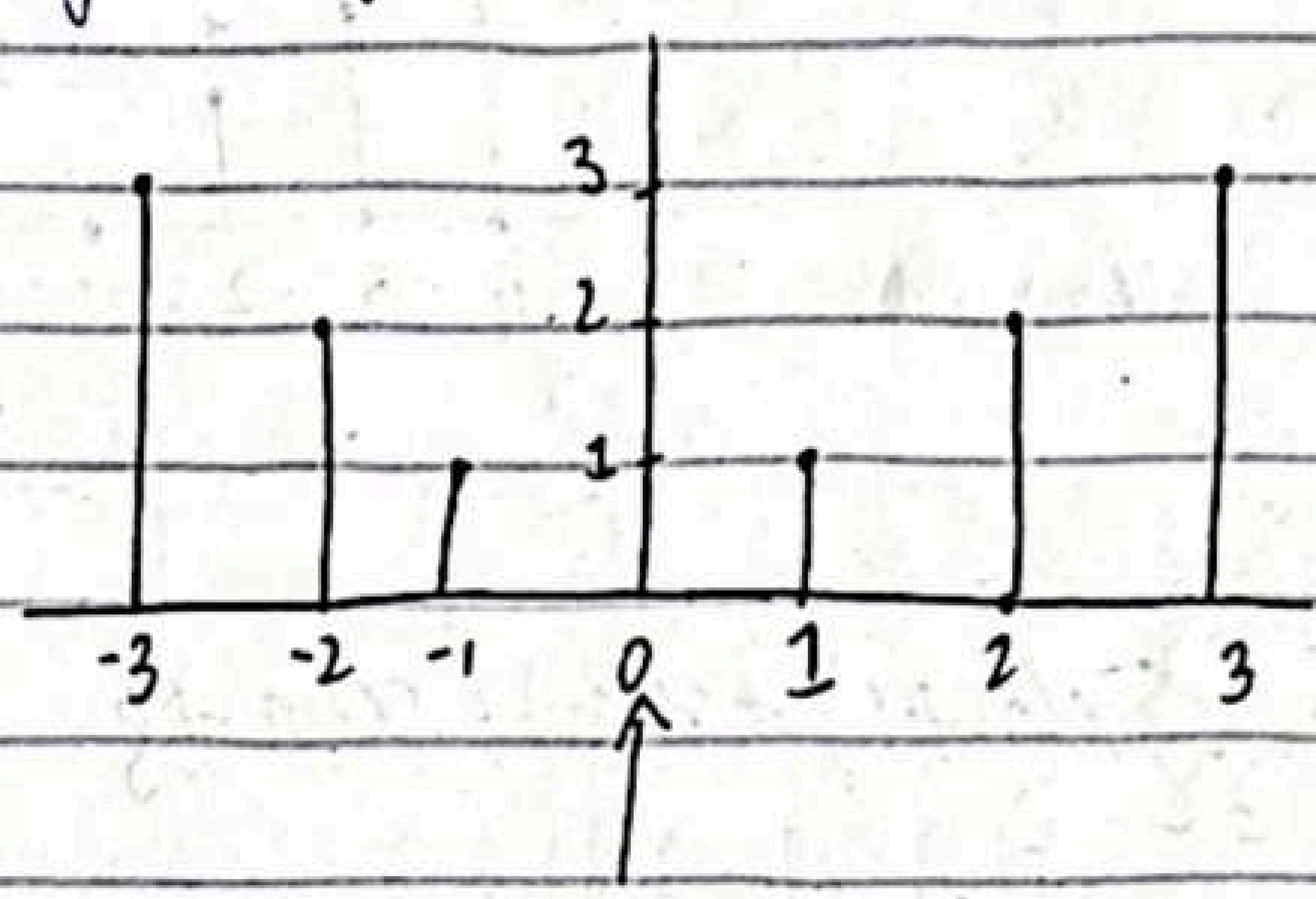
$x(n)$ slides

Example 2.2.1:-

$$x(n) = \begin{cases} |n| & -3 \le n \le 3 \\ 0 & \text{otherwise} \end{cases}$$

(a) $y(n) = x(n)$

(a) In this case the output is exactly, the same as the input signal. Such system is known as the identity system.



$= x(n)$

$x(n-1) = 1$

(b) $y(n) = x(n-1)$

This system simply delays the input by one sample, output is

$$x(n) = \{ \ldots\ldots 0, 3, 2, \underset{\uparrow}{1}, 0, 1, 2, 3, 0, \ldots \}$$

(c) $y(n) = x(n+1)$

system advances the input ~~or~~ one same into the future e.g. $n = 0$, $y(0) = x(1)$

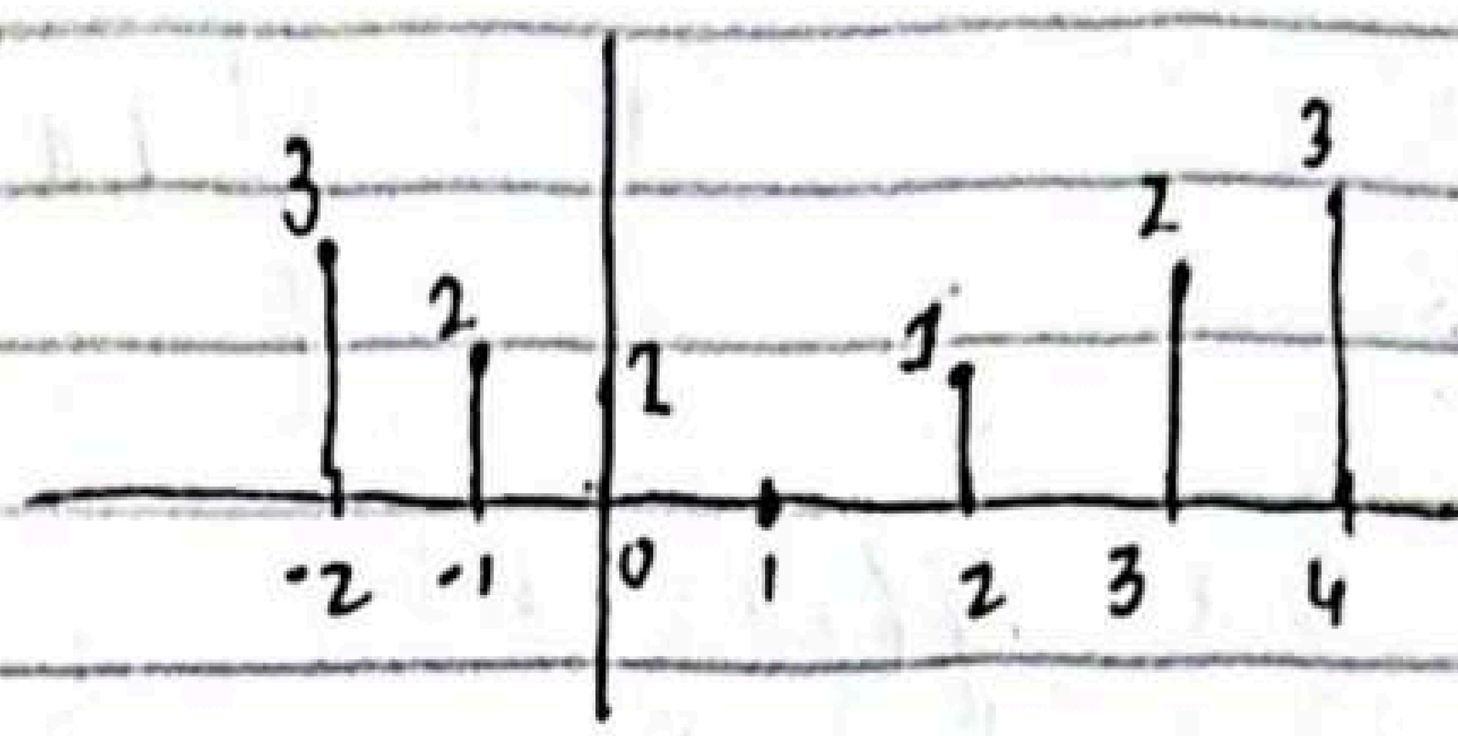
$$x(n) = \{ \ldots 0, 3, 2, 1, 0, \underset{\uparrow}{1}, 2, 3, 0, \ldots \}$$

$n-1=0$

$\Rightarrow n=1$

-2 to 4

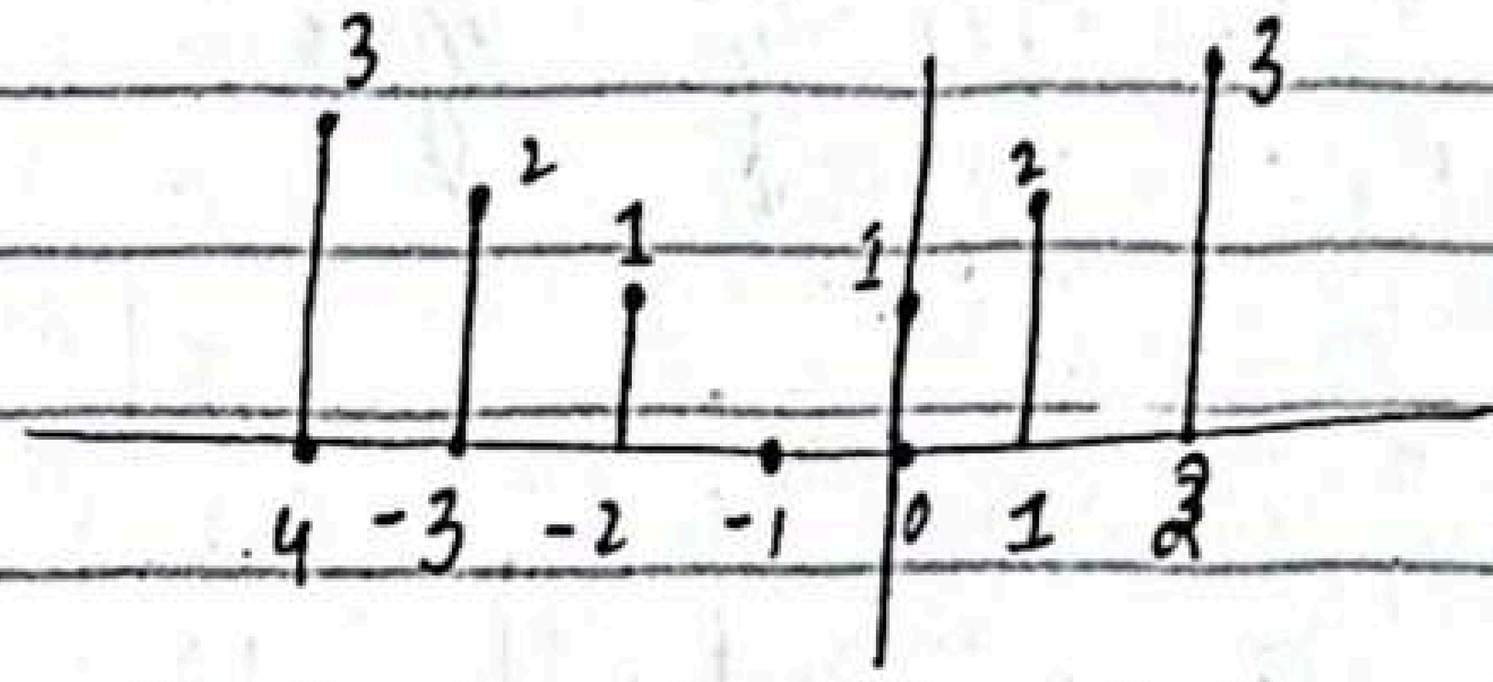
$y(n) = x(n-1)$



Now,

-3	-2
-2	-1
-1	0
0	1
1	2
2	3
3	4
old axis	New axis

$x(n+1)$



(d) $y(n) = \frac{1}{3}\{x(n+1) + x(n) + x(n-1)\}$

The output of the system at any time is mean value of present, immediate past and immediate future samples.

$y(0) = \frac{1}{3}\{x(-1) + x(0) + x(1)\} = \frac{1}{3}\{1+0+1\} = \frac{2}{3}$

$y(n) = \{\ldots 0, 1, \frac{5}{3}, 2, 1, \underset{\uparrow}{\frac{2}{3}}, 1, 3, \frac{5}{3}, 1, 0 \ldots\}$

(e) $y(n) = \max[x(n+1), x(n), x(n-1)]$

(f) $y(n) = \sum_{k=-\infty}^{n} x(k) = x(n) + x(n-1) + x(n-2) + \dots$

(e) The system selects as its output at time n the maximum value of three input samples $x(n-1)$, $x(n)$ and $x(n+1)$. Thus the response of this system to the input signal $x(n)$ is

$$x(n) = \{0, 3, 3, 3, 2, \underset{\uparrow}{1}, 2, 3, 3, 3, 0, \dots\}$$

(f)

This system is basically an accumulator that computes the ~~a~~ running sum of all the past input values up to present time

slides.

Example 2.2.2:-

$x(n) = nu(n)$. Determine its output

(a) if it infinity relaxed [i.e. $y(-1) = 0$]

(b) Initially $y(-1) = 1$

$$y(n) = \sum_{k=-\infty}^{n} x(k) = \sum_{k=-\infty}^{-1} x(k) + \sum_{k=0} x(k)$$

$$= y(-1) + \sum_{k=0}^{n} x(k)$$

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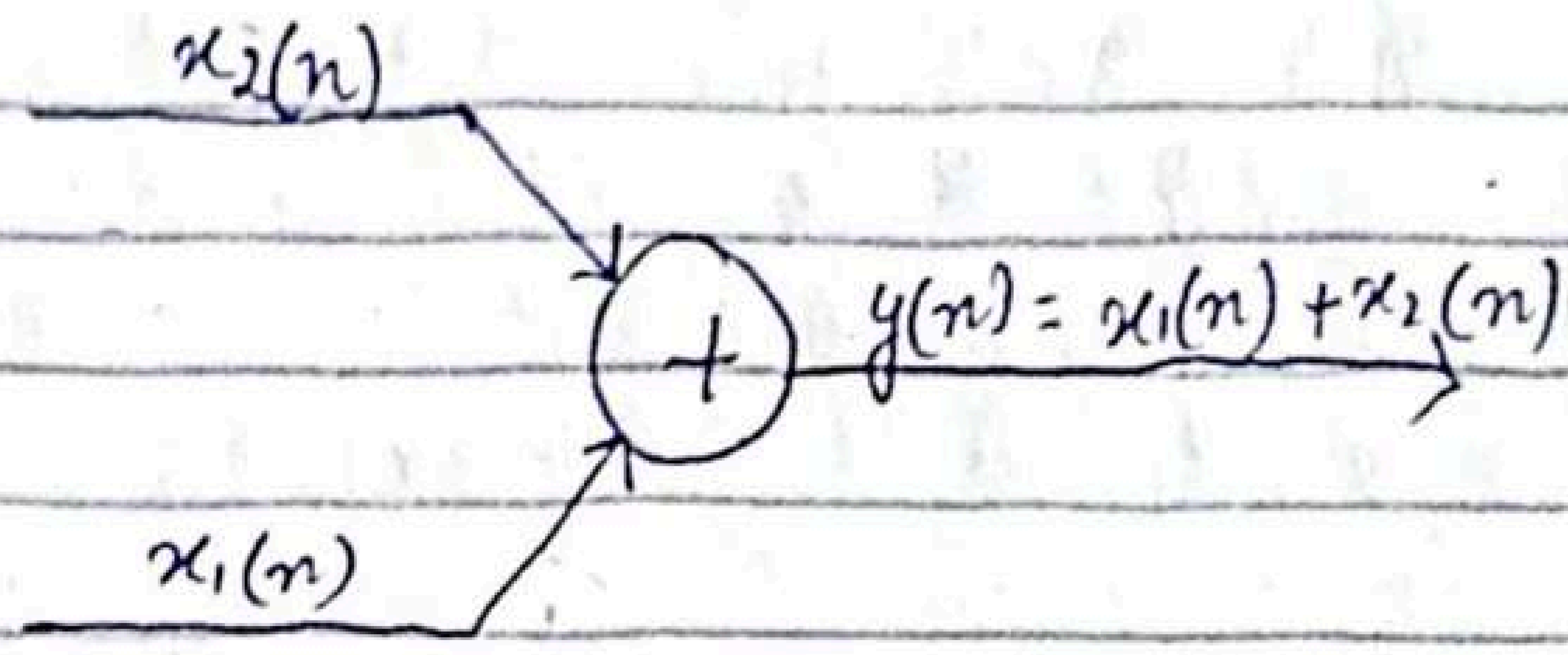
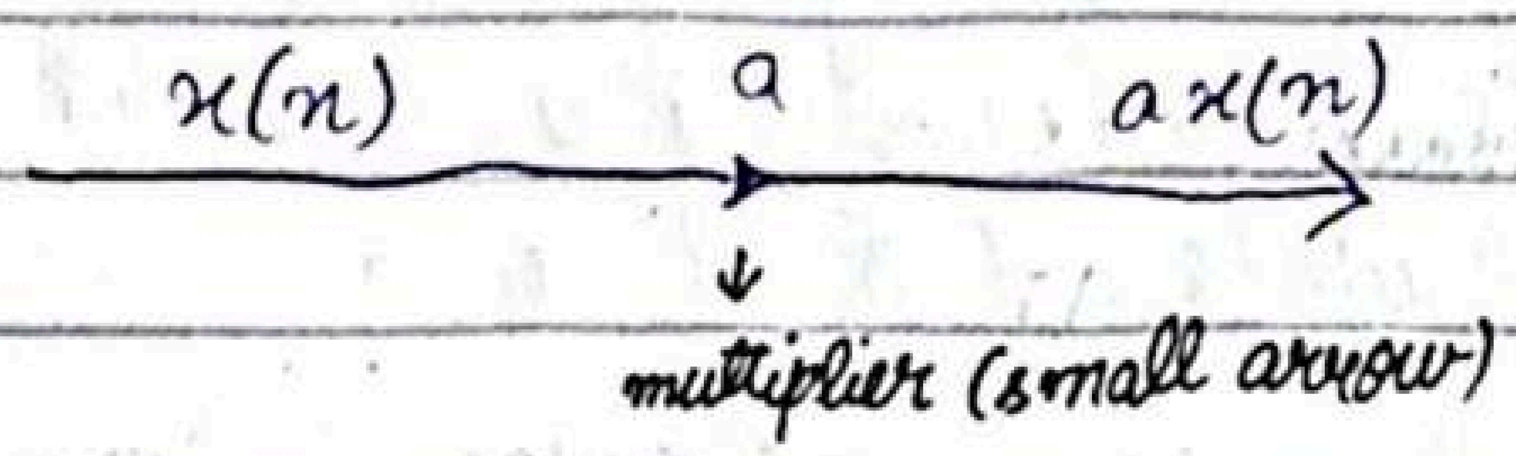
Block Diagram Representation of Discrete Time signals systems

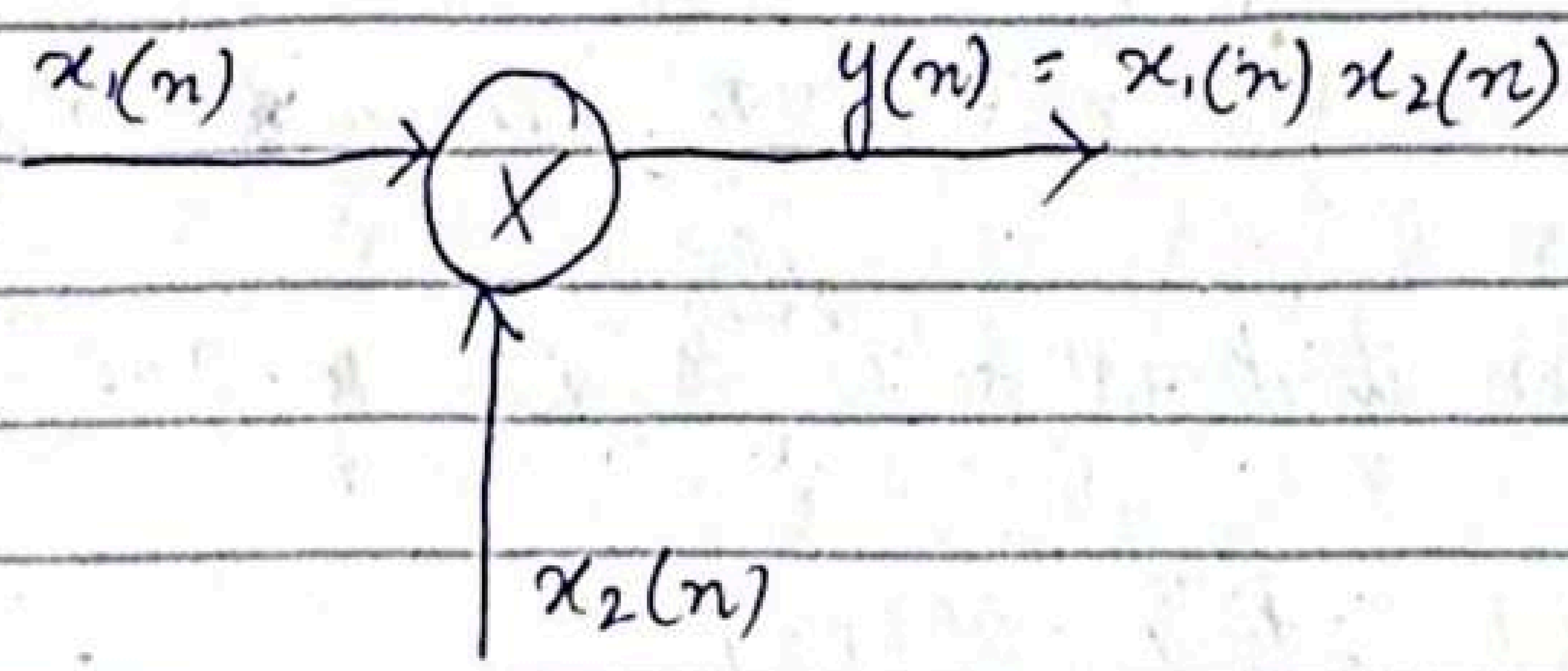
An adder:-



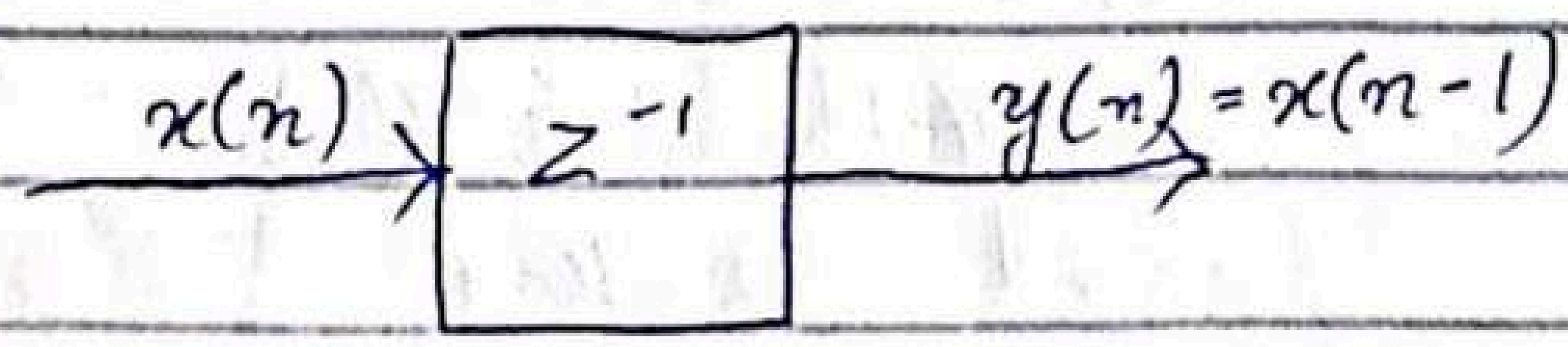
A constant Multiplier:-



A signal Multiplier:-



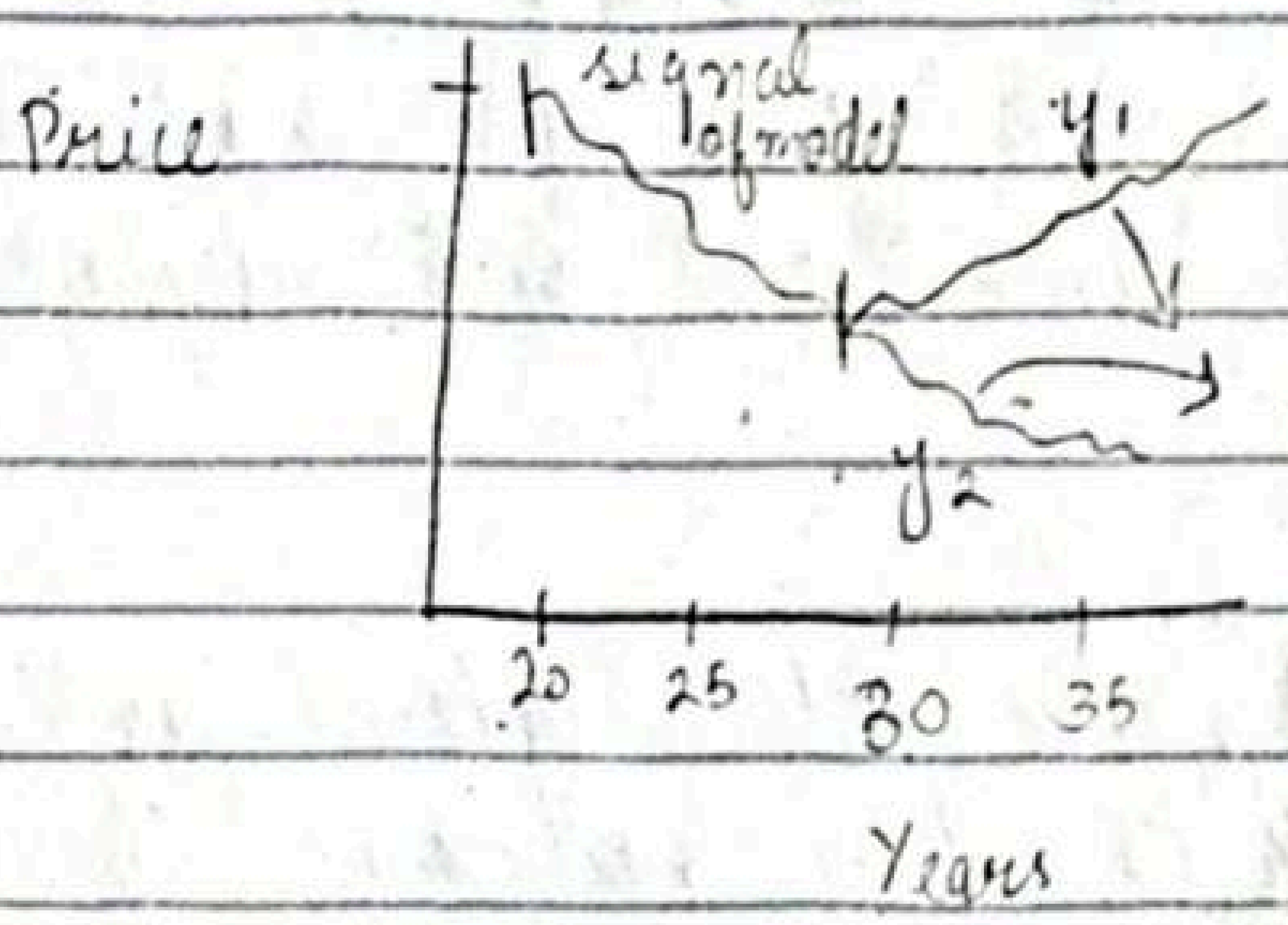
A unit delay Element:-



A unit Advance Element:-

$$x(n) \rightarrow \boxed{z} \rightarrow y(n) = x(n+1)$$

We observe that any such advance is physically impossible in real time since, in fact, it involves looking into the future of signal. On other hand, if we store the signal in memory of computer, we can recall any sample at an time. In such a non-real time application it is possible to advance the signal $x(n)$ in time.



this signal is future for model

y_2 signal is better than y_1 because it is closer to trend

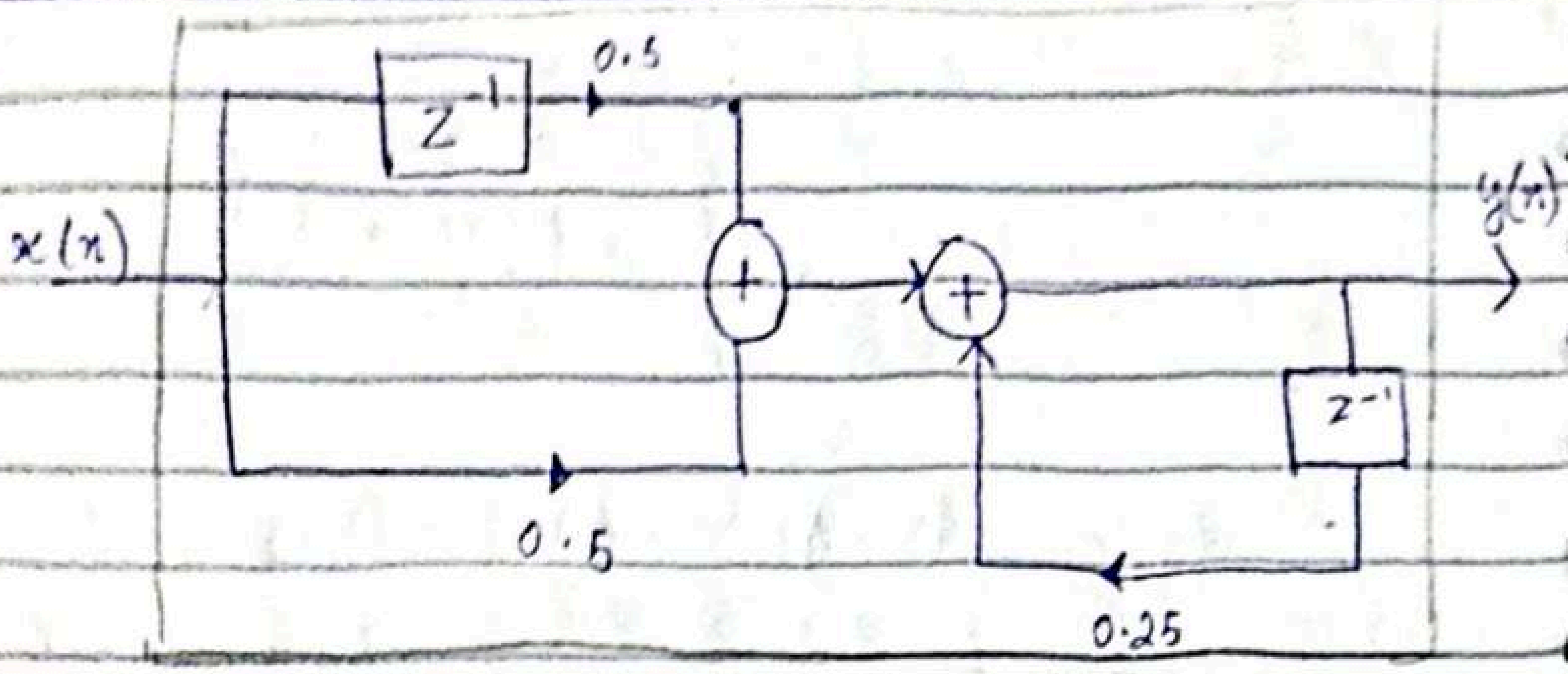
Example 2.2.3

sketch block Diagrams of :

$$y(n) = \frac{1}{4} y(n-1) + \frac{1}{2} x(n) + \frac{1}{2} x(n-1)$$

$x(n)$ is input and $y(n)$ is output

Black Box



Classification of Discrete time ~~signal~~ systems:-

Static vs Dynamic :-

A discrete-time sig. system is called static or memoryless if its output at any instant n depends at most on input sample at same time, but not on past or future samples.

In dynamic the system is said to be dynamic or to have memory. If the output of a system at time n is completely determined by the input samples in the interval from $n-N$ to n ($N \geq 0$) the system is said to have memory of duration N. → slides

The systems described by following:-

$$y(n) = ax(n)$$
$$y(n) = nx(n) + bx^3(n)$$

They both are static or memoryless. Note that there is no need to store any past inputs

$$y(n) = x(n) + 3x(n-1)$$

$$y(n) = \sum_{k=0}^{n} x(n-k)$$

$$y(n) = \sum_{k=0}^{\infty} x(n-k) = y$$

They are dynamic systems or systems with memory.

Time - Invarient verses Time variant:-

A system is called time invarient if its input - output don't change with time.

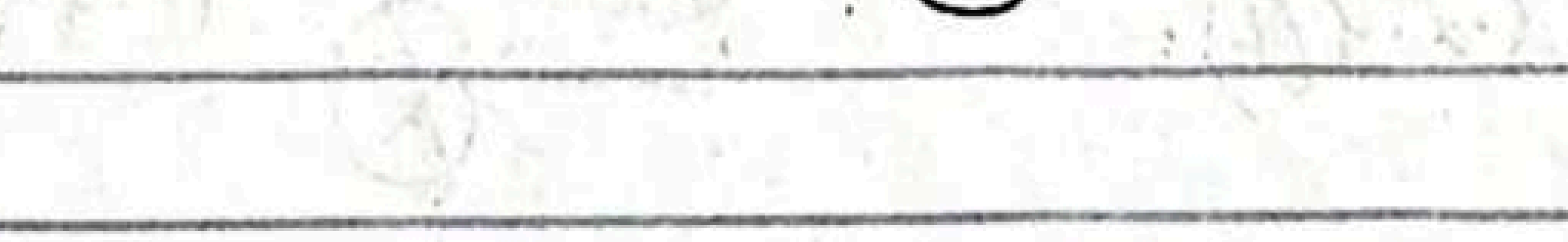
Ex.

$$x(n) \rightarrow y(n)$$

$$x(n-k) \rightarrow y(n-k)$$

If paths are same then time invarient.

Example 2.2.4



$$y(n) = x(n) - x(n-1)$$

Let us assume of form

$$y_1(x) = x_1(x) - x_1(n-1) \quad \text{—— (i)}$$

Let us introduce time delay 'k' in the input signal in eq (i)

$$x_2(n) = x_1(n-k)$$

the delay input therefore results in output :-

$$y_2(n) = x_2(n) - x_2(n-1) \rightarrow \text{delay by } (n-1)$$

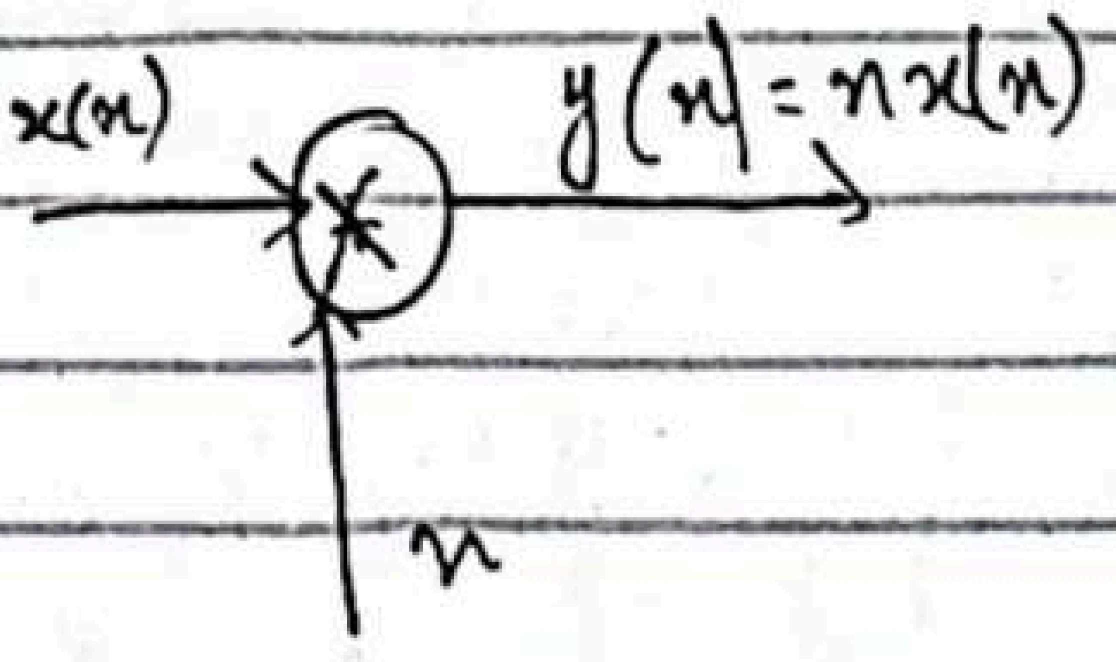
$$y_2(n) = x_1(n-k) - x_1(n-1-k) \quad \text{—— (ii)}$$

Now we introduce the same time delay 'k' in the output of eq

$$y_1(n-k) = x_1(n-k) - x_1(n-k-1) \quad \text{—— (iii)}$$

(ii) and (iii) are same so it is time invariant.

Example 2.2.4 (b) :-



$$y(n) = nx(n)$$

step 1 :-

$$y_1(x) = nx_1(n)$$

static → present
Dynamic past, present, future
causal → present, past
Non-causal, f, p, p

$x_2(n) = n x_1(n-k)$

$y_2(n) = n x_2(n) = n x_1(n-k)$ — (i)

delay 'k' in output

$y(n-k) = (n-k) x_1(n-k)$

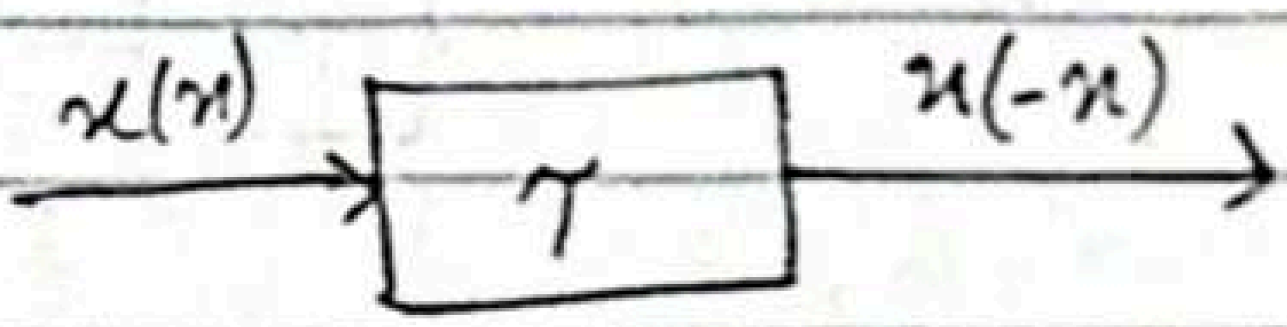
$y_1(n-k) = n x_1(n-k) - k x_1(n-k)$ — (ii)

$y_2(n) \neq y_1(n-k)$

Therefore it is time variant

Example:-

$y(n) = x(-n)$



$y_1(n) = x_1(-n)$

Delay in input

$x_2(n) = x_1(n-k)$ → always write this compulsory

$y_2(n) = x_2(n)$

$y_2(n) = x_1(-n-k)$ — (ii)

Now for output

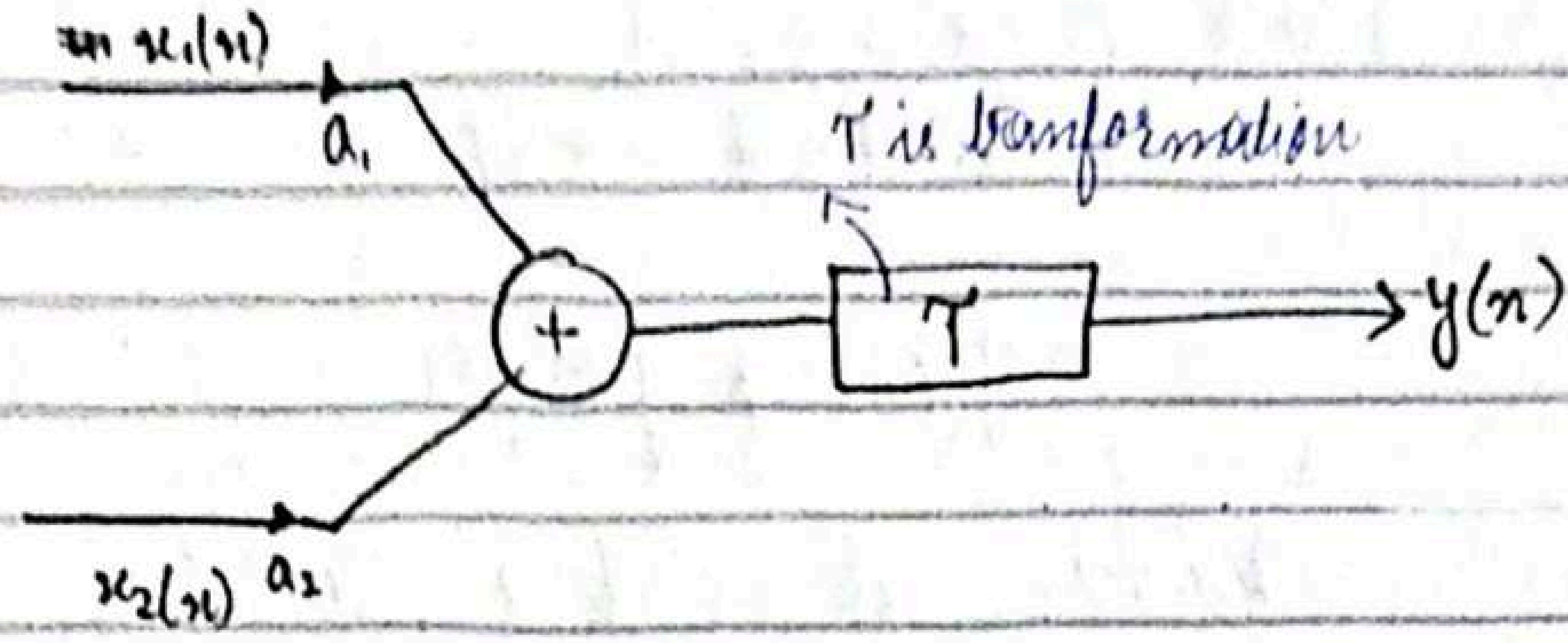
$y_1(n-k) = x_1(-n-k) = x_1(-n+k)$

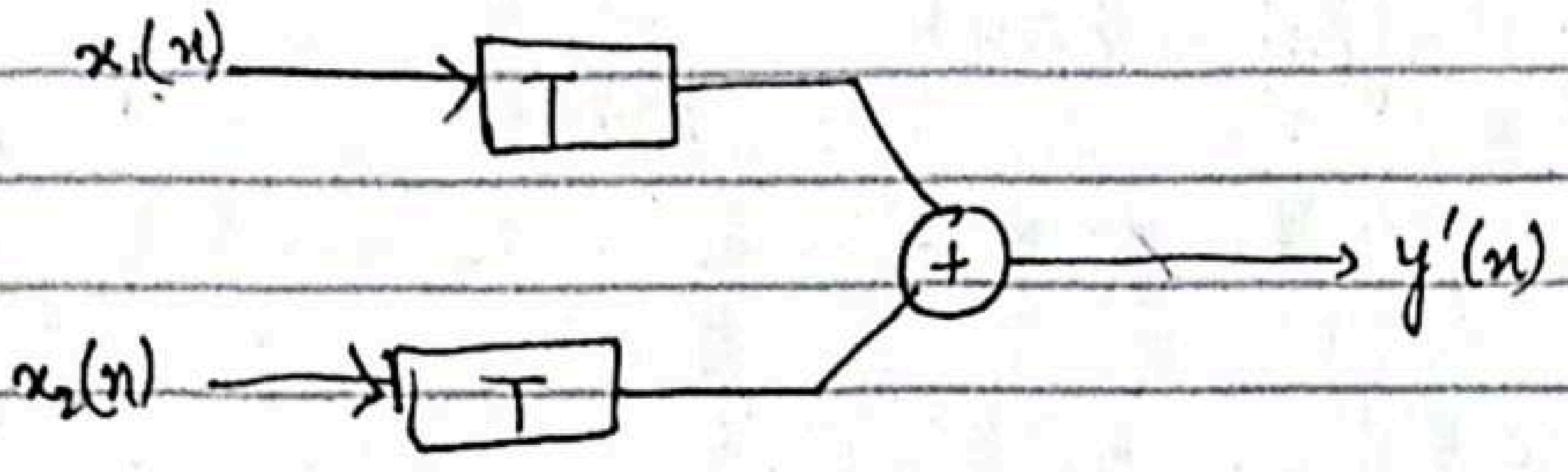
$y_1(n-k) \neq y(n)$ $\boxed{\text{Time variant}}$

Linear vs Non-Linear:-

A system that satisfies the superposition principle. superposition requires that response of system to a weighted

slides





additive and Multiplicative properties are superposition.

Example 2.2.5:-

(a) $y(n) = n x(n)$

(b) $y(n) = x(n^2)$

(c) $y(n) = x^2(n)$

(d) $y(n) = Ax(n) + B$

(e) $y(n) = e^{x(n)}$

(a) For input $x_1(n)$, $x_2(n)$

step 1:- $y_1(n) = n\, x_1(n)$

$y_2(n) = n\, x_2(n)$

step 2:-

$$y_3(n) = T[(a_1 x_1(n) + a_2 x_2(n)]$$

$$= n[(a_1 x_1(n) + a_2 x_2(n)]$$

$$= a_1 n x_1(n) + a_2 n x_2(n)$$

linear combination of two outputs

$$a_1 y_1(n) + a_2 y_2(n) = a_1 n x_1(n) + a_2 n x_2(n)$$

$\downarrow$

It is linear

solve rest of parts.

P, P

04/02/25

Causal versus non-causal systems :-

A system is said to be causal if the output of system at any time n.] slides.

e.g. = $y(n) = F[x(n), x(n-1), x(n-2) \ldots$

Example :-

Determine if the systems described by following input - output eq. are causal or non - causal.

(a) $y(n) = x(n) - x(n-1)$

(b) $y(n) = \sum_{k=-\infty}^{n} x(k)$

(c) $y(n) = ax(n)$

(d) $y(n) = x(n) + 3x(n+4)$

(e) $y(n) = x(n^2)$

(f) $y(n) = x(2n)$

(g) $y(n) = x(-n)$

a, b, c are causal since output depends only on present and past inputs. On other hand, the part (d), e, f and non - causal, since output depends on future values of input. g is also non - causal, eg. $n = -1$ yields $y(-1) = x(1)$ thus output at $n = -1$ depends on input at $n = 1$ which is two units of time

time into the future.

Stable vs Unstable system:-

Unstable systems usually exibit erratic and extreme behaviour and cause overflow in any practical implementation ↓ slides

$Mx \rightarrow$ input
$My \rightarrow$ output

$$|x(n)| \leq M_x < \infty \qquad |y(n)| \leq M_y < \infty$$

for all n. If for some bounded input sequence $x(n)$, the output is unbounded (infinite) the system is classified as unstable.

P,?

Example 2.2.7

Consider the non-linear sytem described by the input - output eq. Dy. causa.

$$y(n) = y^2(n-1) + x(n)$$

As an input sequence we select the bounded signal

$$x(n) = C\delta(n)$$

C is constant. We also assume that $y(-1) = 0$. Then the output sequence is

$$y(0) = C, \quad y(1) = C^2, \quad y(2) = C^4 \quad \ldots \quad y(n) = C^{2^n}$$

$$y(0) = y^2(0-1) + x(0) = y^2(-1) + 0$$

$x(n) = C\delta(n)$

assume $y(-1) = 0$

$y(0) = C$, $y(1) = C^2$, $y(2) = C^4$

$y(0) = 0 + x(0)$

$= C\delta(n)$, $n = 0$ $\qquad \delta(0) = 1$

$y(0) = C$

(Analysis)

Resolution of Discrete-Time Signal into impulses:-

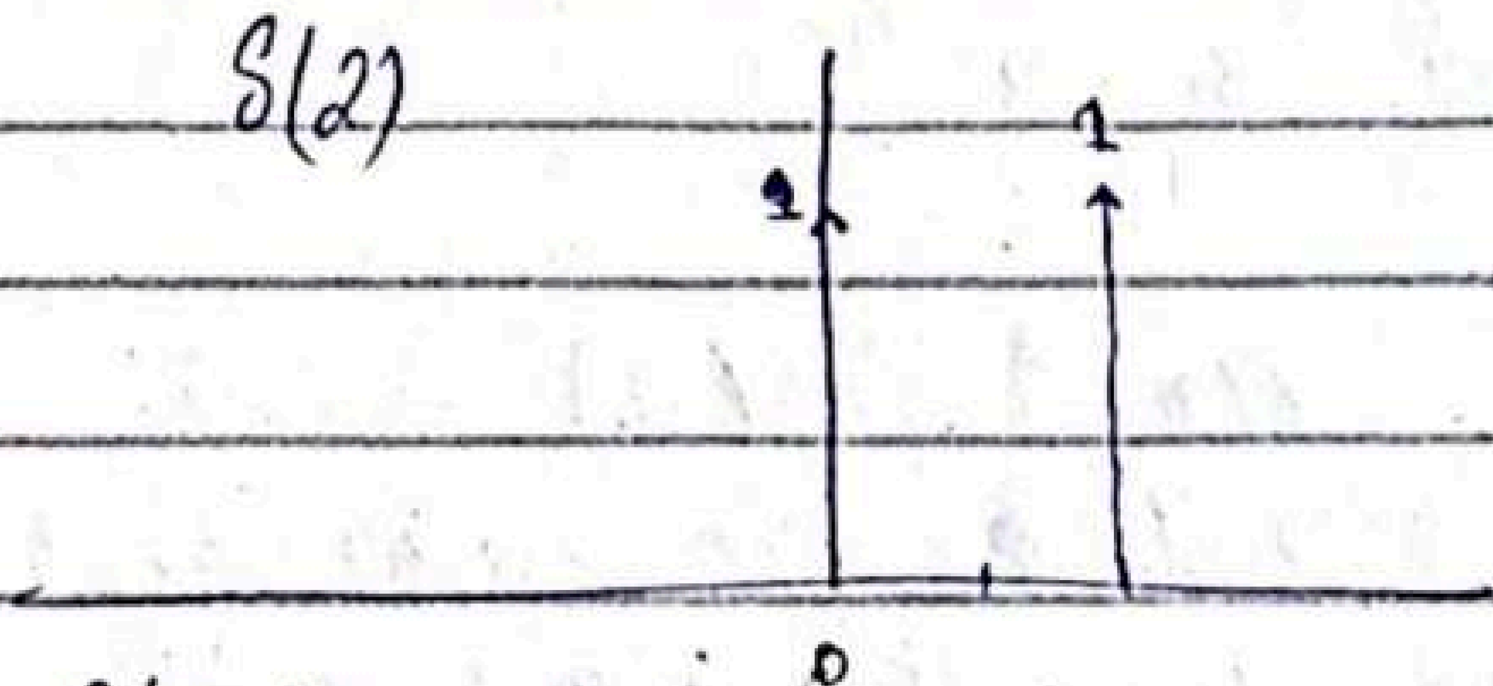
$$x_k(n) = \delta(n-k)$$

k represents the delay of unit sample sequence.] slides

Now suppose that we multiply the two sequences $x(n)$ and $\delta(n-k)$. Since $\delta(n-k)$

Example

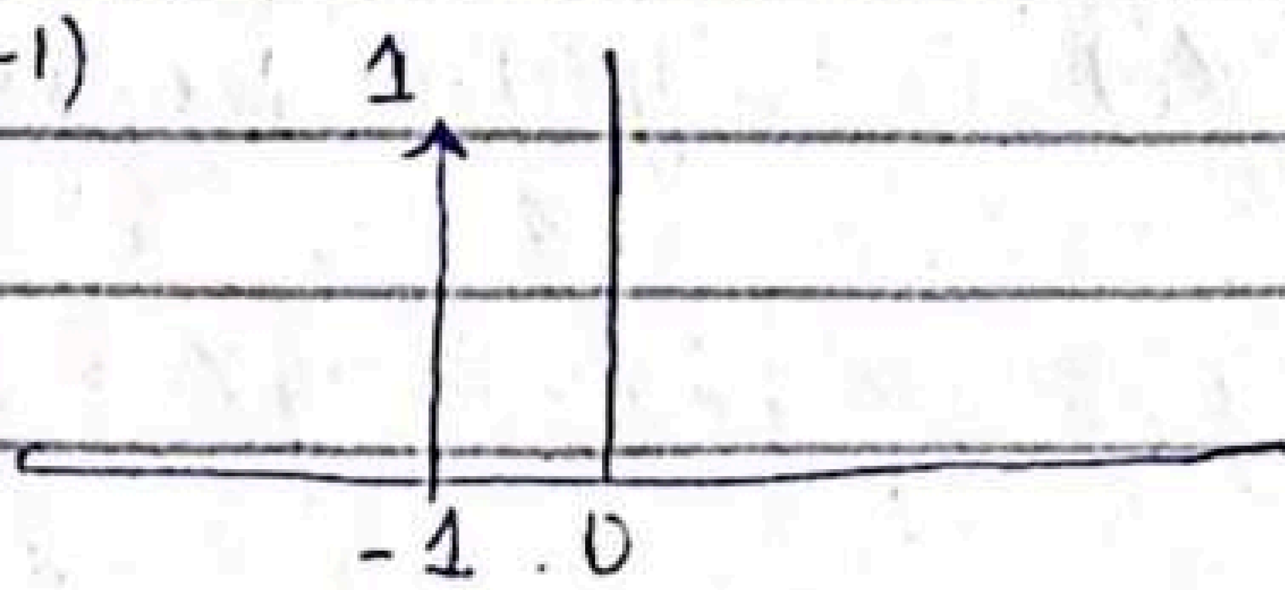
$\delta(2)$



$\delta(n-k)$

$\delta(2-3)$ $\quad k = 3$

$\delta(-1)$



$$x(n) = \sum_{k=-\infty}^{\infty} x(k)\,\delta(n-k)$$

Example:-

$$x(n) = [2, \underset{\uparrow}{4}, 0, 3] \qquad n+1$$

$n = -1, 0, 1, 2$

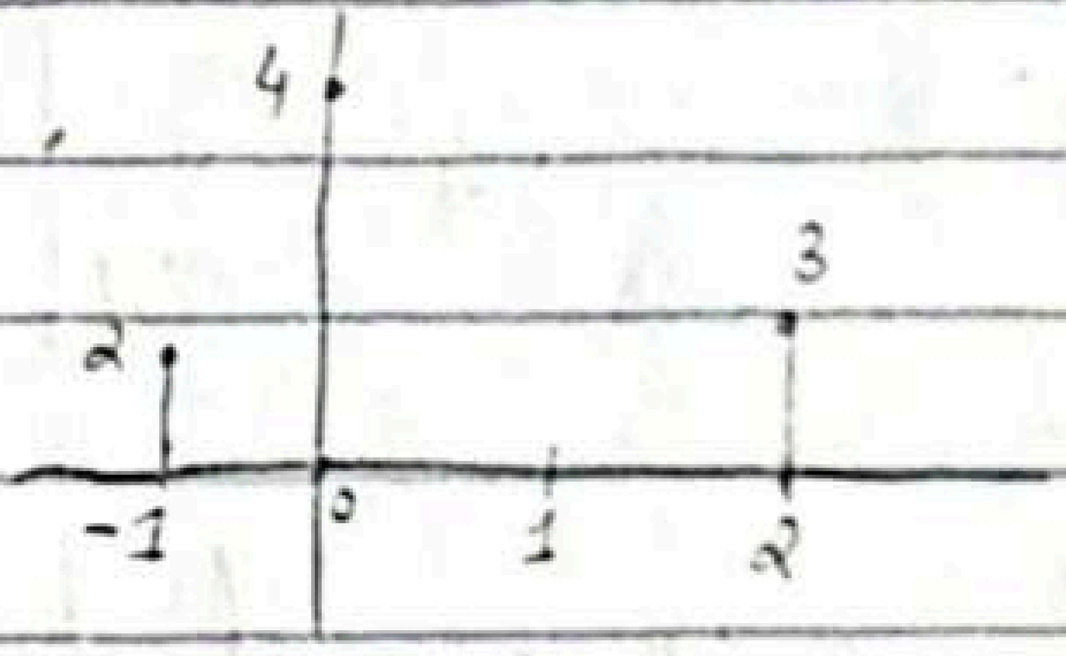
Resolve sequence $x(n)$ into a sum of weighted impulse sequences.

$x(n)$ is non-zero for time instants $n = -1, 0, 2$ we need three impulses at delays $k = -1, 0, 2$

$$x(n) = \underbrace{2\delta(n+1)}_{2\times 1} + 4\delta(n) + 3\delta(n-2)$$

$\delta(n-k)$ shift to right because of k

$\delta(n+k)$ shift to left because of k



$x(-1) = 2\delta(n+1)$

$x(0) = 4\delta(n)$

$x(2) = 3\delta(n-2)$

$x(n) = 2\delta(n+1) + 4\delta(n) + 3\delta(n-2)$

→ slides

$y(n) = \sum_{k=-\infty}^{\infty} x(k)h(n-k)$ convolution sum.

Example 2.3.2

The impulse response of LTI is

$$h(n) = [1, \underset{\uparrow}{2}, 1, -1]$$

Determine response of system to input signal

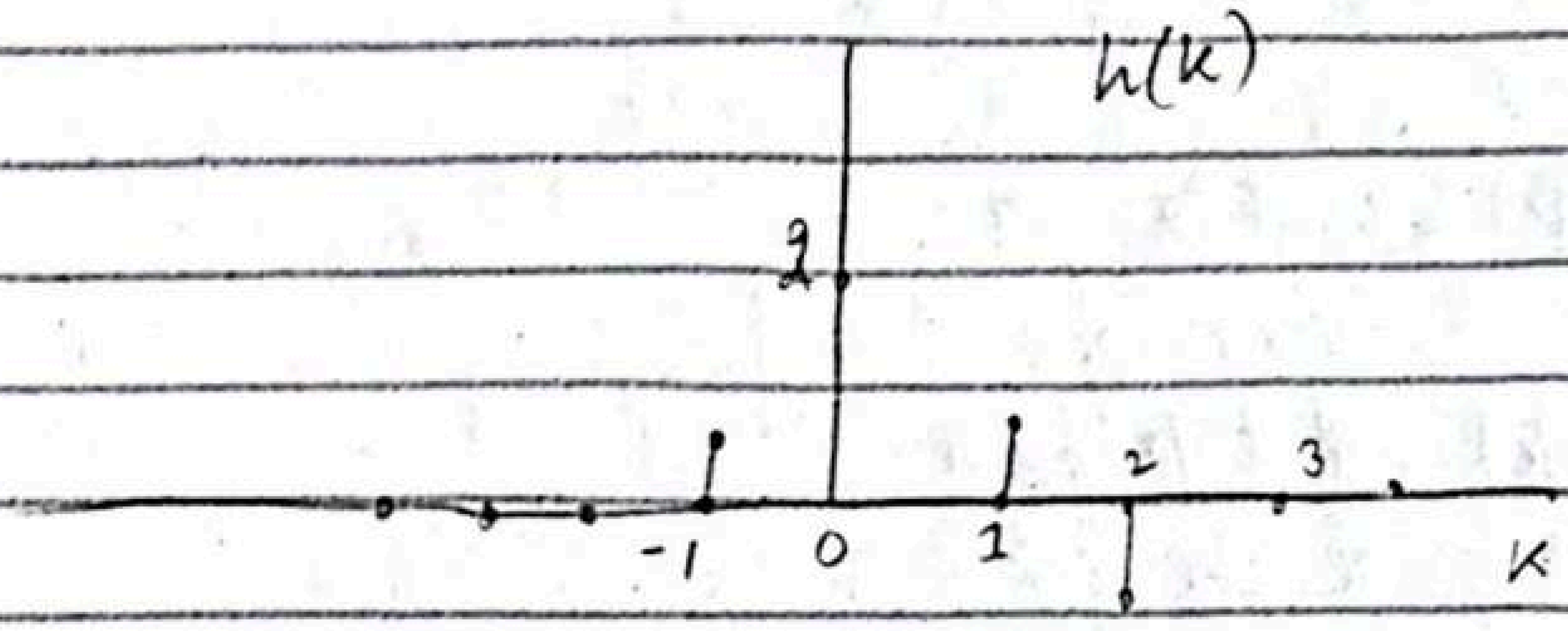
$$x(n) = [\underset{\uparrow}{1}, 2, 3, 1]$$

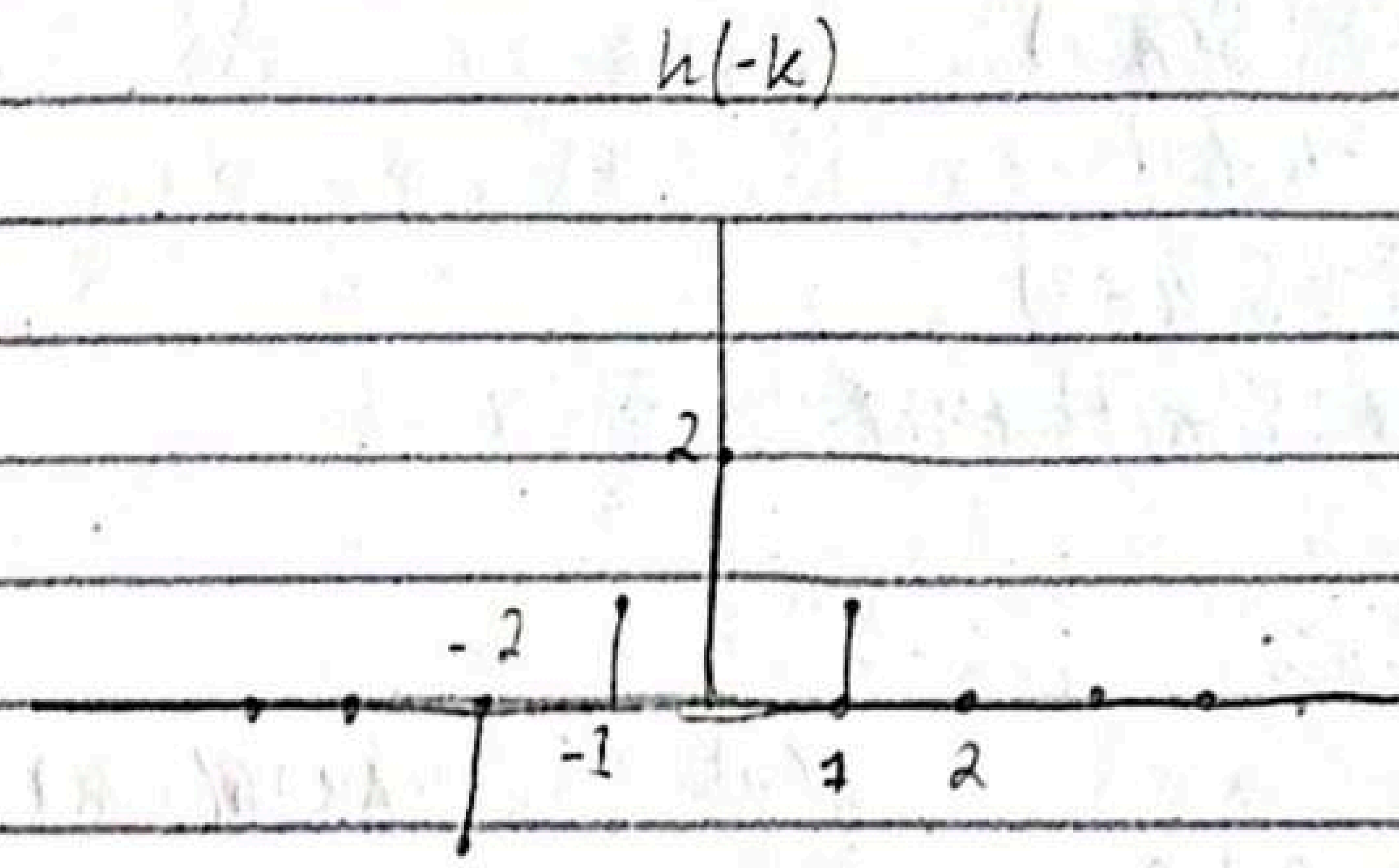
solution :-

$$y(n) = \sum_{k=-\infty}^{\infty} x(k)h(n-k)$$

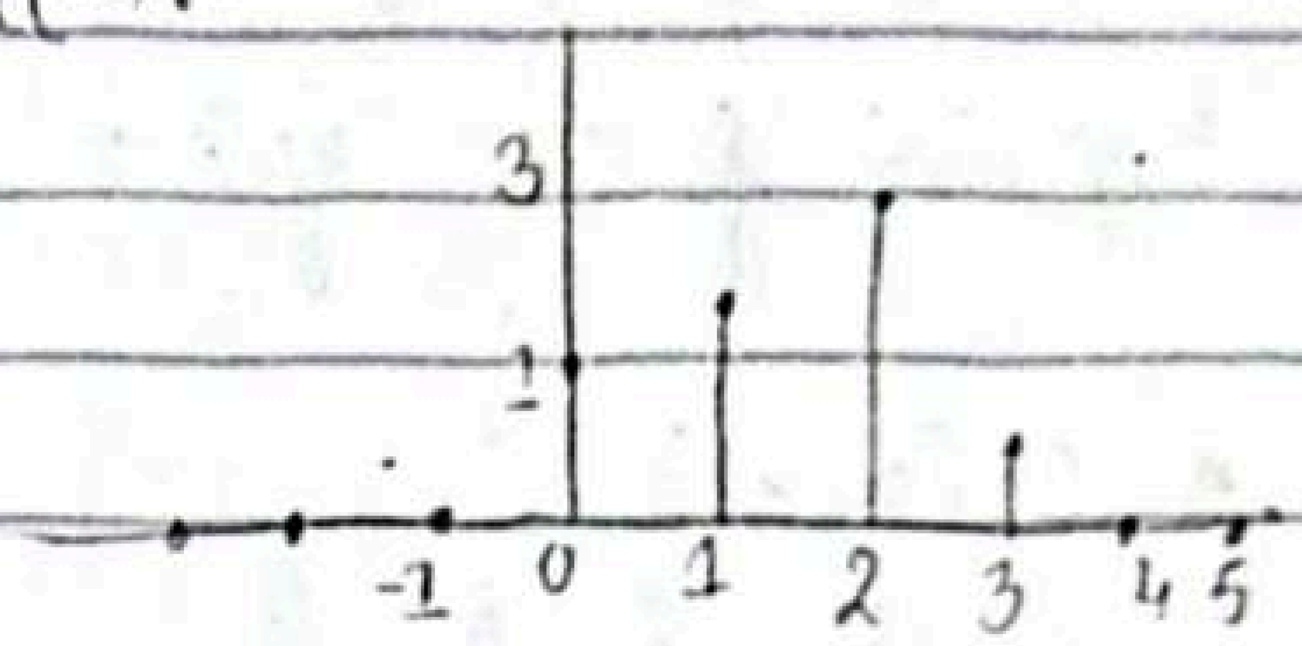
put $y(0)$ in eq

$$y(0) = \sum_{k=-\infty}^{\infty} x(k)\,h(-k)$$





$x(k)$:-



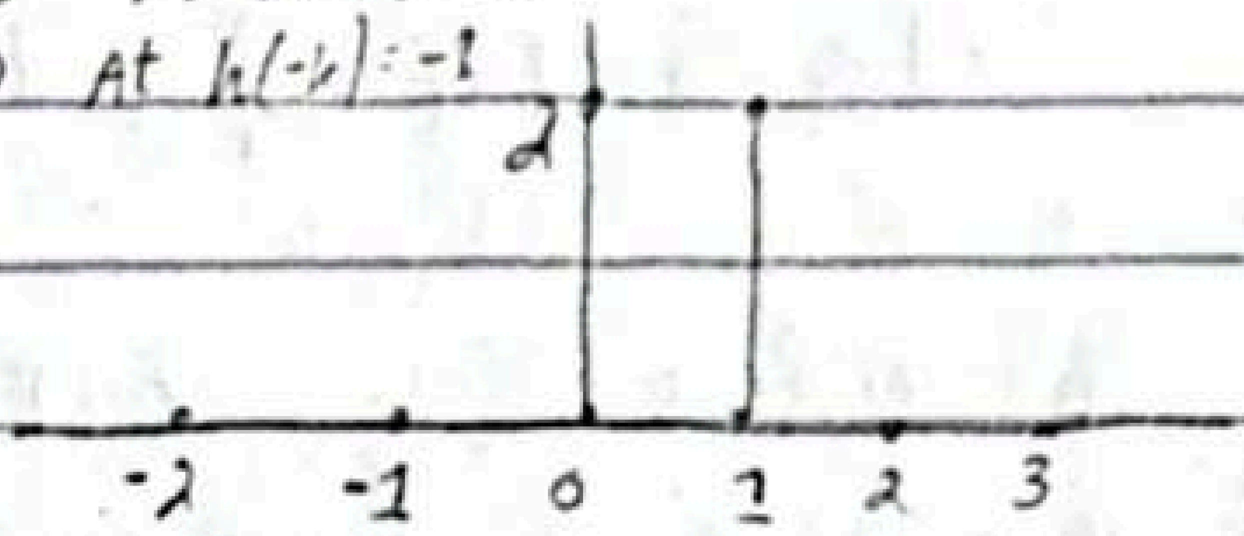
$x(k)h(-k)$

$0 \times -2 = 0$ At $h(-k) = -2$

$0 \times 2 = 0$ At $h(-k) = -1$

$1 \times 2 = 2$

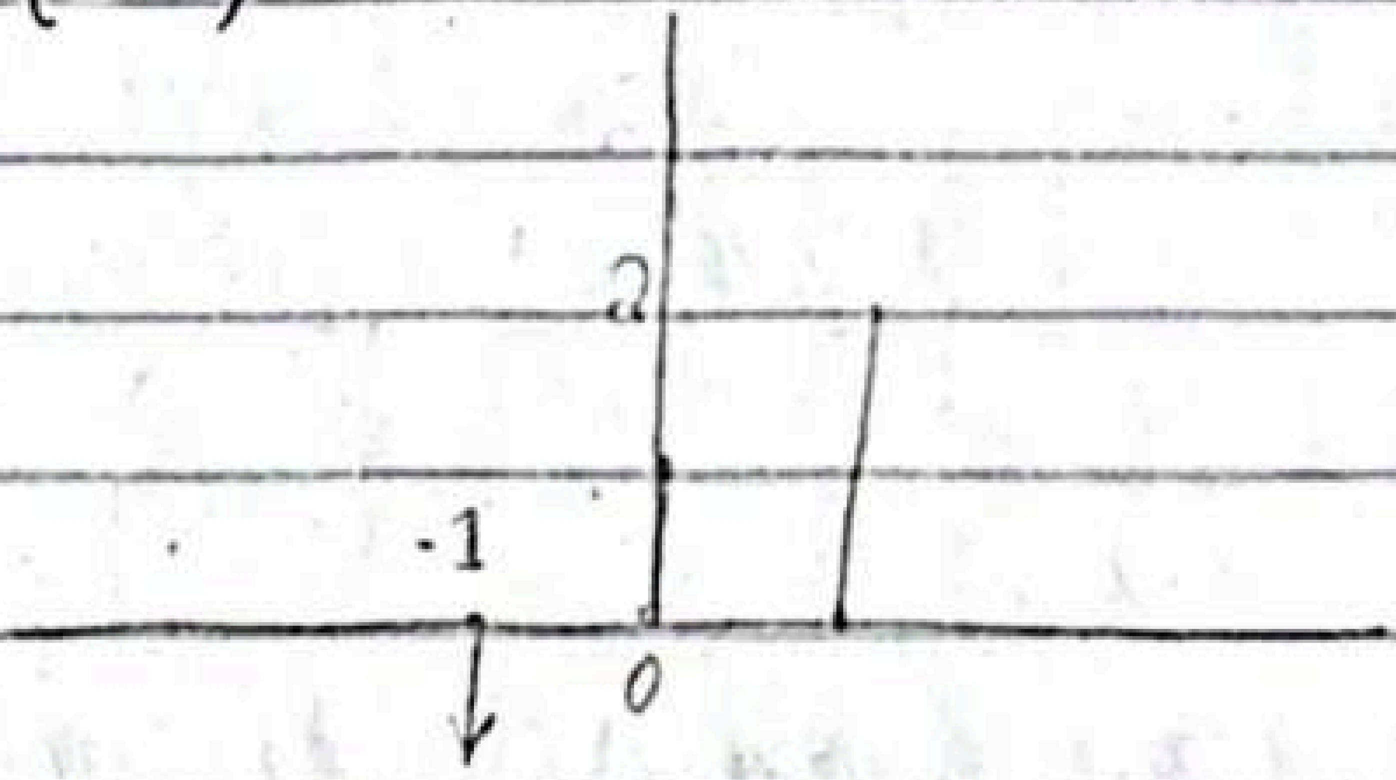
$2 \times 1 = 2$



$y(0) = 2 + 2 = 4$ sum of amplitude

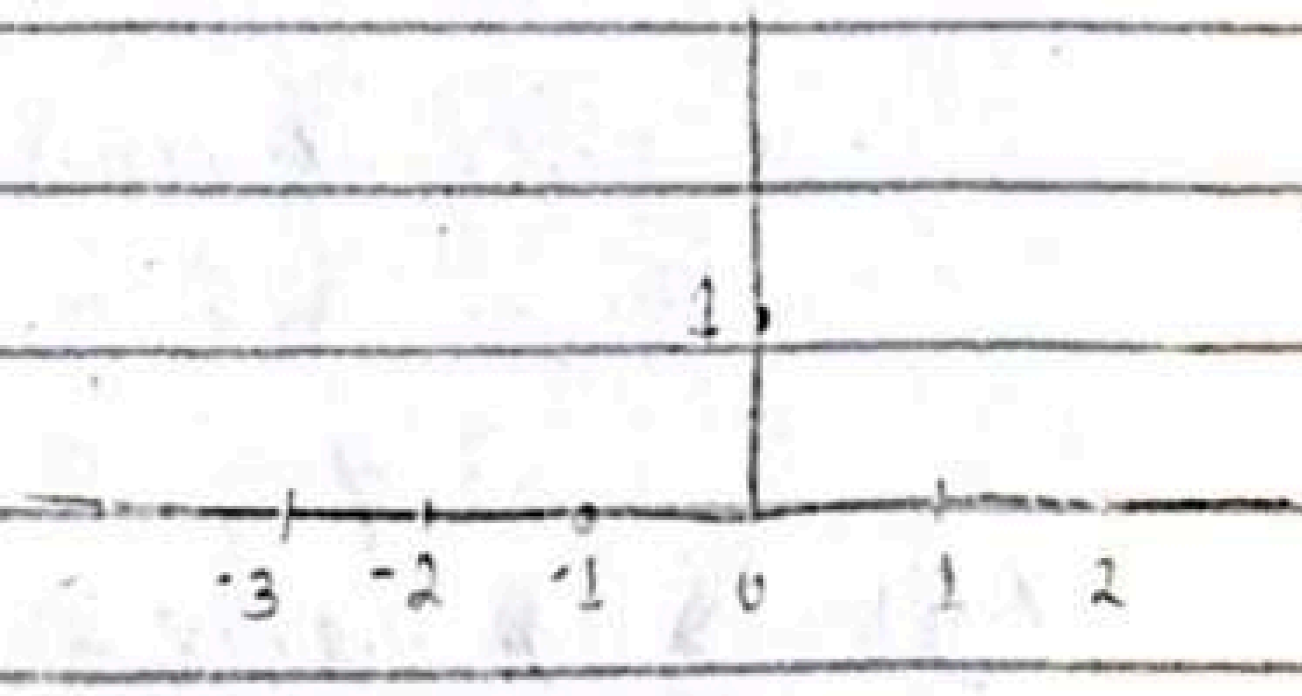
Example:-

$h(1-k)$



shift $h(-k)$ by 1

$h(1-k)\,x(k)$



06/02/25

Example :-

Explain if the system is stable or not?

$y(n) = \cos[x(n)]$

As we know that

$$-1 \leq \cos[x(n)] \leq 1$$

for all values of $x(n)$. Hence, system is stable irrespective of value of $x(n)$

~~Auto~~corelation :-

Corelation resembles convolution, signals are similar. It is used to check if 2 signals are similar. → slides

We can represent the recieved signal

$y(n) = \alpha x(n-D) + w(n)$

α is some attenuation factor representing → slides

$x(n) \rightarrow$ reference signal, transmitted signal

$y(n) \rightarrow$ recieved signal

The signal recieved is written as

$y(n) = x_i(n) + w(n)$, $i = 0, 1$

, $0 \leq n \leq L-1$

no. of samples

Crosscorelation And AutoCorelation:-

$x(n)$ and $y(n)$ is a sequence $r_{xy}(l)$, is defined as.

remain same; delayed, shifted by l

$$r_{xy}(l) = \sum_{n=-\infty}^{\infty} x(n)\, y(n-l) \qquad l = 0, \pm 1, \pm 2, \cdots$$

l is (time) shift or lag parameter and subscription represent autocorelation or corelation by x and y.

Example 2.6.1

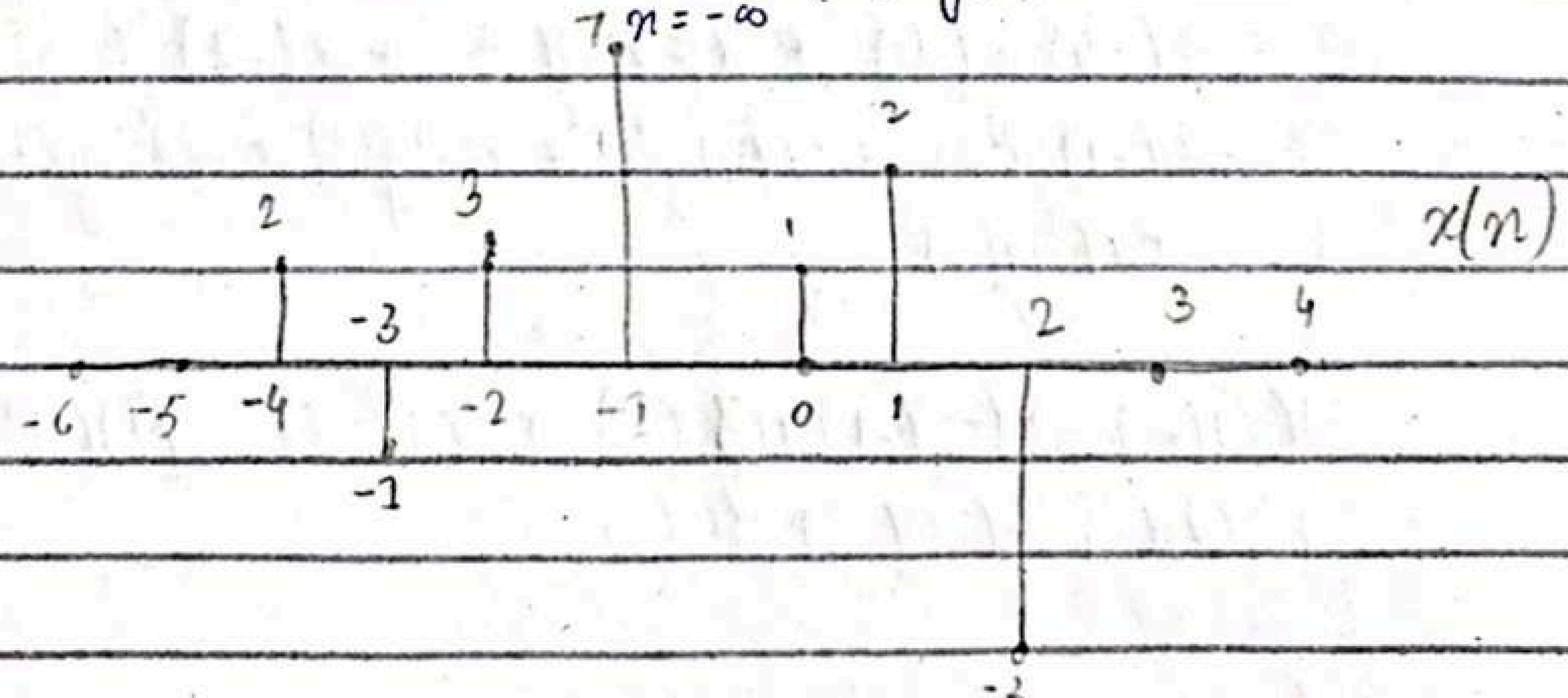
$$x(n) = \{\cdots 0, 0, 2, -1, 3, 7, \underset{\uparrow}{1}, 2, -3, 0, 0 \cdots\}$$

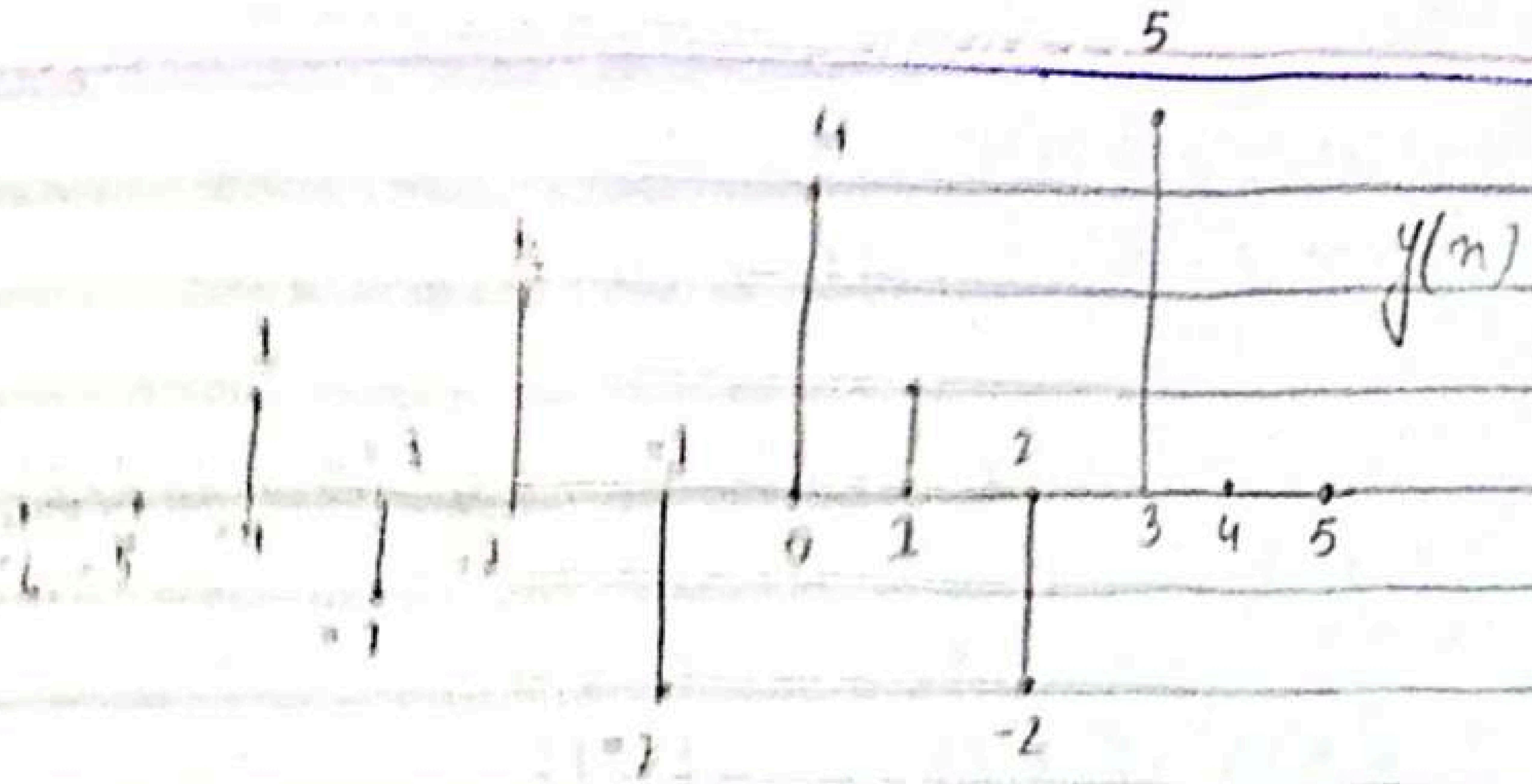
$$y(n) = \{\cdots 0, 0, 1, -1, 2, -2, \underset{\uparrow}{4}, 1, -2, 5, 0, 0 \cdots\}$$

compute $r = 0$

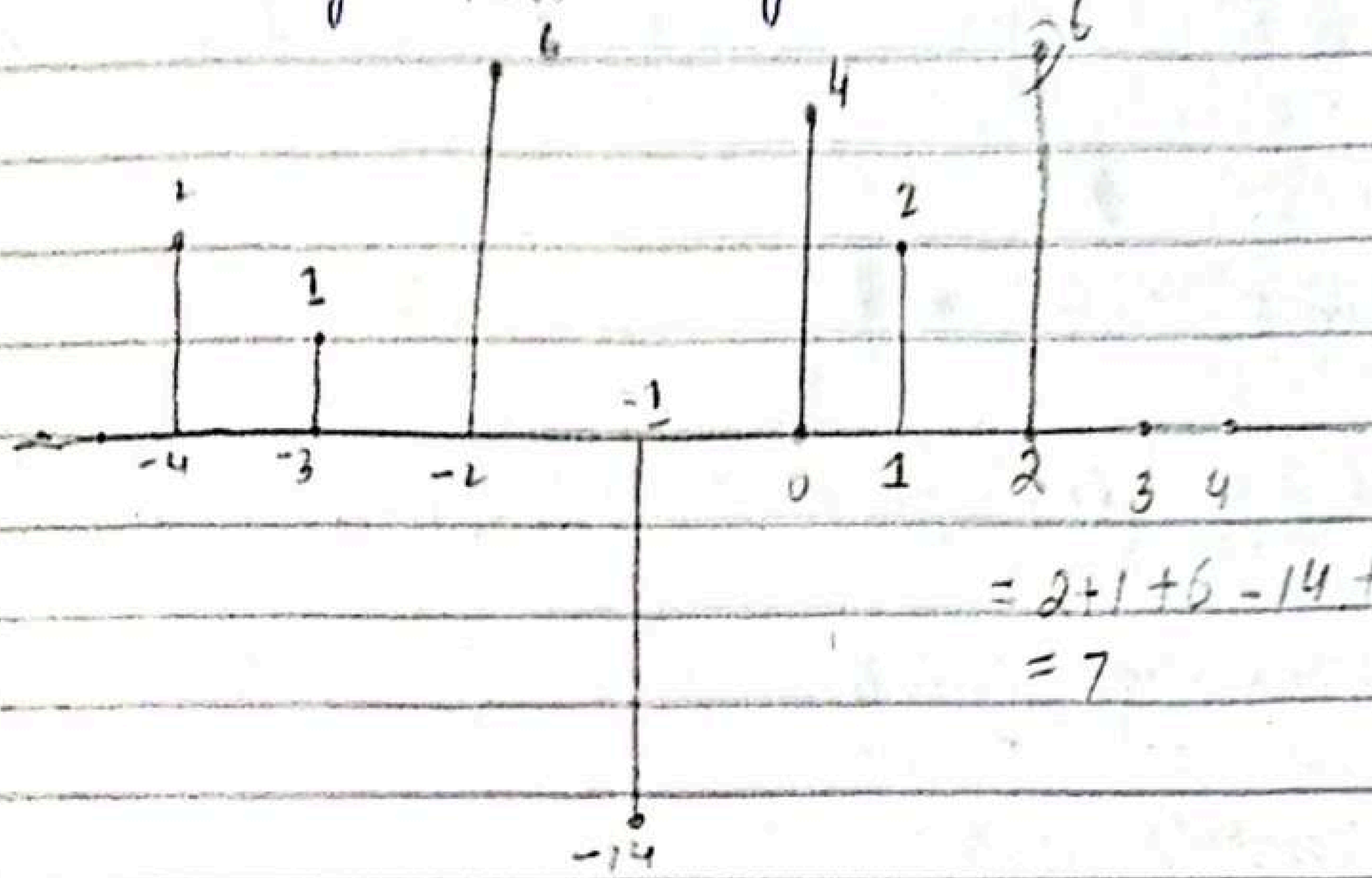
$$r_{xy}(0) = \sum_{n=-\infty}^{\infty} x(n)\, y(n-0)$$

$$= \sum_{n=-\infty}^{\infty} x(n)\, y(n)$$





$$r_{xy}(0) = \sum_{n=-\infty}^{\infty} x(n)\, y(n)$$



$-4 \le n \le 3$
Take the large range sig

$= 2+1+6-14+4+2+6$
$= 7$

$$r_{xy}(0) = \sum_{n=-4}^{3} x(n)y(n)$$

$= x(-4)\,y(-4) + x(-3)\,y(-3) + x(-2)\,y(-2) +$
$x(-1)y(-1) + x(0)y(0) + x(1)y(1) + x(2)\,y(2)$
$+ x(3)y(3)$

$= (2)(-1) + (-1)(-1) + (3)(2) + (7)(-2) + (1)(4)$
$+ (2)(1) + (-3)(-3) + (0)(5)$

$= 7$

Chapter #4

Frequency Analysis of signals and systems

Fourier Transform:

It is a mathematical tool that is useful for analysis and design of LTI systems. These signals include decomposition of signals → slides

Fourier series for Discrete-Time periodic signals:

suppose $x(n) = x(n+N)$ it is related exponentially

$$e^{j2\pi kn/N} \qquad k = 0, 1 \ldots N-1$$

$$x(n) \quad \sum_{k=0}^{N-1} \underbrace{c_k}_{\text{co-efficent}} e^{j2\pi kn/N}$$

To derive expression for fourier co-efficients

$$\sum_{n=0}^{N-1} e^{j2\pi kn/N} = \begin{cases} N & k = 0, \pm N, \pm 2N \ldots \\ 0 & \text{otherwise} \end{cases}$$

first $k = 0$

$$\sum_{n=0}^{N-1} e^{j2\pi(0)n/N} = \sum_{n=0}^{N-1} e^0 = N$$

$k = N = 3$

If k is multiple of N then ans = N otherwise 0

$$\sum_{n=0}^{2} e^{j2\pi 3n/3} = \sum_{n=0}^{2} e^{j2\pi n}$$

put $n = 0, 1, 2$

$$= e^0 + e^{j2\pi} + e^{j4\pi}$$
$$= 1 + (\cos 2\pi + j\sin 2\pi) + (\cos 4\pi + j\sin 4\pi)$$
$$= 1 + 1 + 0 + 1 + 0$$
$$= 3$$

Let $k = 2$, $N = 3$

$$\sum_{n=0}^{2} e^{j2\pi \frac{2n}{N}} = \sum_{n=0}^{2} e^{j\frac{2\pi 2n}{3}} = e^0 + e^{j2\pi 2(1)/3} + e^{j\frac{2\pi 2(2)}{3}}$$

$$= 1 + \left(\cos\frac{4\pi}{3} + j\sin\frac{4\pi}{3}\right) + \left(\cos\frac{8\pi}{3} + j\sin\frac{8\pi}{3}\right)$$

$$= 1 - 0.5 - j0.866 + (-0.5 + j0.866)$$

$$= 1 - 1 + 0 = 0$$

find value of c_k

Multiply b/s by $e^{j2\pi ln/N}$ and summing the product from $n=0$ to $n=N-1$

$$\sum_{n=0}^{N-1} x(n)\, e^{-j2\pi ln/N} = \sum_{n=0}^{N-1}\sum_{k=0}^{N-1} c_k\, e^{j2\pi(k-l)n/N}$$

$$\cancel{\sum_{n=0}^{N-1} x(n)\, e^{-j2\pi ln/N}}$$

$$e^{2\pi k} \cdot e^{-2\pi l} = e^{2\pi(k-l)}$$

$$\sum_{n=0}^{N-1} e^{j2\pi(k-l)n/N} = \begin{cases} N & \text{when } k=l \\ 0 & \text{when } k\neq l \end{cases}$$

① $k=l$

$$\sum_{n=0}^{N-1} e^{j2\pi(0)n/N} = N$$

② $k \neq l$

$$= \sum_{n} e^{j2\pi(k-l)n/N} \rightarrow \alpha$$

$$= \sum_{n=0}^{N-1} \alpha^{n}$$

By geometric series:-

$$\sum_{n=0}^{N-1} \alpha^{n} = \frac{1-\alpha^{N}}{1-\alpha}$$

$$= \frac{1-\alpha^{N}}{1-\alpha} \qquad \alpha = e^{j2\pi(k-l)/N}$$

$$= \frac{1 - e^{j2\pi(k-l)nN/N}}{1 - e^{j2\pi(k-l)/N}}$$

$$= 0$$

Therefore, right hand side reduces to

$$C_k = \frac{1}{N} \sum_{n=0}^{N-1}$$

When $k = l$

$$= NC_k = NC_l = \frac{1}{N} \sum_{n=0}^{N-1} x(n) e^{-j2\pi kn/N}$$

synthesis eq:-

$$x(n) = \sum_{k=0}^{N-1} C_k e^{j2\pi kn/N}$$

Analysis eq: $x(n) = \frac{1}{N} \sum_{n=0}^{N-1} x(n) e^{-j2\pi kn/N}$

11/02/25

Convolution

Example :-

Determine the output $y(n)$ of a relaxed LTI system with impulse response

$$h(n) = a^n u(n) \quad ; -1 < |a| < 1$$

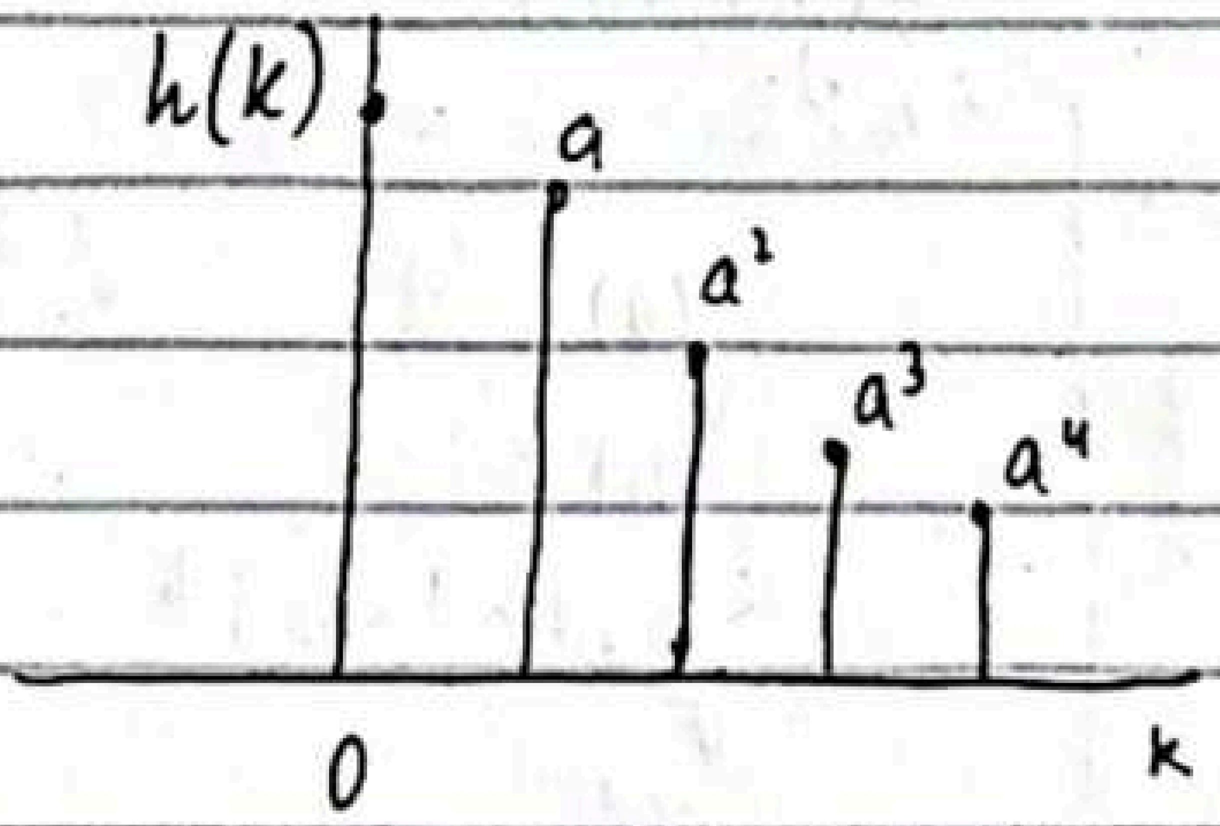
when the input is a unit step sequence that is $x(n) = u(n)$

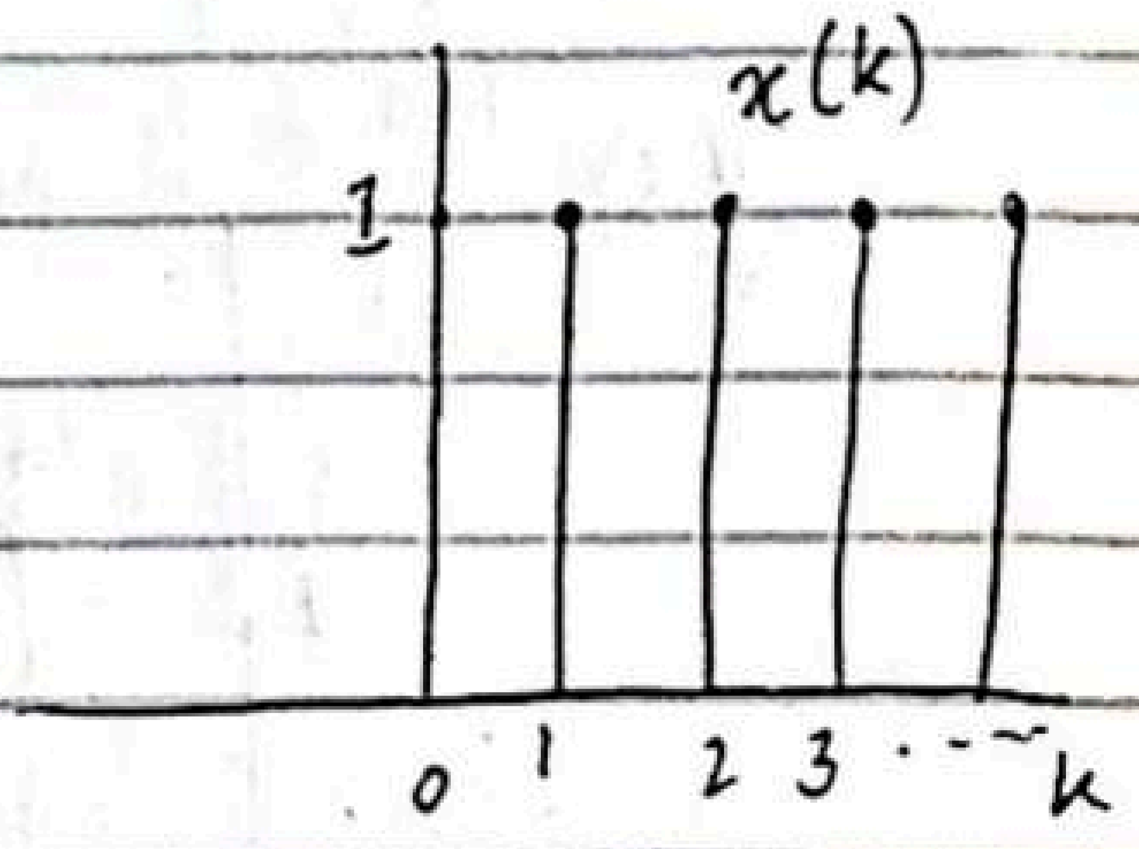
solution :-

In this case both $h(n)$ and $x(n)$ are infinite duration sequence we use following form of convolution.

$$\cancel{y(n) = \sum_{k=-\infty}^{\infty} x(k)\, h(n-k)}$$

$$y(n) = \sum_{k=-\infty}^{\infty} x(n-k)\, h(k)$$ in which input $x(k)$ is folded.





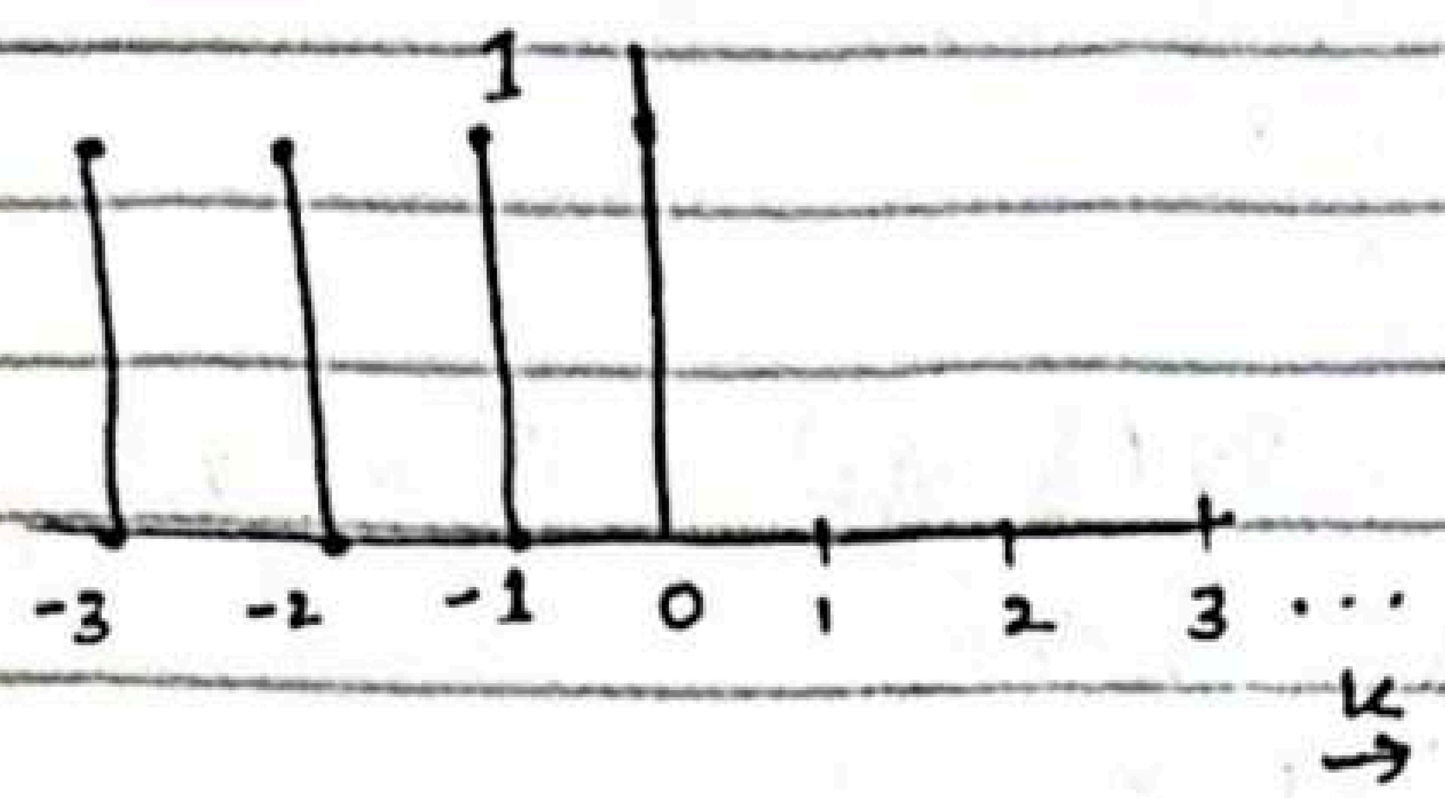
$(n-2) \rightarrow$ rig

$(-2-k)$

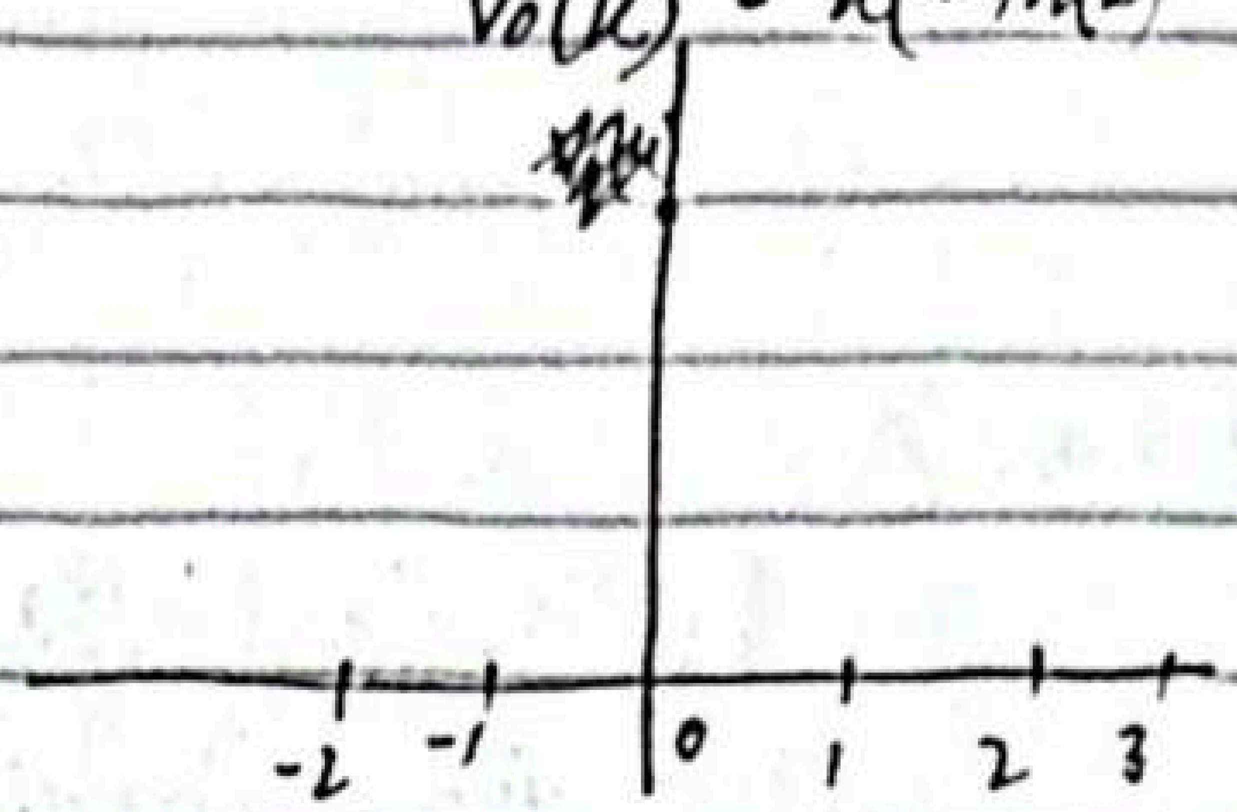
$2-k=0$

$-2=k$

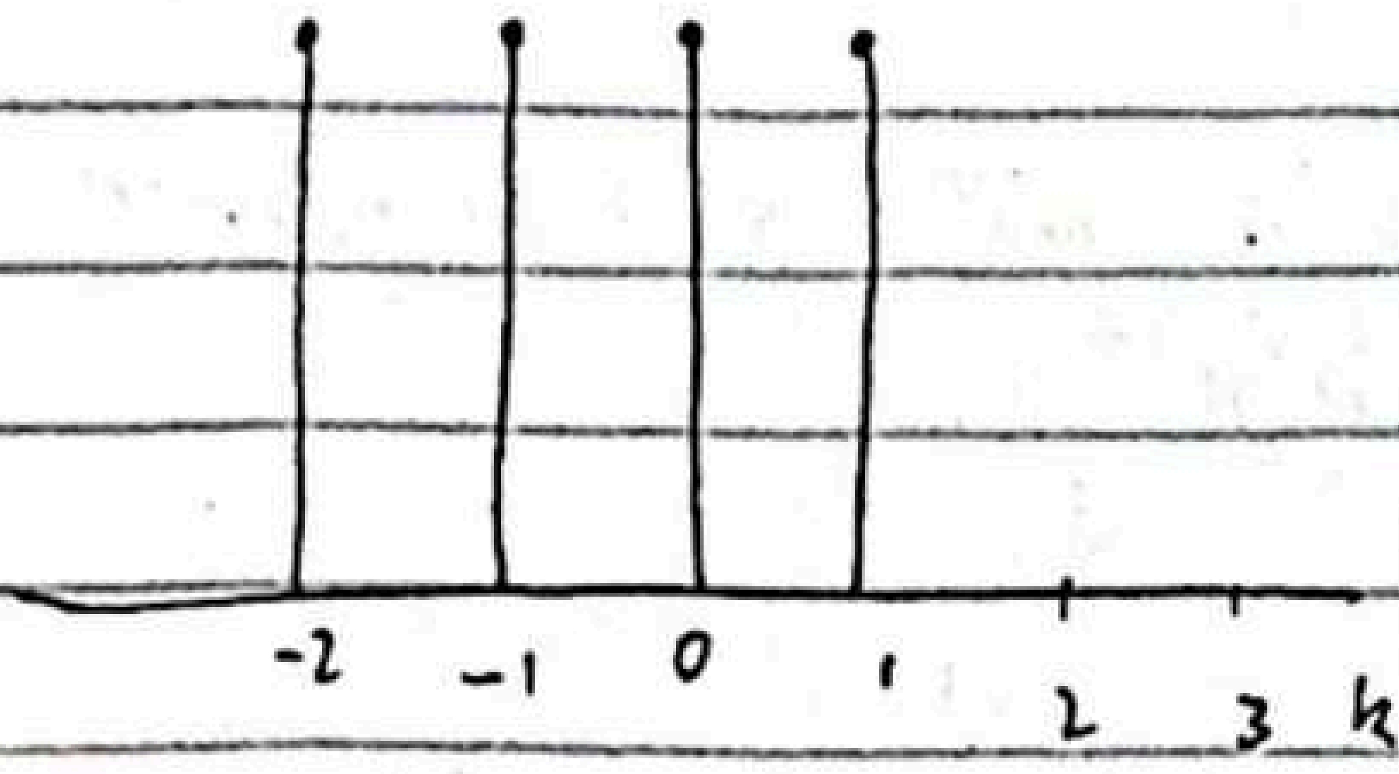
$x(-k)$



$V_0(k) = x(-k)h(k)$

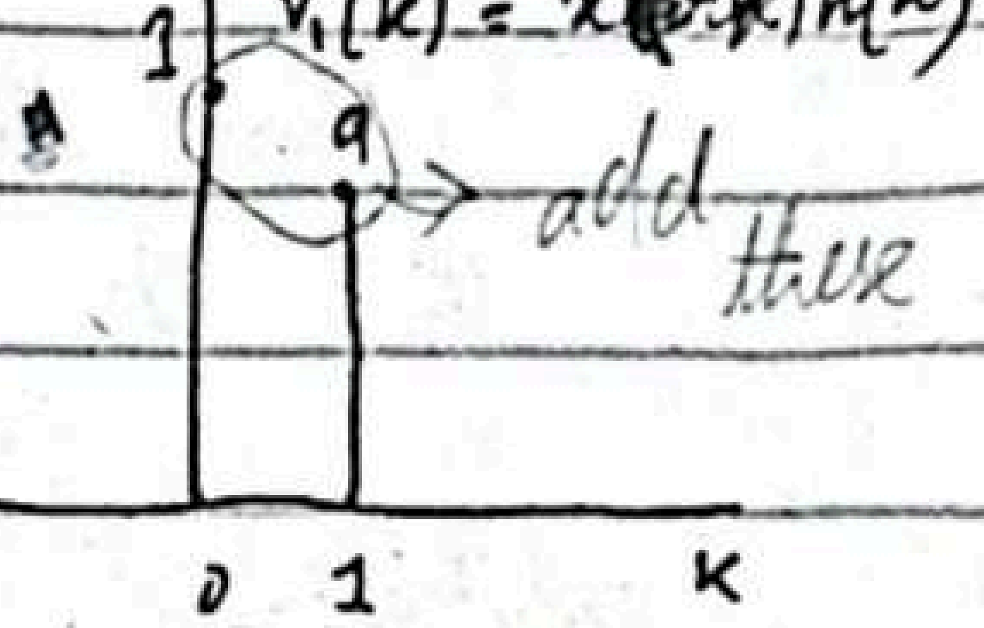


$x(1-k)$

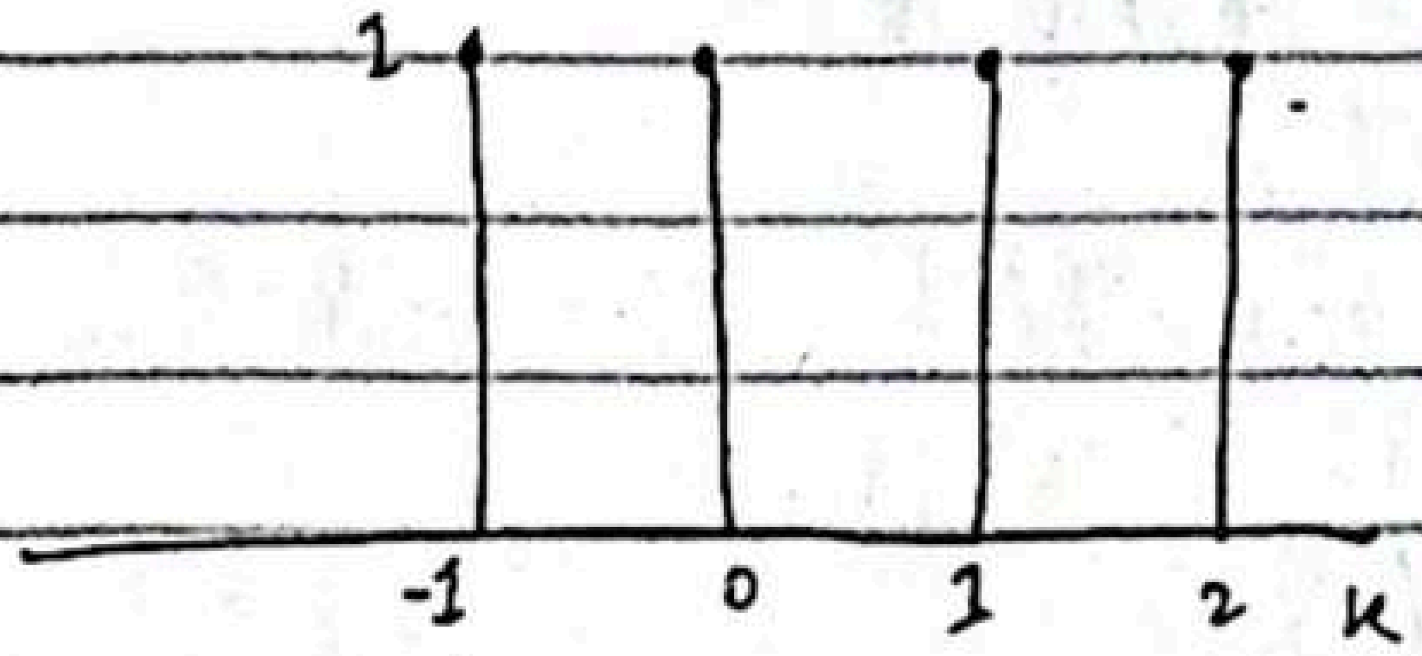


$x(1-k)$

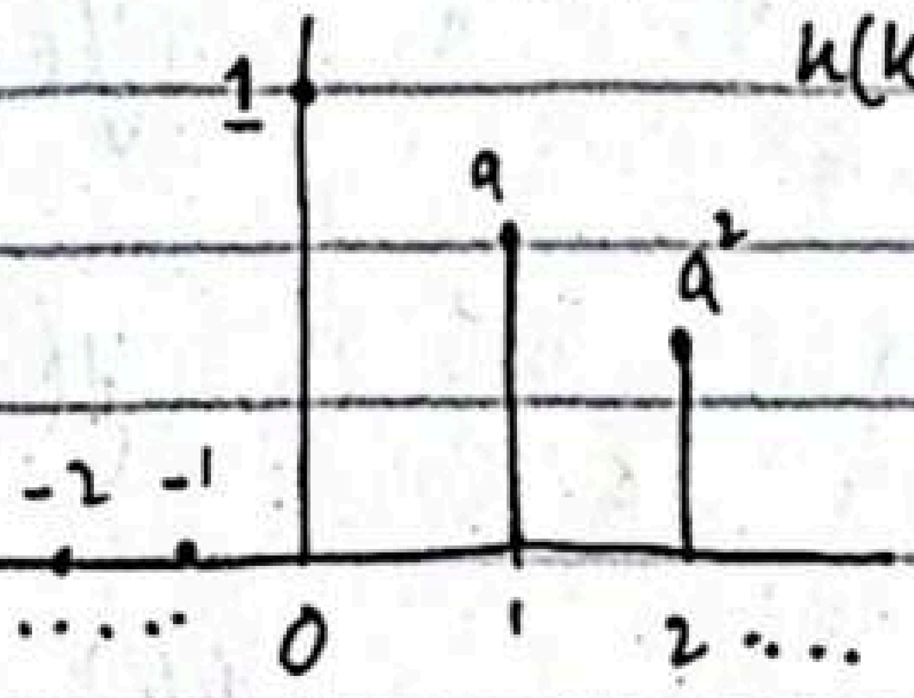
$V_1(k) = x(1-k)h(k)$

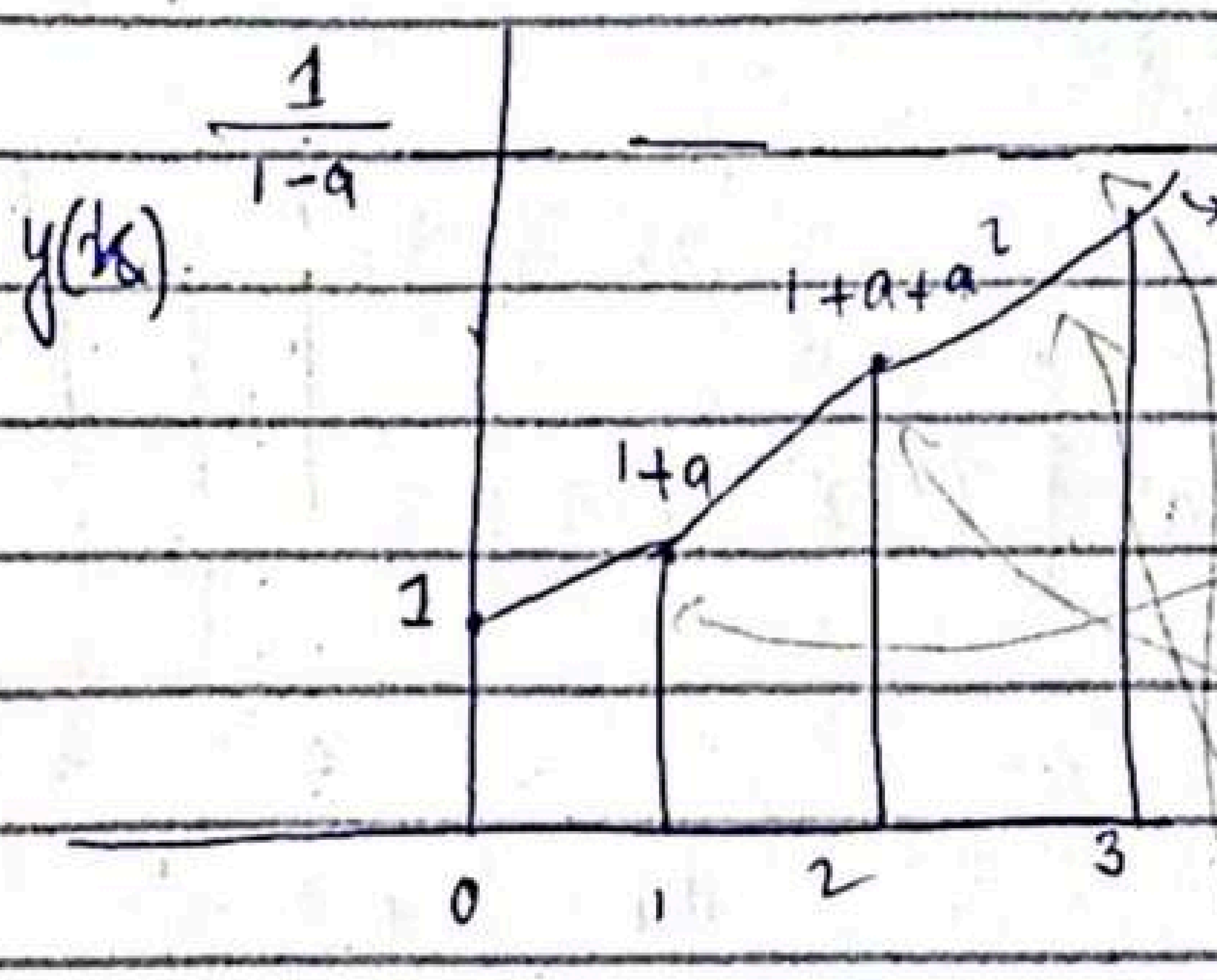


$x(2-k)$



$V_2(k) = x(2-k)h(k)$





This will touch asymptote when y approaches ∞.

$y(0) = 1$

$y(1) = 1+a$

$y(2) = 1+a+a^2$

$y(3)$

$y(n) = 1+a+a^2+a^3 \ldots + a^n$

$y(\infty) = \dfrac{1}{1-a}$

Homework -

Determine the impulse response for & cascade of two LTI systems having impulse response

$$h_1(n) = \left(\frac{1}{2}\right)^n u(n)$$

$$h_2(n) = \left(\frac{1}{4}\right)^n u(n)$$

18/02/25

Power Density Spectrum of Periodic Signals

The average power of a discrete-time periodic signal with period N was defined

$$P_x = \frac{1}{N} \sum_{n=0}^{N-1} |x(n)|^2$$

P_x in terms of Fourier co-efficent C_k

$$P_x = \frac{1}{N} \sum_{n=0}^{N-1} x(n)x^*(n)$$

$$= \frac{1}{N} \sum_{n=0}^{N-1} x(n)\left(\sum_{k=0}^{N-1} C_k^* e^{-j2\pi kn/N} \right)$$

$$x(n) = [\underset{\uparrow}{1} \quad -2 \quad 1 \quad -2 \quad 1 \quad -2]$$

if arrow is not given then left most value of sequence gives $n=0$

we get 2 from: if we look into sequence then it is repeating after 2.

$$= \frac{1}{2} \sum_{n=0}^{1} |x(n)|^2$$

$$= \frac{1}{2} \left[|x(0)|^2 + |x(1)|^2 \right]$$

$$= \cancel{\frac{1}{2} [|x|}$$

$$= \frac{1}{2} \left[|1|^2 + |-2|^2 \right]$$

$$= \frac{1}{2} [1+4]$$

$$P_x = \frac{5}{2}$$

Now, we can interchange the order of two summations

$$P_x = \sum_{k=0}^{N-1} c_k^* \left[\frac{1}{N} \sum_{n=0}^{N-1} x(n) e^{-j2\pi kn/N} \right]$$

$$P_x = \sum_{k=0}^{N-1} |c_k|^2 = \frac{1}{N} \sum_{n=0}^{N-1} |x(n)|^2$$

This is known as Parseval's (theorem) relation for discrete-Time periodic signals. The sequence $|c_k|^2$ for $k = 0, 1, \ldots N-1$ is distribution of power as a function of frequency and is called the power density spectrum of periodic signal.

Energy of sequence $x(n)$ over a single period.

$$E_N = \sum_{n=0}^{N-1} |x(n)|^2$$

$$E_N = N \sum_{k=0}^{N-1} |c_k|^2$$

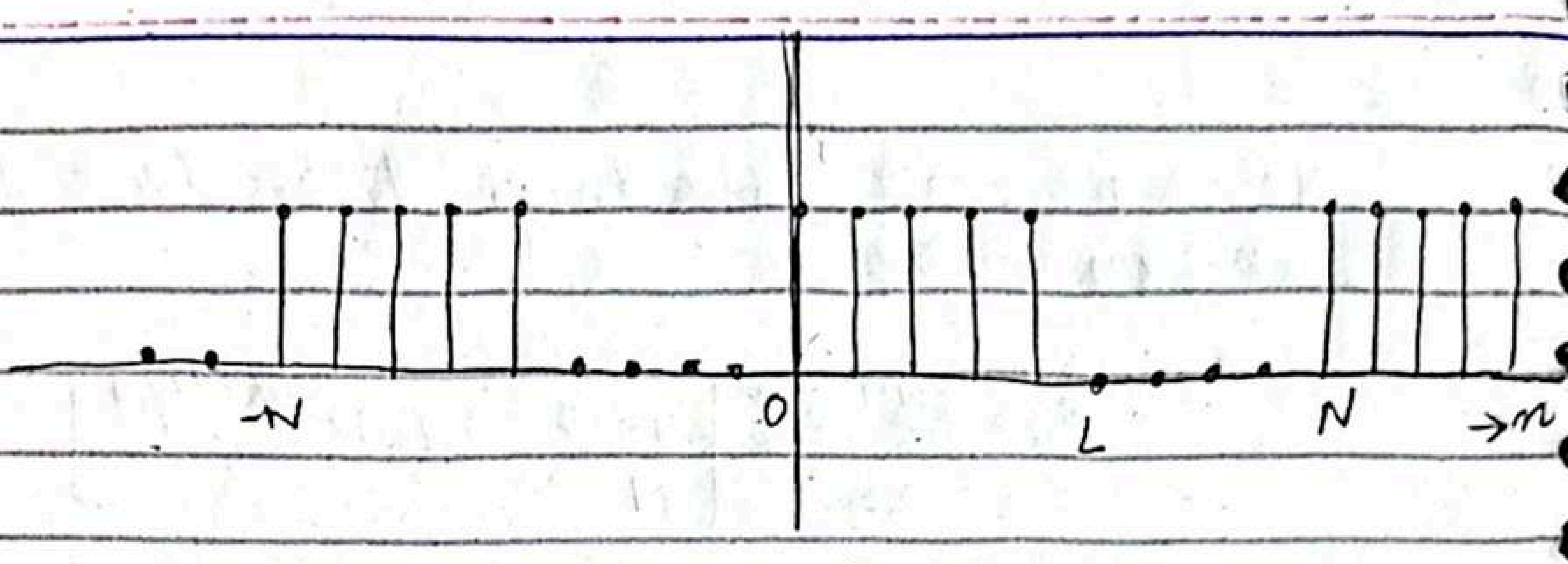
Example 4.2.2.

Periodic 'square wave' signal.

Determine:-

(i) Fourier series co-efficients

(ii) Power density spectrum of following signal.



By applying analysis

$$c_k = \frac{1}{N} \sum_{n=0}^{N-1} x(n) e^{-j2\pi kn/N} \qquad k = 0, 1 \cdots N-1$$

which is geometric series

$$c_k = \frac{A}{N} \sum_{n=0}^{L-1} (e^{-j2\pi L/N})^n = \begin{cases} \frac{AL}{N} & , k = 0 \\ \frac{A}{N} \frac{1 - e^{-j2\pi kL/N}}{1 - e^{-j2\pi k/N}} & k = 1, 2, \cdots N-1 \end{cases}$$

Partial summation Formula:-

$$\sum_{n=0}^{N-1} \alpha^n = \sum_{n=0}^{\infty} \alpha^n - \sum_{n=N}^{\infty} \alpha^n = \frac{1 - \alpha^N}{1 - \alpha}$$

The last expression can be simplified

$$\frac{1 - e^{-j2\pi kL/N}}{1 - e^{-j2\pi k/N}} = \frac{e^{-j\pi kL/N}}{e^{-j\pi k/N}} \frac{e^{j\pi kL/N} - e^{-j\pi kL/N}}{e^{j\pi k/N} - e^{-j\pi k/N}}$$

$$= e^{-j\pi k(L-1)/N} \frac{\sin(\pi k L/N)}{\sin(\pi k/N)}$$

Therefore,

$$c_k = \begin{cases} \dfrac{AL}{N} & k = 0, +N, \pm 2N, \ldots \\ \dfrac{A}{N} e^{-j\pi k(L-1)/N} \dfrac{\sin(\pi k L/N)}{\sin(\pi k/N)}, & \text{otherwise} \end{cases}$$

$$|c_k|^2 = \begin{cases} \left(\dfrac{AL}{N}\right)^2 & , k = 0, +N, \pm 2N \\ \left(\dfrac{A}{N}\right)^2 \dfrac{\sin(\pi k L/N)}{\sin(\pi k/N)} & , \text{otherwise} \end{cases}$$

The Fourier Transform of discrete time signals (Aperiodic)

Fourier Transform of a finite energy discrete time signal $x(n)$

$$X(\omega) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n}$$

$X(\omega)$ represents the frequency content signal $x(n)$. In other words, $X(\omega)$ is periodic with 2π

$$X(\omega + 2\pi k) = \sum_{n=-\infty}^{\infty} x(n)\, e^{-j(\omega + 2\pi k)n}$$

$$= \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n} e^{-j2\pi K n}$$

Synthesis equation
inverse transform $\quad x(n) = \frac{1}{2\pi} \int_{2\pi} X(\omega) e^{j\omega n} d\omega$

Analysis equation $\quad X(\omega) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n}$

Energy Density spectrum of Aperiodic signals

$$E_x = \sum_{n=-\infty}^{\infty} |x(n)|^2$$

$$E_x = \sum_{n=-\infty}^{\infty} x(n)\, x^*(n) = \sum_{n=-\infty}^{\infty}$$

slides.

$$S_{xx} = |X(\omega)|^2$$

Example 4.2.3

Determine and sketch energy spectrum Density $x = a^n u(n)$

$$\sum_{n=-\infty}^{\infty} |x(n)| = \sum_{n=0}^{\infty} |a|^n = \frac{1}{1-|a|} < \infty$$

$$X(\omega) = \sum_{n=0}^{\infty} a^n e^{-j\omega n} = \sum_{n=0}^{\infty} (a e^{-j\omega})^n$$

$|ae^{-j\omega}| = |a| < 1$, by using geometric formula:-

$$X(\omega) = \frac{1}{1-ae^{-j\omega}}$$

$$S_{xx} = |X(\omega)|^2 = X(\omega) X^*(\omega)$$

$$S_{xx} = \frac{1}{(1-ae^{-j\omega})(1-ae^{j\omega})}$$

20/02/25

Find Discrete Time Fourier Transform of signal:-

$$x(n) = \begin{cases} 1 & 0 \le n \le 4 \\ 0 & \text{otherwise} \end{cases}$$

$$X(w) = \sum_{0}^{4} 1(e^{-jwn})$$

$x(e^{jw})$

$$x(w) = e^{0} + e^{-jw} + e^{-jw2} + e^{-3jw} + e^{-4jw}$$

$$x(w) = 1 + e^{-jw} + e^{-2jw} + e^{-3jw} + e^{-4jw}$$

$$X(w) = |X(w)| e^{j\theta}$$

find $|X(w)| e^{j\theta}$

$$\sum_{n=0}^{4} x(n) e^{-jwn} = \sum_{n=0}^{N-1} (e^{-jw})^{n}, \quad N = 5$$

$$= \frac{1 - e^{-jw5}}{1 - e^{-jw}} = \frac{e^{-jw5/2}}{e^{-jw/2}} \cdot \frac{e^{jw5/2} - e^{-jw5/2}}{e^{jw/2} - e^{-jw/2}}$$

$$\Rightarrow e^{-2jw} \cdot \frac{\sin 5w/2}{\sin w/2} = e^{j\theta} \cdot |X(w)|$$

As $e^{-j2w} = e^{j\theta}$

$$|X(w)| = \frac{\sin 5w/2}{\sin w/2}$$

hence, $\angle X(w) = -2w$

$$|X(w)| = \frac{\sin 5w/2}{\sin w/2}$$

Find Discrete fourier Transform of signal :-

$$x(n) = \{ a^n u(n) \quad , |a|<1$$

$$X(w) = \sum_{n=0}^{\infty} a^n u(n) e^{-jwn}$$

$-1 < a < 1$

$$= a^{(0)} e^{-jw(0)} + a \cancel{g} e^{jw}$$

$$X(w) = 1 + ae^{jw}$$

formula.

$$\therefore \sum_{k=a}^{\infty} x^k = \left\{ \frac{x^a}{1-x} \right.$$

$$= \frac{(ae^{-jw})^0}{1 - ae^{-jw}} = \frac{1}{1 - ae^{jw}}$$

$$= \frac{1}{1 - a(\cos w - j\sin w)} = \frac{1}{1 - a\cos w + aj\sin w}$$

$$|X(w)| = \frac{1}{\sqrt{\underset{\text{real}}{(1 - a\cos w)^2} + \underset{\text{img}}{(a\sin jw)}}}$$

$$= \frac{1}{\sqrt{(1 + a^2\cos^2 w - 2a\cos w) + a^2\sin^2 w}}$$

$$= \frac{1}{\sqrt{1 - 2a\cos w + a^2(\cos^2 w + \sin^2 w)}}$$

Lets find minima and maxima for plotting by derivative.

let $g(w) = 1 - 2a\cos w + a^2$

$$\frac{dg}{dw} = 0 - 2a(-\sin w) + 0 = 0$$

$$g(w) = 2a \sin w = 0$$

$\sin w = 0$ at $0, \pi, 2\pi$

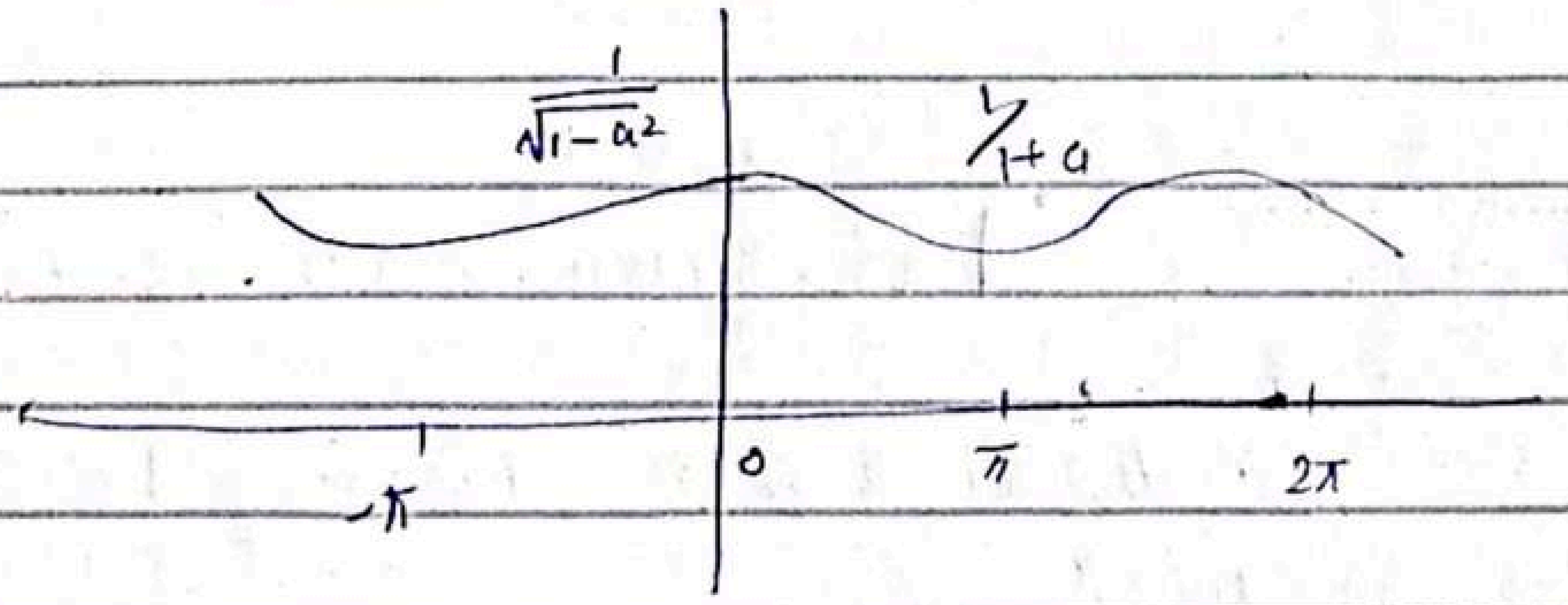
Put $w = 0$

$$|X(w)| = \frac{1}{\sqrt{1 - 2a + a^2}} = \frac{1}{\sqrt{(1-a)^2}}$$

$$|X(0)| = \frac{1}{1-a}$$

$$|X(\pi)| = \frac{1}{\sqrt{1 + 2a + a^2}} = \frac{1}{\sqrt{(1+a)^2}} = \frac{1}{1+a}$$

$$X(2\pi) = \frac{1}{1-a}$$



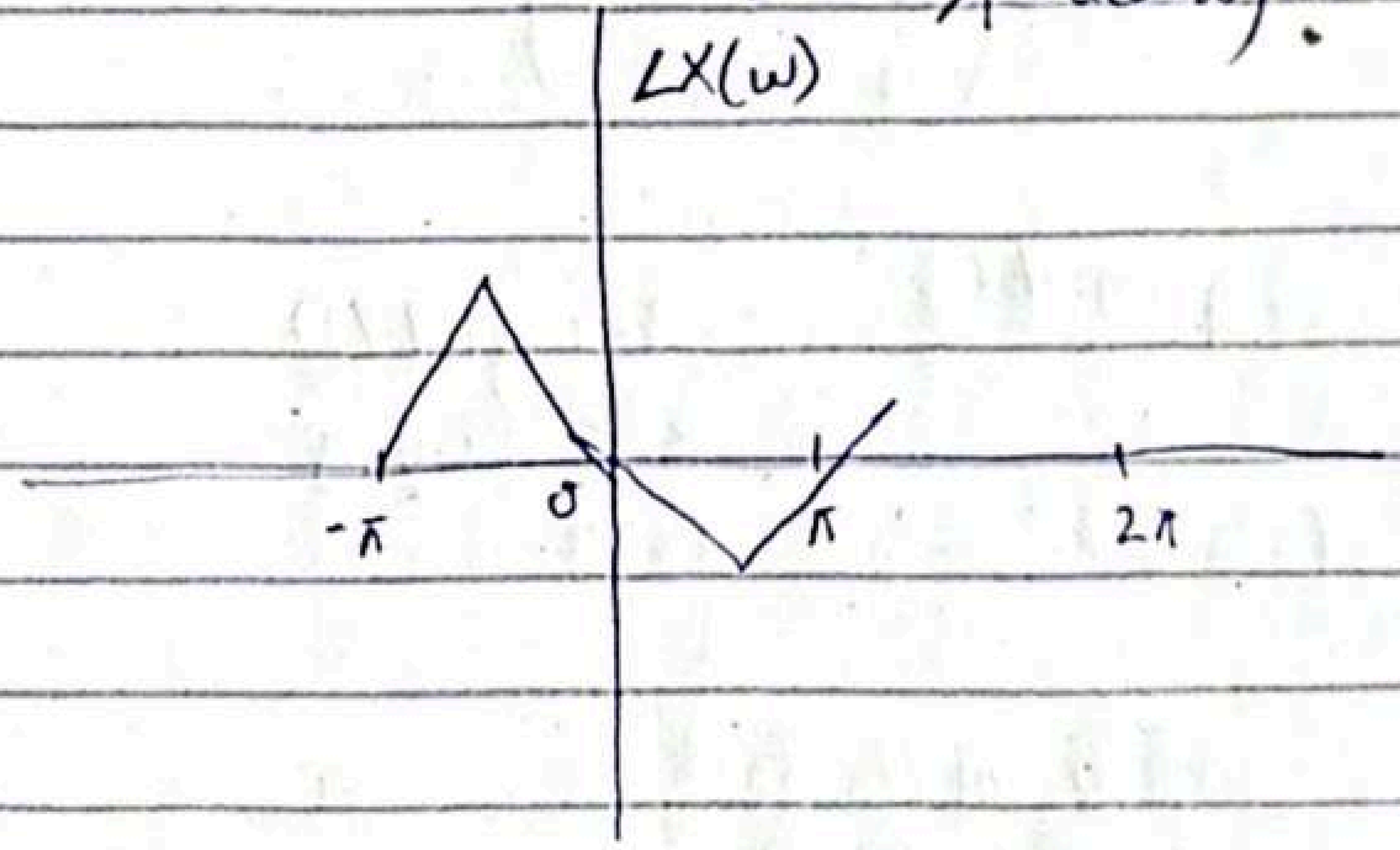
$\angle X(\omega) = \angle num - \angle den$

$= \angle 1 - \angle(1 - a\cos\omega + ja\sin\omega)$

$= \tan^{-1}\left(\frac{0}{1}\right) - \tan^{-1}\left(\frac{a\sin\omega}{1 - a\cos\omega}\right)$

$= 0 - \tan^{-1}\left(\frac{a\sin\omega}{1 - a\cos\omega}\right)$

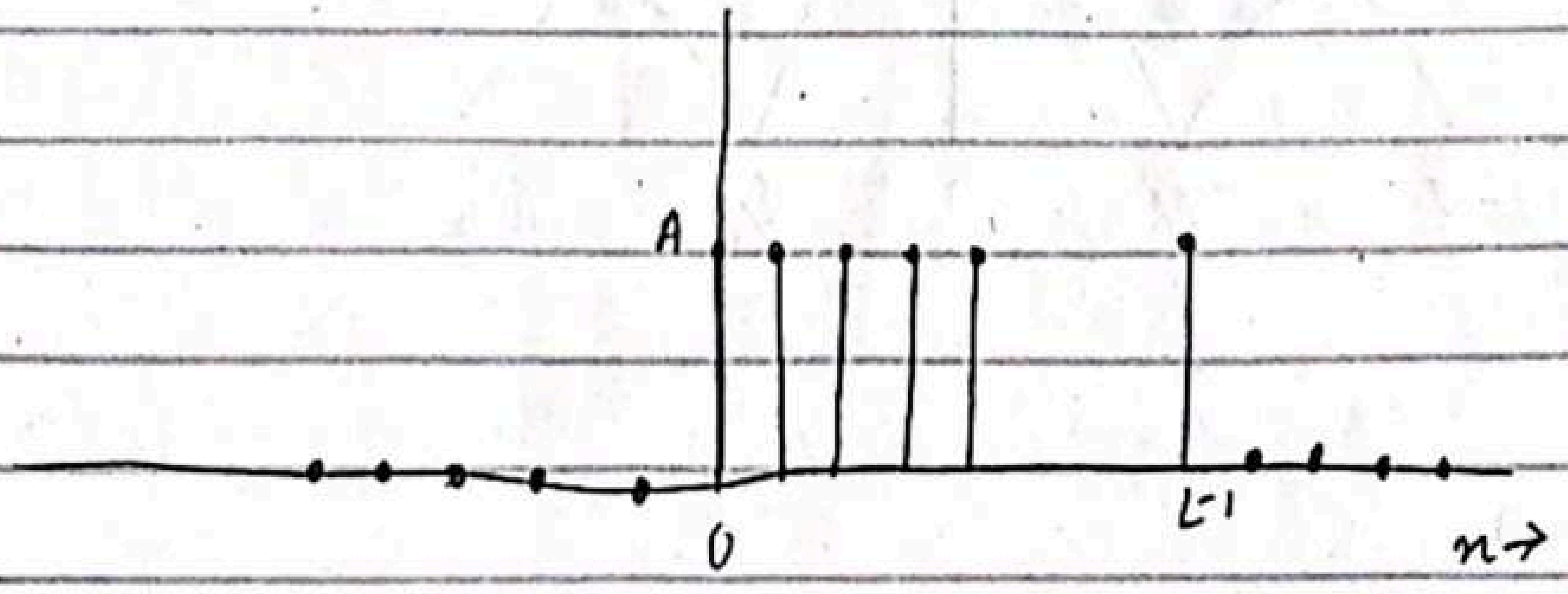
$= -\tan^{-1}\left(a\sin\omega / 1 - a\cos\omega\right).$



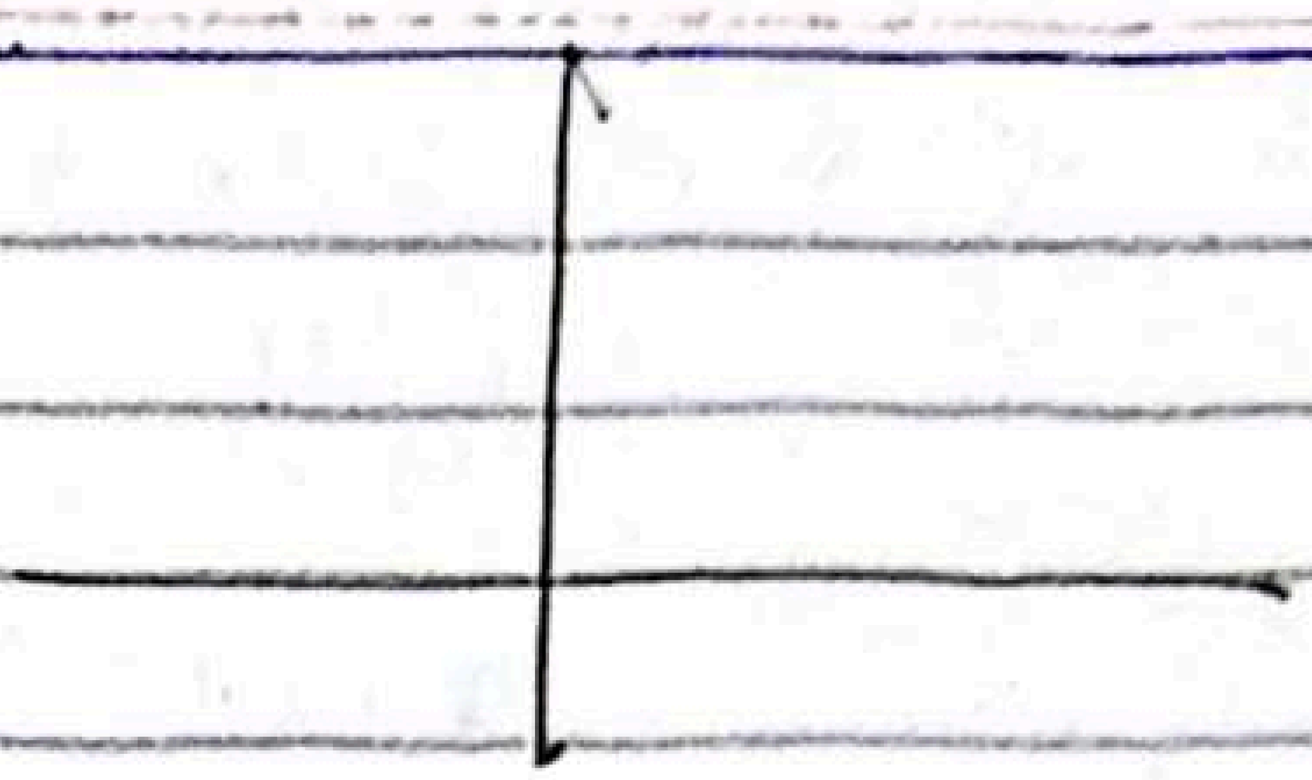
Example 4.2.4

Determine the Fourier Transform and energy density spectrum of the sequence.

$$x(n) = \begin{cases} A & 0 \le n \le L-1 \\ 0 & \text{otherwise} \end{cases}$$



$$X(w) = \sum_{0}^{L-1} A e^{-jwn}$$



$$\sum_{0}^{L-1} a^n = \frac{1-a^N}{1-a}$$

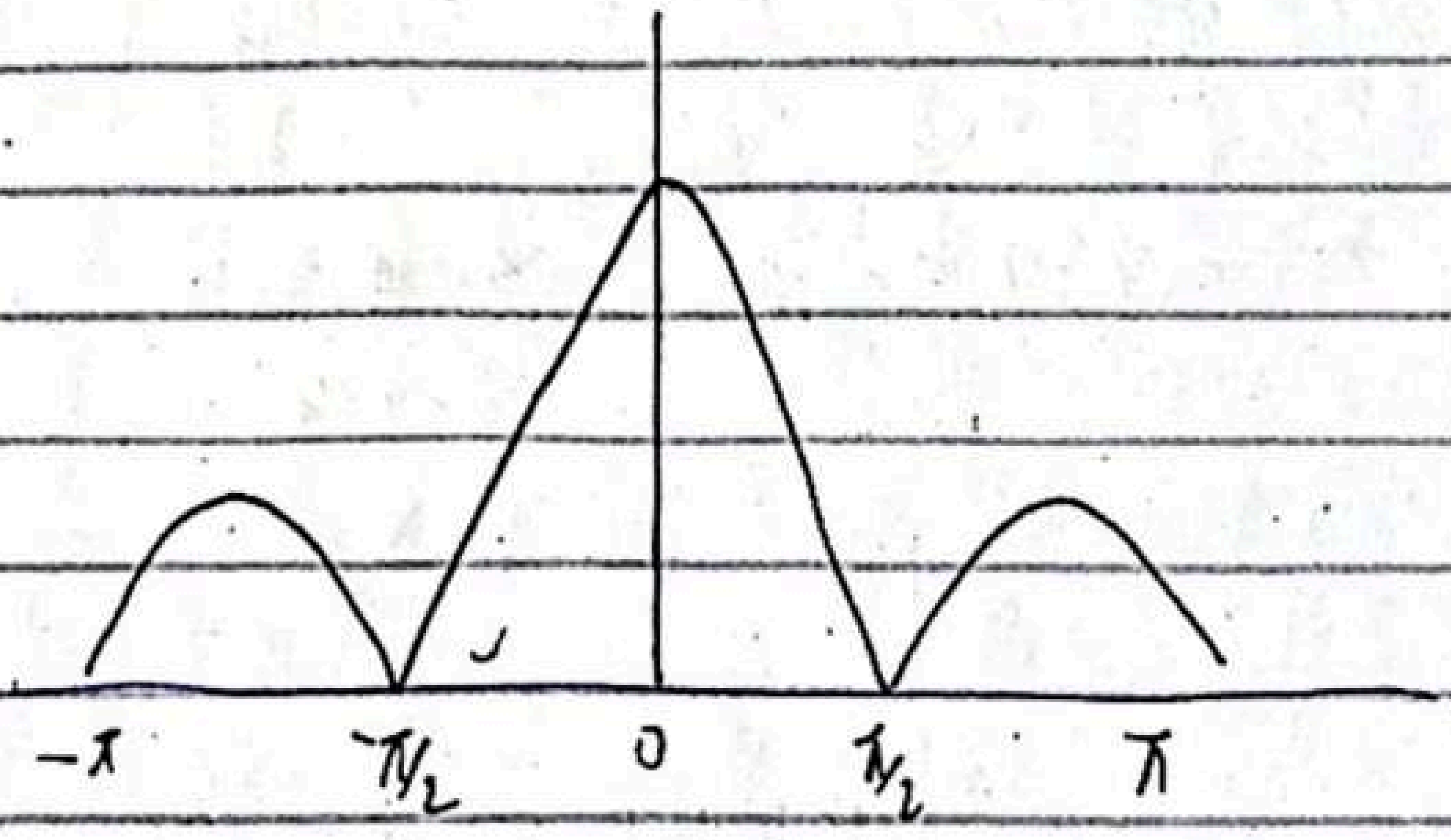
$$X(w) = A\left(\frac{1-e^{-jwL}}{1-e^{-jw}}\right)$$

$$= A e^{-jw/2(L-1)} \cdot \frac{\sin(wL/2)}{\sin(w/2)}$$

$w = 0$

$X(w) = |A| L.$ $\quad$ $w = 0$

$$X(w) = \frac{|A| \sin wL/2}{\sin w/2}$$ otherwise



Example 4.4.4

$h(n) = ba^n u(n)$

A linear time ....

$y(n) = ay(n-1) + bx(n) \qquad 0<a<1$

(a) determine magnitude of freq response.

(a) frequency response is

$$H(\omega) = \sum_{n=-\infty}^{\infty} h(n) e^{-j\omega n}$$

$$= \frac{b}{1 - ae^{-j\omega}}$$

$1 - ae^{-j\omega} = (1 - a\cos\omega) + j(a\sin\omega) \rightarrow$ slides.

Concept of Band width :- → slides.

submission { after mid, class 1 (Tuesday) }

Google Scholar

DTFT (Power, speech etc)

→ Abstract, → conclusion] read.

04/03/25

Question : 1

Find the DTFT of following signal :-

$$x(n) = \begin{cases} 1 & 0 \le n \le 4 \\ 0 & \text{otherwise} \end{cases}$$

$$x(\omega) = \sum_{0}^{4} 1(e^{-j\omega n})$$

$$x(\omega) = e^{0} + e^{-1j\omega} + e^{-2j\omega} + e^{-3j\omega} + e^{-4j\omega}$$

$$X(\omega) = |X(\omega)| e^{j\theta}$$

$$\sum_{n=0}^{4} x(n) e^{-j\omega n} = \sum_{n=0}^{N-1} x(n) e^{-j\omega n}, \quad N=5$$

$$= \frac{1-e^{-5j\omega}}{1-e^{-j\omega}} = \frac{e^{-j\omega 5/2}}{e^{-j\omega/2}} \cdot \frac{e^{j\omega 5/2} - e^{-j\omega 5/2}}{e^{j\omega/2} - e^{-j\omega/2}}$$

$\boxed{e^{-j\theta}}$

$$= e^{-2j\omega} \cdot \frac{\sin 5\omega/2}{\sin \omega/2}$$

$$|X(\omega)| = \frac{\sin 5\omega/2}{\sin \omega/2}$$

$$\angle X(\omega) = -2\omega$$

Next example solved previously

$$x(n) = a^{n} u(n) \quad |a| < 1$$

Example 4.4.4

A ~~LTI~~ linear time-invariant system is described by the following difference equation

$$y(n) = ay(n-1) + bx(n) \quad a<a<1$$

(a) Determine the magnitude and phase of frequency response of

$$h(n) = ba^n u(n)$$

The system is BIBO → stable and hence $H(w)$ exists, The frequency response is

$$H(w) = \sum_{n=-\infty}^{\infty} h(n) e^{-jwn}$$

$$\boxed{H(w) = \frac{b}{1-ae^{-jw}}} = \frac{b}{1-(a\cos w + aj\sin w)}$$

magnitude : $1 - ae^{-jw} = (1 - a\cos w) + ja\sin w$

$$|1 - ae^{-jw}| = \sqrt{(1 - a\cos w)^2 + (a\sin)^2}$$

$$= \frac{|b|}{\sqrt{1 + a^2 - 2a\cos w}}$$

$$\measuredangle = \measuredangle b - \tan^{-1}\left(\frac{a\sin w}{1 - a\cos w}\right)$$

06/03/25

Multirate Digital Signal Processing

In many practical applications of DSP one is faced with the problem of changing the sampling rate of a signal, either increasing it or decreasing it.

Example:-

(i) In telecommunication systems that transmit and recieve different type of signals (e.g. speech, video etc), there is a requirement to process the various signals at different rates according to the corresponding bandwidths of the signals. The process of converting a signal from a given rate to different rate is called "sampling rate conversion". In turn, systems that employ multiple sampling rates in the processing of digital signals are called multirate digital signal Processing systems.

(ii) Audio - Processing:- Audio signals can be converted from 44.1KHz (for CD's) to 48kHz (DVD's)

(iii) Image Processing:- Images/Videos can be upscaled to higher resolution

sampling is process of converting signal into discrete.

(iv) Seismic Data Processing:-

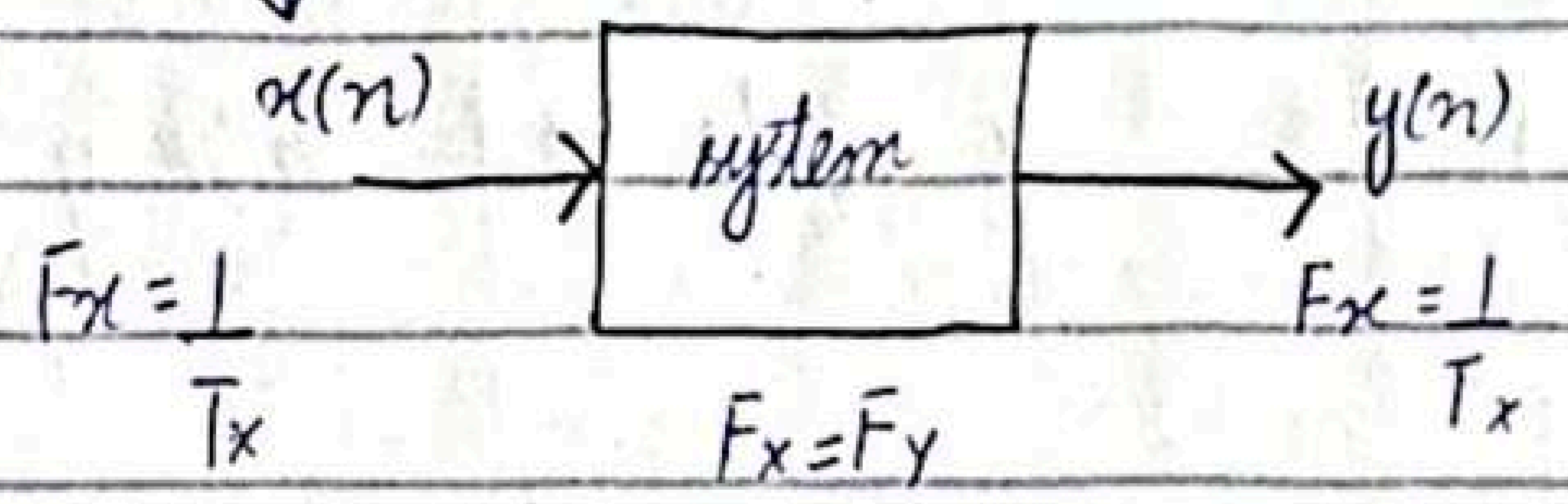
The size of seismic data can be reduced for storage and transmission.

(V) Reduced Bandwidth:-

By reducing the sampling rate of signal (decimation) the amount of data that needs to be transmitted/stored is decreased. This directly reduces the required bandwidth.

down sampling

$x_a(t) = x(nT_x)$

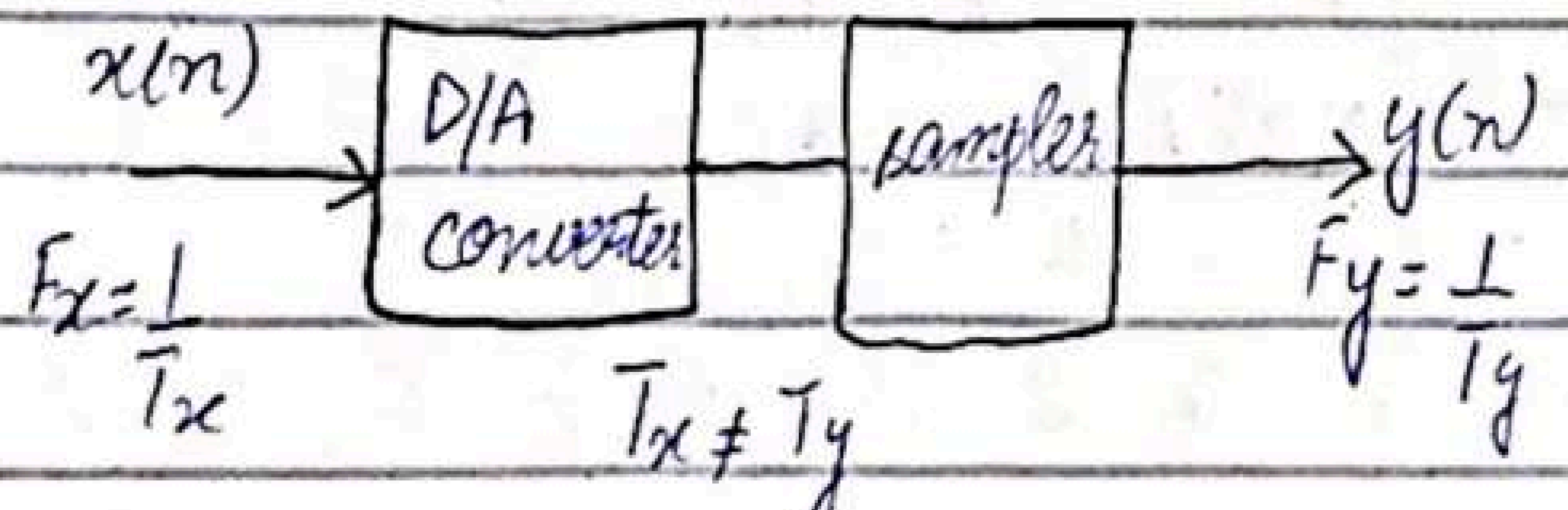


if $F_x \neq F_y$ (Then system is called multirate)

Sampling rate conversion of a digital signal:-

It can be accomplished in one of two general methods. One method is to pass the digital signal through a D/A converter, filter it if necessary and then to resample the resulting analogue signal at the desired rate. The second method is ~~the~~ to perform the sampling rate conversion entirely in the digital domain.

(i) First Method:-



Pros And Cons:-

→ The benefit of this method is that Fx and Fy can have any arbitary value

→ The Problem in this method is extensive approximation.

Digital Domain Method:-

→ In this method, sampling rate conversion is performed entirely in digital domain.

→ In this method, Fx and Fy will have some relationship. However, we need not to perform D/A conversion.

$$F_y = IF_x$$

$$F_x \longrightarrow \boxed{\uparrow I} \longrightarrow F_y$$

$F_y = IF_x$

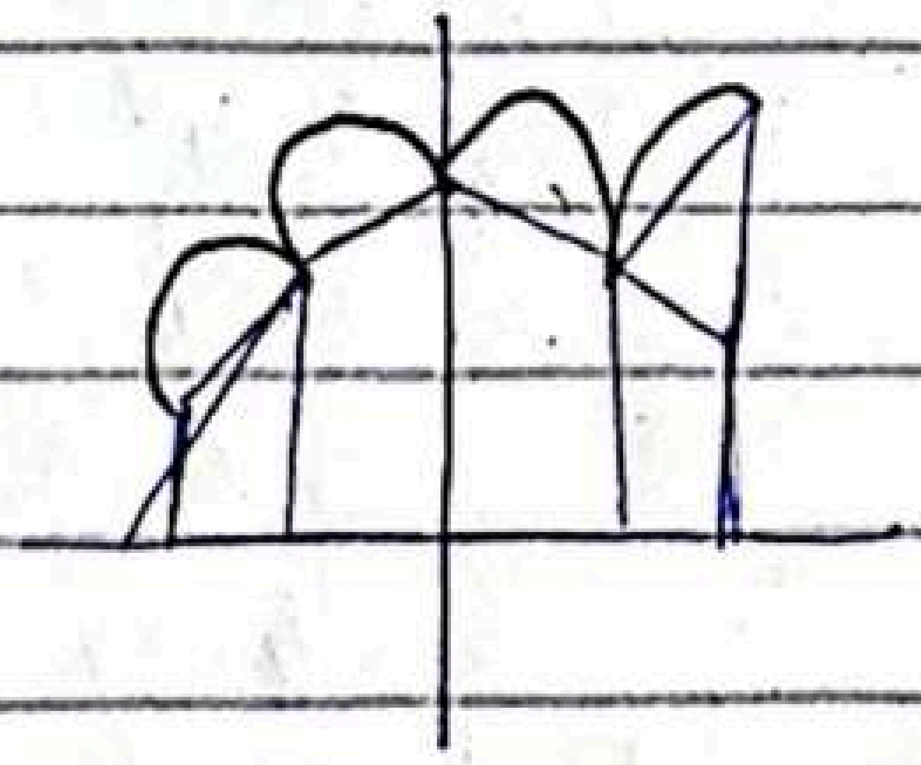
$(I-1)$ zeros will be added between points.

Up sampler:-

$$\underset{F_x}{x(n)} \longrightarrow \boxed{\text{Discrete system}} \longrightarrow \underset{F_y}{y(n)}$$

$F_y > F_x$

Distortion



let

$x(n) = \{1 \quad 2 \quad 3 \quad 4\}$

$n = \quad 0 \quad 1 \quad 2 \quad 3$

D/A (Digital to Analogue Converter)

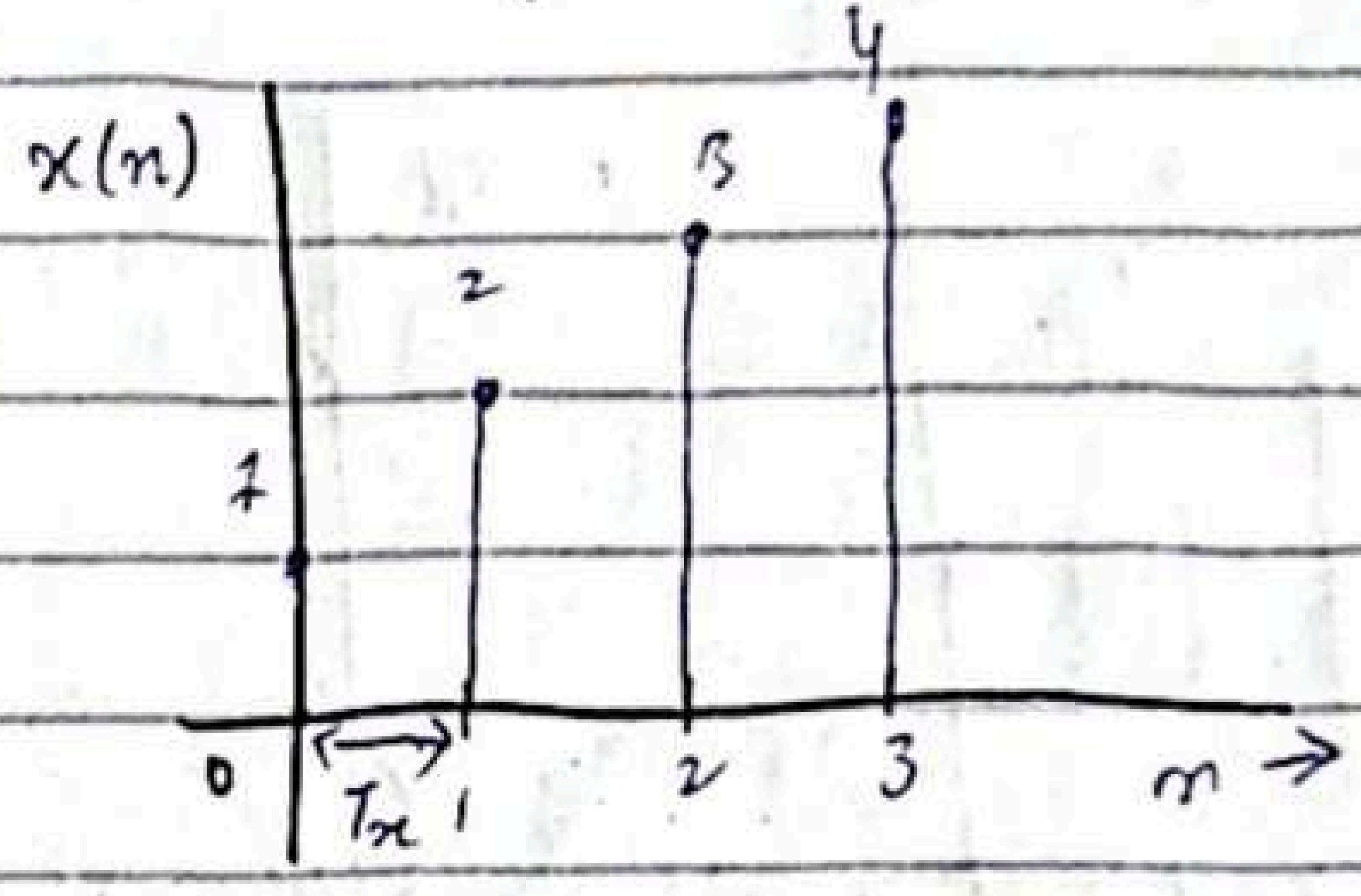
If sampling rate is increased, it means T_x is reduced and F_x is increased.

Let's assume we up sample the signal by a factor of 2.

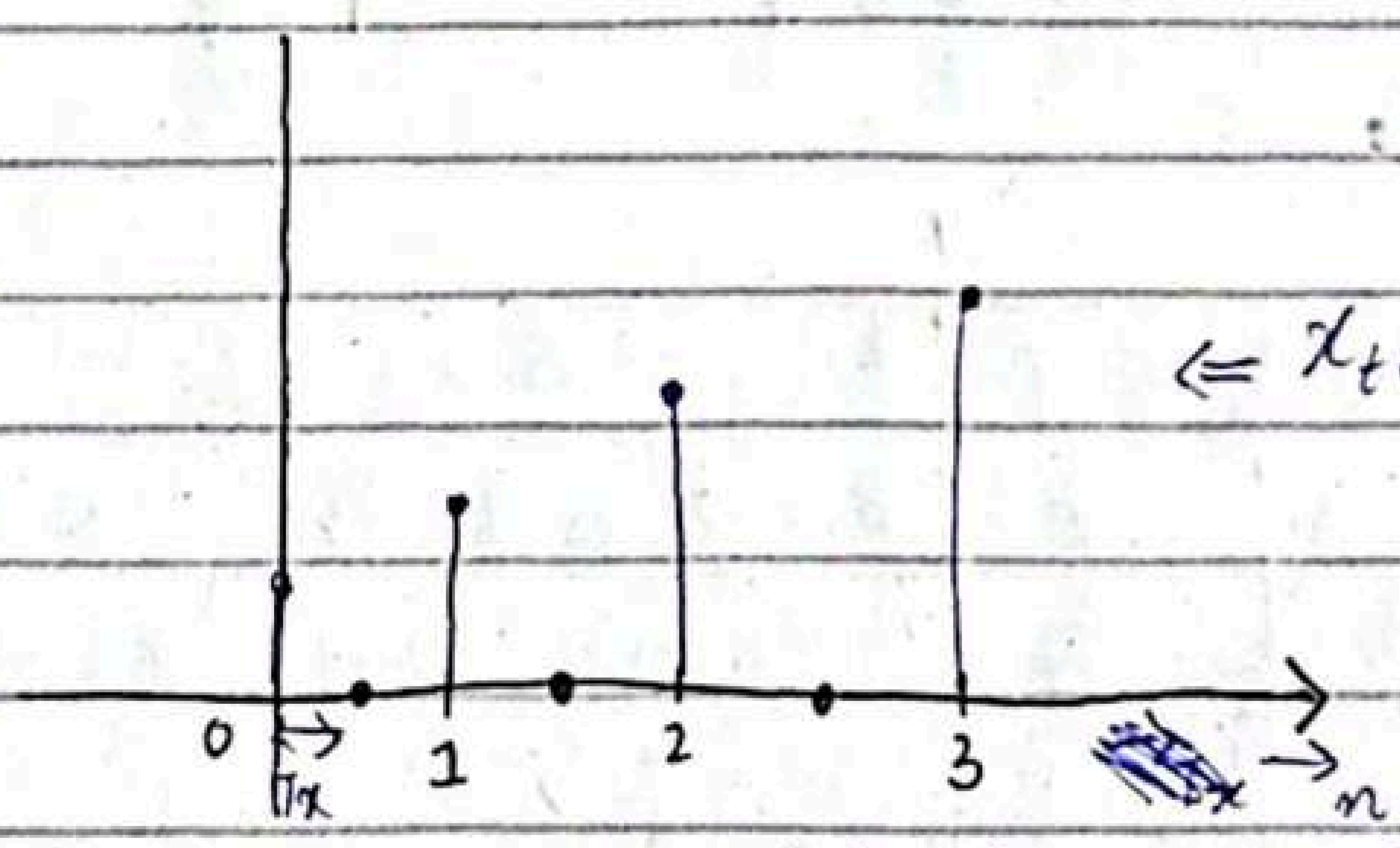
$$x(n) \rightarrow \boxed{\uparrow 2} \rightarrow y(n)$$

$F_y = 2F_x$

$(2-1) = 1$ add 1 zero between each sample



After every T_x we are sampling



$\Leftarrow x_t(n)$

$x_t(n) = \{1\,0\,2\,0\,3\,0\,4\}$

we add '0' between samples.

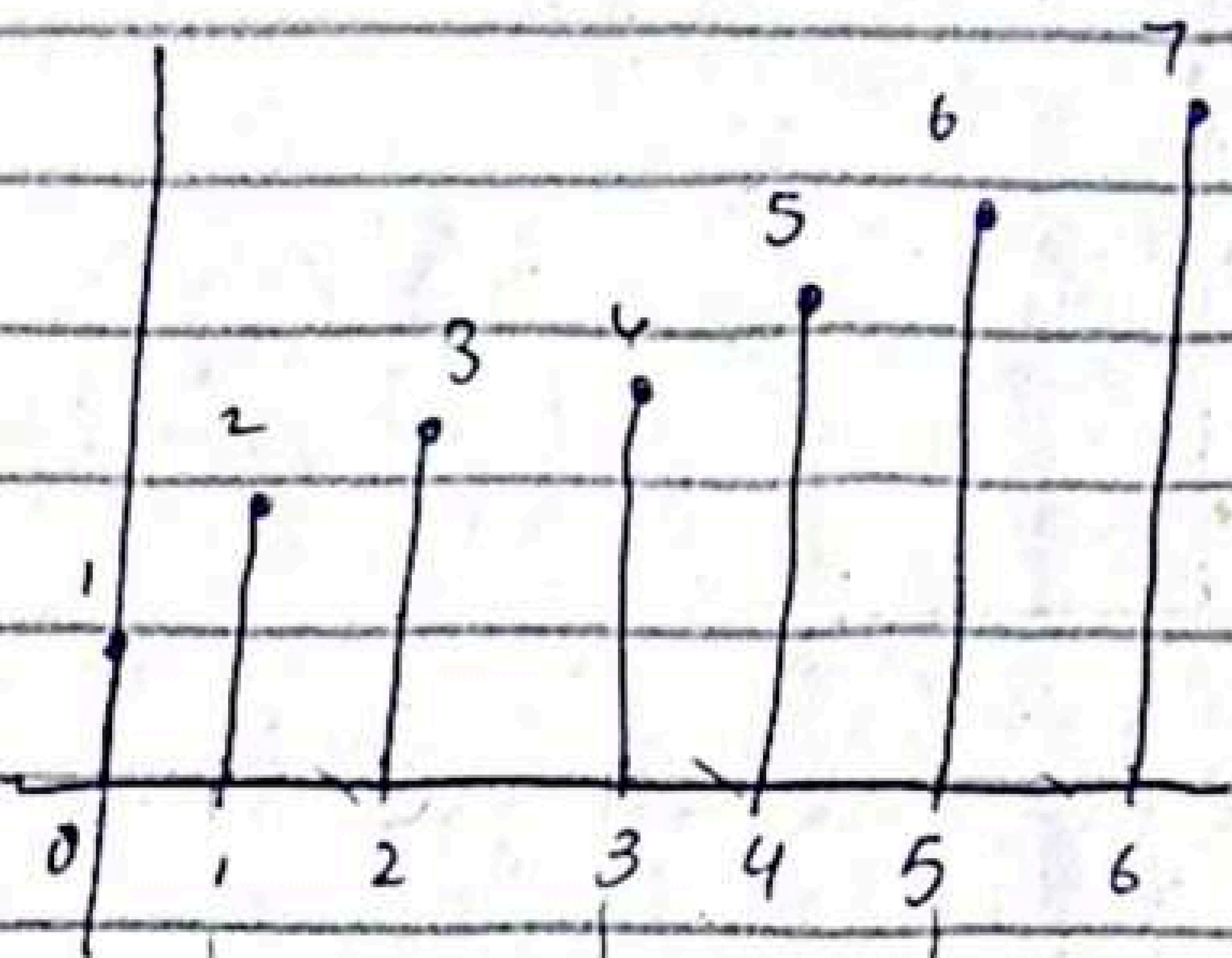
11/03/25

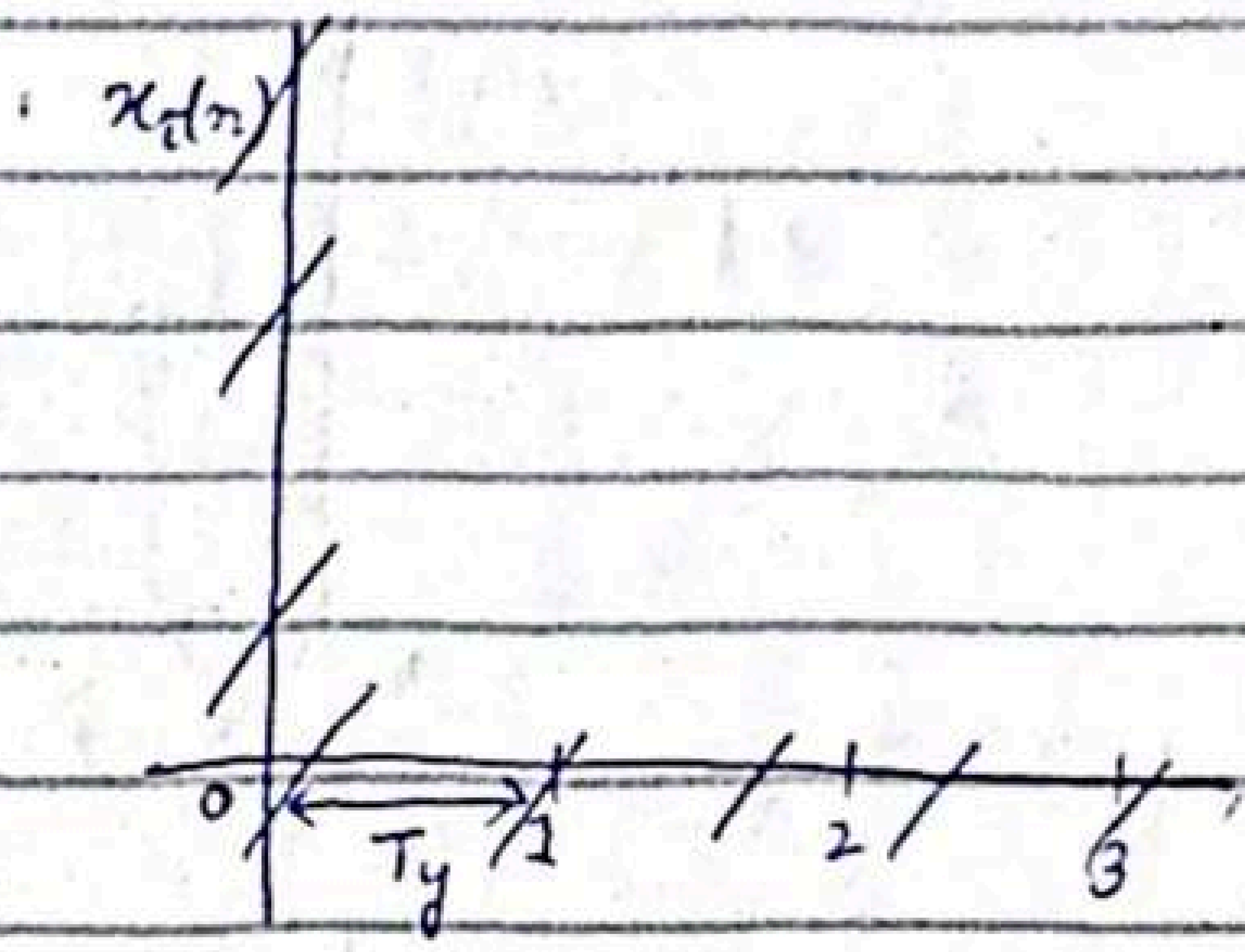
Down Sampler

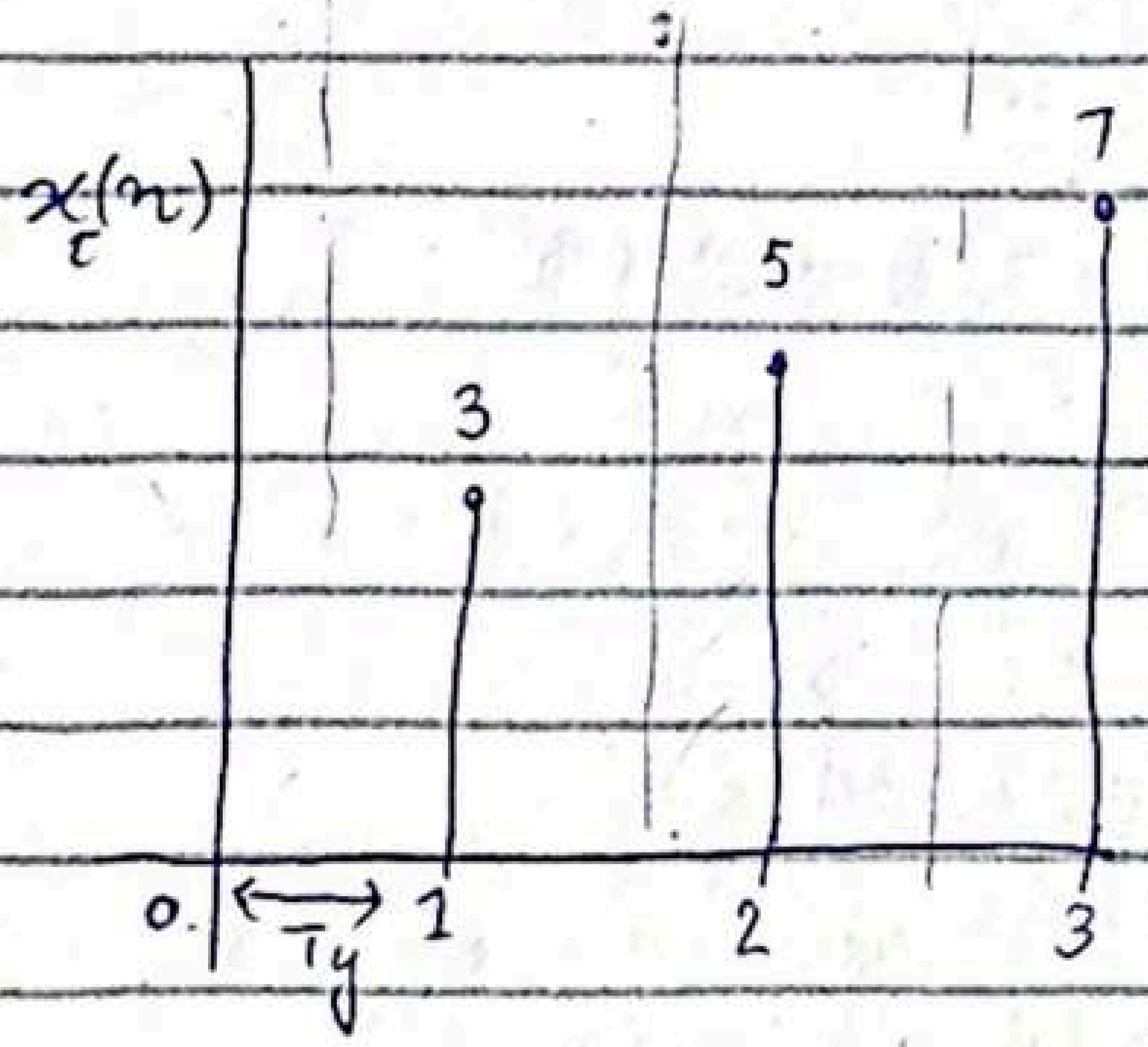
$$F_x \longrightarrow \boxed{\downarrow 2} \longrightarrow F_y$$

$$F_x > F_y$$

Let $x(n) = \{1, 2, 3, 4, 5, 6, 7\}$
$n = \{0, 1, 2, 3, 4, 5, 6\}$







Decimation by factor D

$$F_x \longrightarrow \boxed{D} \longrightarrow F_y$$

$$F_y = \frac{F_x}{D}$$

$(D-1)$ samples will be dropped.

Chapter #7

Discrete Fourier Transform

Frequency analysis of a discrete time signals is usually perfomed on a digital signal processing. To perform frequency analysis on a discrete time signal $\{x(n)\}$ we convert the time domain to an equivalent frequency domain, representation.

We know that such a representation is given by the fourier transform $X(w)$ of the sequence $x(n)$. However, $X(w)$ is continous function of frequeny and therefore it is not a computationally convenient representation of the sequence $\{x(n)\}$.

Therefore we now consider representation of a sequence $x(n)$ by samples of its spectrum $x(n)$ such representation is called discrete fourier transform (DFT)

$$\sin 50wt = \delta(w_0 - w) + \delta(w_0 + w)$$

if $X(w)$ is continous DTFT

if $X(w)$ is discrete DFT

7.1
Frequency - Domain sampling, The Discrete Fourier Transform

7.1.1
Frequency domain sampling of Discrete Time signals

Recall that

$$X(\omega) = \sum_{n=-\infty}^{\infty} x(n)\, e^{-j\omega n} \quad —1.1$$

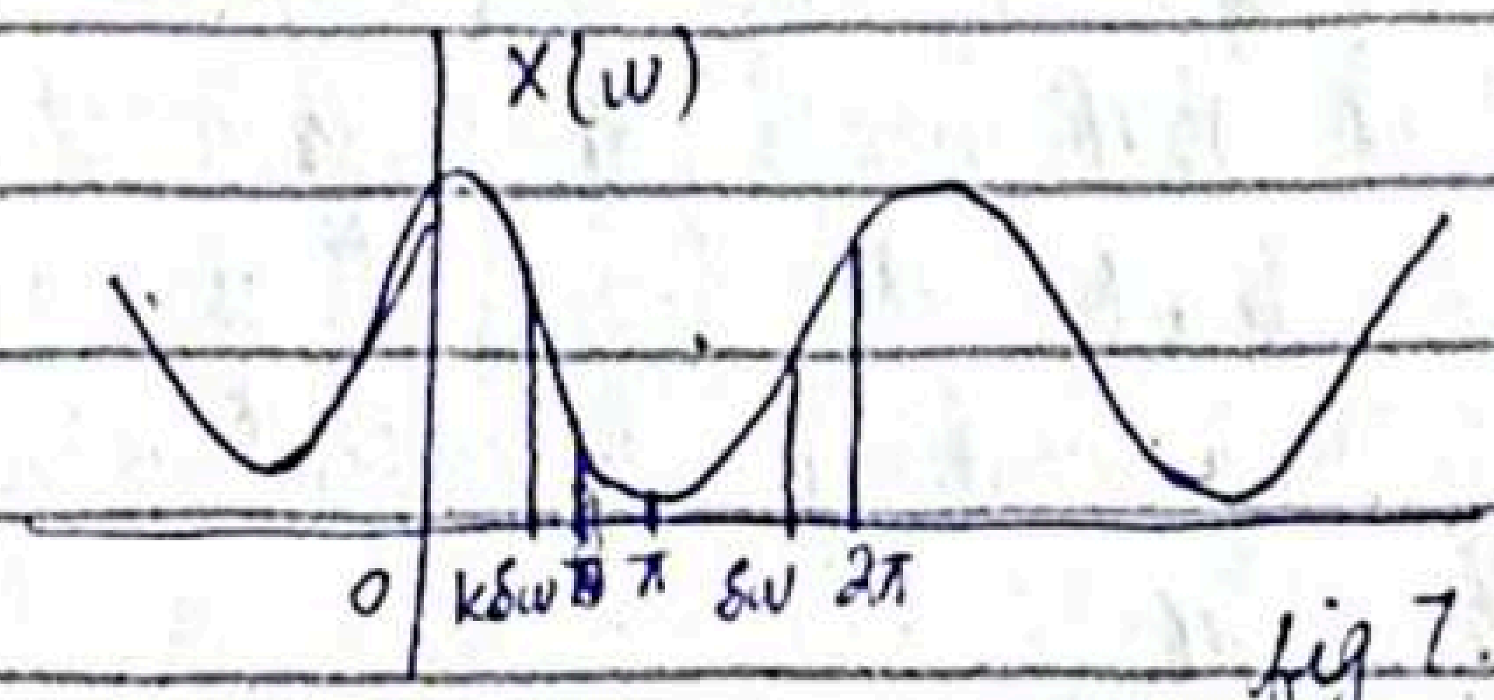


fig 7.1

Suppose that we sample $X(\omega)$ periodically in frequency at spacing of $\delta\omega$ radians between successive samples, where $X(\omega)$ is periodic with period 2π.

⇒ For convienience, we take N equidistant samples in the interval $0 \le \omega < 2\pi$ with spacing $\delta\omega = \frac{2\pi}{N}$ as shown in fig 7.1

⇒ If we evaluate (7.1.1) at $\omega = \frac{2\pi k}{N}$, we obtain

$$X\left(\frac{2\pi}{N}k\right) = \sum_{n=-\infty}^{\infty} x(n)\, e^{-j2\pi kn/N}$$

$k = 0, 1, \ldots N-1$

12/03/25

Discrete Fourier Transform

DFT:-

$$X(k) = \sum_{n=0}^{N-1} x(n)\, e^{\frac{-j2\pi kn}{N}} \quad \text{—(i)} \qquad k = 0, 1, \ldots, N-1$$

IDFT:-

$$x(n) = \frac{1}{N} \sum_{k=0}^{N-1} X(k)\, e^{\frac{j2\pi kn}{N}} \quad \text{—(ii)} \qquad n = 0, 1, 2, \ldots N-1$$

The DFT as a Linear Transformations

The formules in eq.(i) may be expressed as

$$X(k) = \sum_{n=0}^{N-1} x(n)\, W_N^{kn} \qquad k = 0, 1, \cdots N-1$$

$$x(n) = \frac{1}{N} \sum_{k=0}^{N-1} x(k)\, W_N^{-kn} \qquad n = 0, 1, \cdots N-1$$

where,

$$W_N = e^{\frac{-j2\pi}{N}}$$

4-point DFT:-

put value of 'k' and 'n'

$N=4$; $k = n = 0:3$

For

$$k=0 \Rightarrow x(0)W_4^0 + x(1)W_4^0 + x(2)W_4^0 + x(3)W_4^0$$
$$k=1 \Rightarrow x(0)W_4^0 + x(1)W_4^1 + x(2)W_4^2 + x(3)W_4^3$$
$$k=2 \Rightarrow x(0)W_4^0 + x(1)W_4^2 + x(2)W_4^4 + x(3)W_4^6$$
$$k=3 \Rightarrow x(0)W_4^0 + x(1)W_4^3 + x(2)W_4^6 + x(3)W_4^9$$

$\Downarrow$ 4point DFT $\downarrow$

$$\underset{4\times4}{W_N}\ \underset{4\times1}{x_N} = X(k)$$

It will be given

$$W_4 = \begin{bmatrix} W_4^0 & W_4^0 & W_4^0 & W_4^0 \\ W_4^0 & W_4^1 & W_4^2 & W_4^3 \\ W_4^0 & W_4^2 & W_4^4 & W_4^6 \\ W_4^0 & W_4^3 & W_4^6 & W_4^9 \end{bmatrix}_{4\times 4}$$

$W_4^0 = ?$

$W_4^1 = ?$

$W_4^2 = ?$

$W_4^3 = ?$

$$W_4^{kn} = e^{-j\frac{2\pi(0)}{4}} = 1$$

$$W_4^1 = e^{-j\frac{2\pi(1)}{4}} = e^{-j\frac{\pi}{2}} = \cos\frac{\pi}{2} - j\sin\frac{\pi}{2} = 0 - j(1) = -j$$

$$W_4^2 = e^{-j\frac{2\pi(2)}{4}} = e^{-j\pi} = \cos\pi - j\sin\pi = -1$$

$$W_4^3 = e^{-j\frac{2\pi(3)}{4}} = e^{-j\frac{3\pi}{2}} = \cos\left(\frac{3\pi}{2}\right) + j\sin\left(\frac{3\pi}{2}\right) = 0 - j(-1) = j$$

Periodicity:-

$$W_N^R = W_N^{R+N}$$

By Periodicity Property:-

$$W_4^0 = W_4^4$$

$$W_4^1 = W_4^5 = W_4^9$$

$$W_4^2 = W_4^6$$

$$W_N = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix}$$

Example 5.1.3 :-

Compute DFT of the 4-Point sequence

$x(n) = [0 \ 1 \ 2 \ 3]$ or if 4-point $x(n) = [0 \ 1]$ then $W_{4\times4} x_{2\times2}$ not possible. Then we do 0 padding $x(n) = [0 \ 1 \ 0 \ 0]$

Since, we have already found W_N

$$W_N = \begin{bmatrix} 0 & 1 & 2 & 3 \\ 0 & -1j & -2 & 3j \\ 0 & -1 & 2 & -3 \\ 0 & 1j & -2 & -3j \end{bmatrix}$$

$$W_N = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix}_{4\times4} \begin{bmatrix} 0 & 1 & 2 & 3 \end{bmatrix}_{1\times4}$$

Now,

$$= \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix}_{4\times4} \begin{bmatrix} 0 \\ 1 \\ 2 \\ 3 \end{bmatrix}_{4\times1} = \begin{bmatrix} 1(0) + 1(1) + 1(2) + 1(3) \\ 1(0) - j(1) - 1(2) + j(3) \\ 1(0) - 1(1) + 1(2) - 1(3) \\ 1(0) + j(1) - 1(2) - j(3) \end{bmatrix}$$

$$= \begin{bmatrix} 6 \\ 2j-2 \\ -2 \\ -2j-2 \end{bmatrix}_{4\times 1}$$ Answer

18/03/25

If we evaluate

$$X(\omega) = \sum_{n=-\infty}^{\infty} x(n)\, e^{-j\omega n}$$

at $\omega = 2\pi k/N$, we obtain

$$X\left(\frac{2\pi}{N}k\right) = \sum_{n=-\infty}^{\infty} x(n)\, e^{-j2\pi kn/N} \qquad k = 0, 1, \ldots N-1$$

The summation can be subdivided into an infinite number of summations, where each sum contains N terms. Thus,

$$X\left(\frac{2\pi}{N}k\right) = \cdots + \sum_{n=-N}^{-1} x(n)\, e^{-j2\pi kn/N} + \sum_{n=0}^{N-1} x(n)\, e^{-j2\pi kn/N} + \sum_{n=N}^{2N-1} x(n)\, e^{-j2\pi k} + \cdots$$

$$= \sum_{l=-\infty}^{\infty} \sum_{n=lN}^{lN+N-1} x(n)\, e^{-j2\pi kn/N}$$

e.g:-

$N = 4$

$$x(4) = [\underbrace{1\ 2}_{l=1}\ \underbrace{3\ 4}_{2}\ \underbrace{5\ 6}_{3}]$$

$$\sum_{-\infty}^{\infty} = \sum_{n=0}^{1} + \sum_{2}^{3} + \sum_{4}^{5}$$

$l=1 \quad l=2 \quad l=3$

$n = N = 4$

$lN + N - 1 = N + N - 1 = 7$ (8)

There should be l blocks with N members.

Here,

$$X\left(\frac{2\pi}{N}k\right) = \dots + \sum_{n=-N}^{-1} x(n)\, e^{-j2\pi kn/N} + \sum_{n=0}^{N-1} x(n)\, e^{-j2\pi kn/N} \dots$$

Here we take $n = n - lN$

$n = lN - lN = 0$

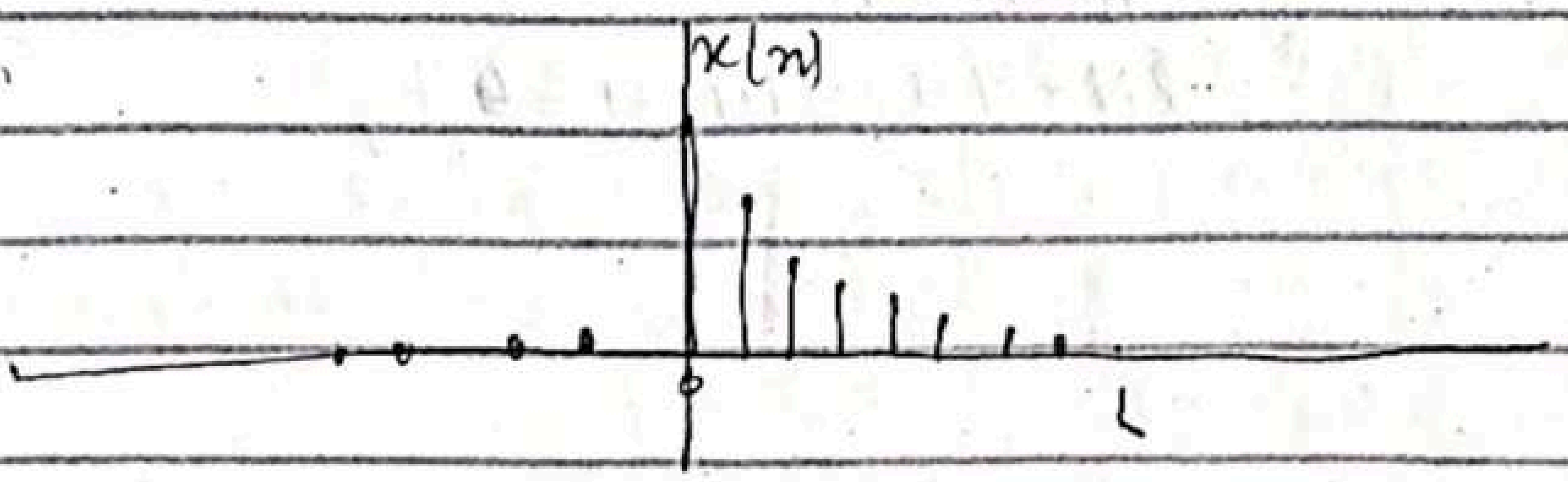
$n = lN + N - 1 - lN \quad = N - 1$

If we change the index in the inner summation from n to $n - lN$ and interchange the order of summation

$$X\left(\frac{2\pi}{N}k\right) = \sum_{n=0}^{N-1} \left[\sum_{l=-\infty}^{\infty} x(n - lN)\right] e^{-j2\pi kn/N}$$

$$x_p(n) = \sum_{l=-\infty}^{\infty} x(n - lN)$$ periodic extention of $x(n)$

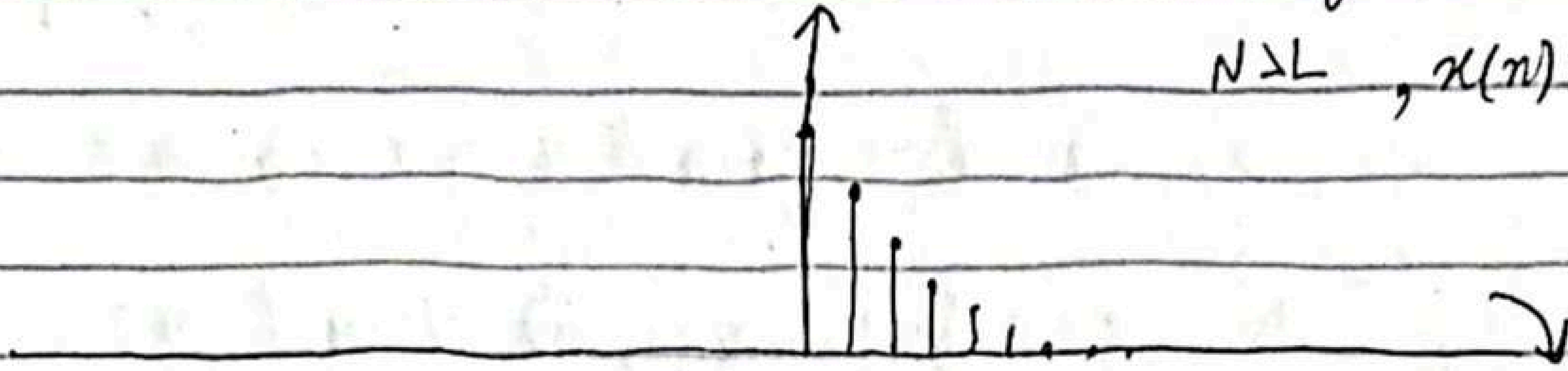
Since $x_p(n)$ is the periodic extension of $x(n)$ as given it is clear that $x(n)$ can be recovered from $x_p(n)$ if there is no aliasing in the time domain, that is if $x(n)$ is time-limited to less than the period N of $x_p(n)$. This is shown



2.0 → 3.25.

Quiz #2 → DFT

$N \geq L \quad , x(n) = x_p(n)$



slides

And if $N < L$ Then aliasing.

It is important to know that zero padding is done for better display of Fourier Transform

sequence $x(n)$ of length L [i.e. $x(n) = 0$ for $n < 0$ and $n \geq L$

$$X(\omega) = \sum_{n=0}^{L-1} x(n) e^{-j\omega n} \qquad 0 \leq \omega \leq 2\pi$$

where the upper and lower indices in the summation reflect the fact that $x(n) = 0$ outside the range $0 \leq n \leq L-1$. When we sample $X(\omega)$ at equally spaced frequencies $\omega_k = 2\pi k/N$, $k = 0, 1, \cdots$

slides.

Example:-

$$x(n) = a^n u(n) \qquad 0 < a < 1$$

The spectrum of the signal

$$\omega_k = \frac{2\pi k}{N} \quad , k = 0, 1, 2, \cdots N-1$$

$a = 0.8$ when $N = 5$, $N = 50$

for $N \geq L$ Then DFT and DTFT will be same.

20/03/25

Circular Symmetries of a sequence

As we have seen, the N-point DFT of a sequence, $x(n)$ of length $N \geq L$ is equivalent to the N-point DFT of a periodic sequence $x_p(n)$, of period N, which is obtained by periodically extending $x(n)$, That is

$$x_p(n) = \sum_{l=-\infty}^{\infty} x(n - lN)$$

Now, suppose we shift the periodic sequence $x_p(n)$ by 'k' units to the right. Thus we obtain another periodic sequence

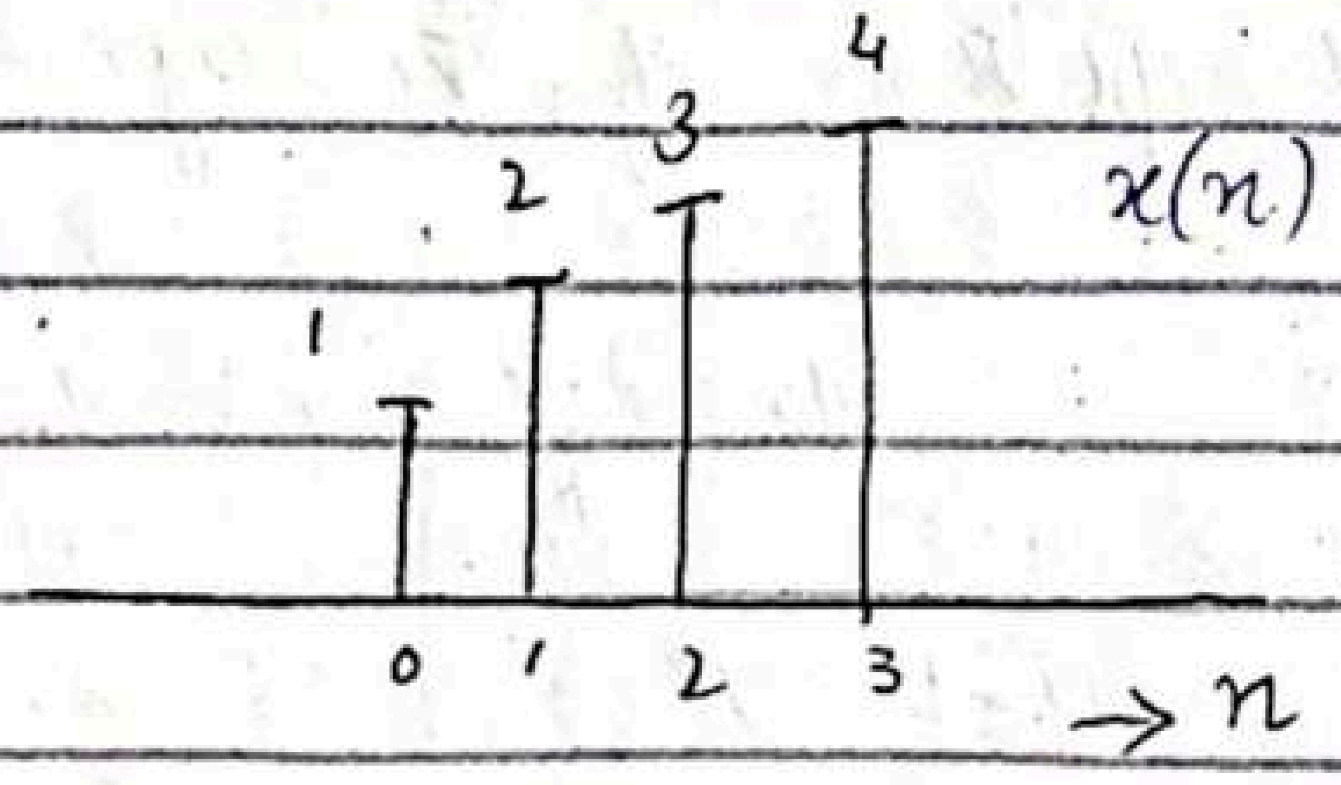
shift in 'n' by 'k'

$$x'_p(n) = x_p(n-k) = \sum_{l=-\infty}^{\infty} x((n-k) - lN)$$

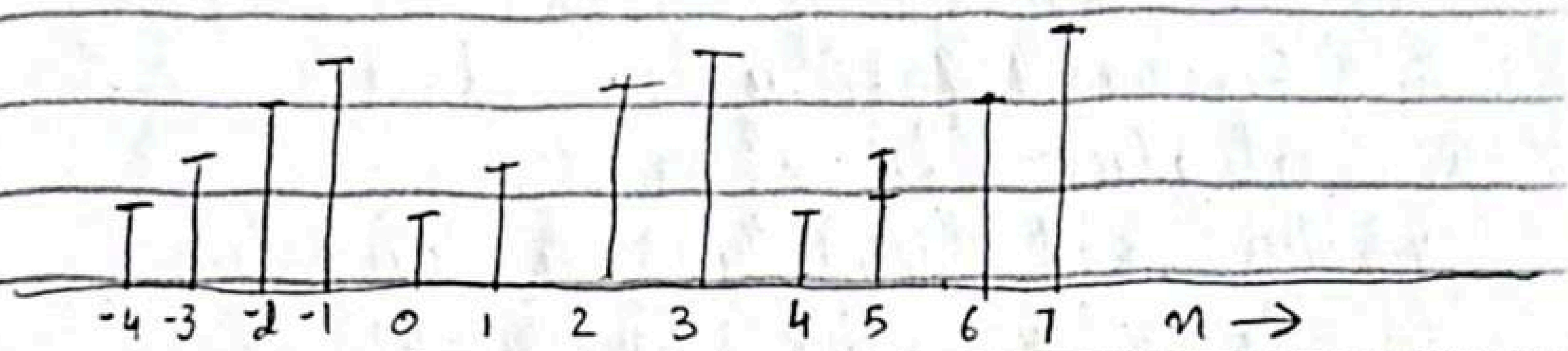
The finit-duration sequence

$$x'(n) = \begin{cases} x'_p(n) & 0 \leq n \leq N-1 \\ 0 & \text{otherwise} \end{cases}$$

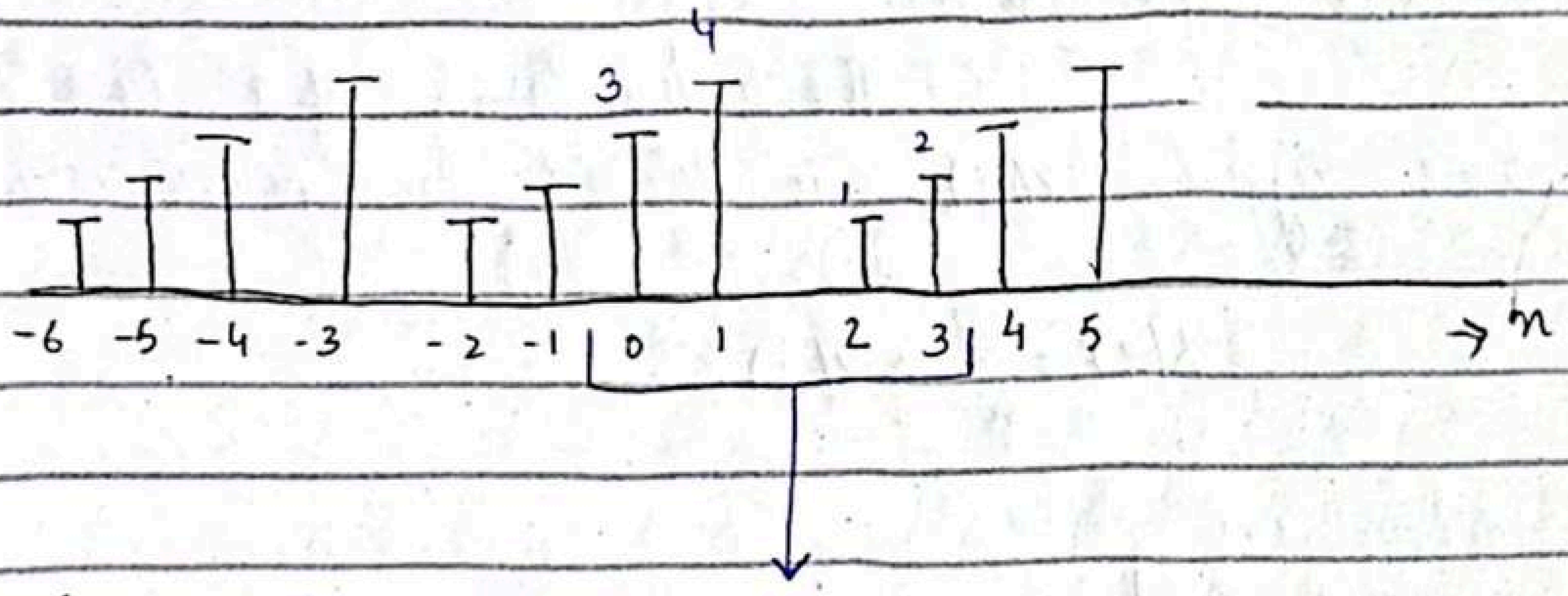
is related to the $x(n)$ by a circular shift As shown in fig 1, for $N=4$.



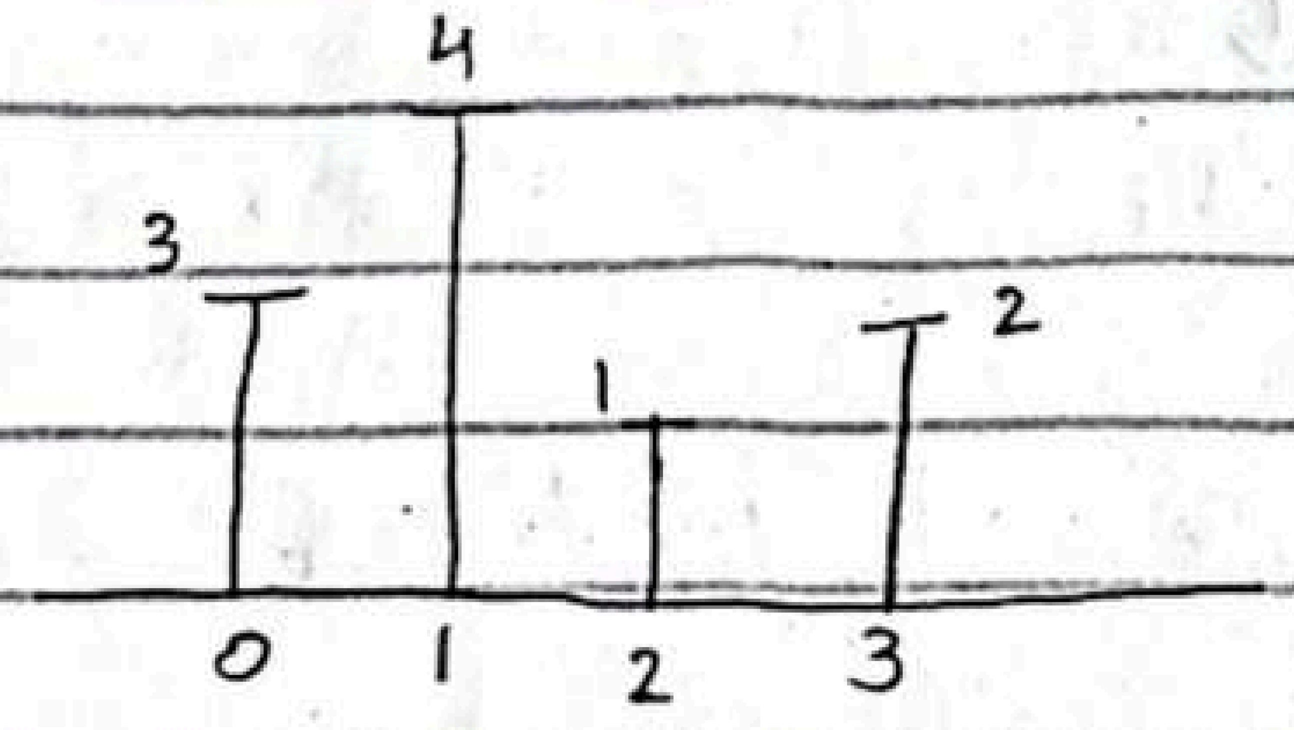
$x_p(n)$

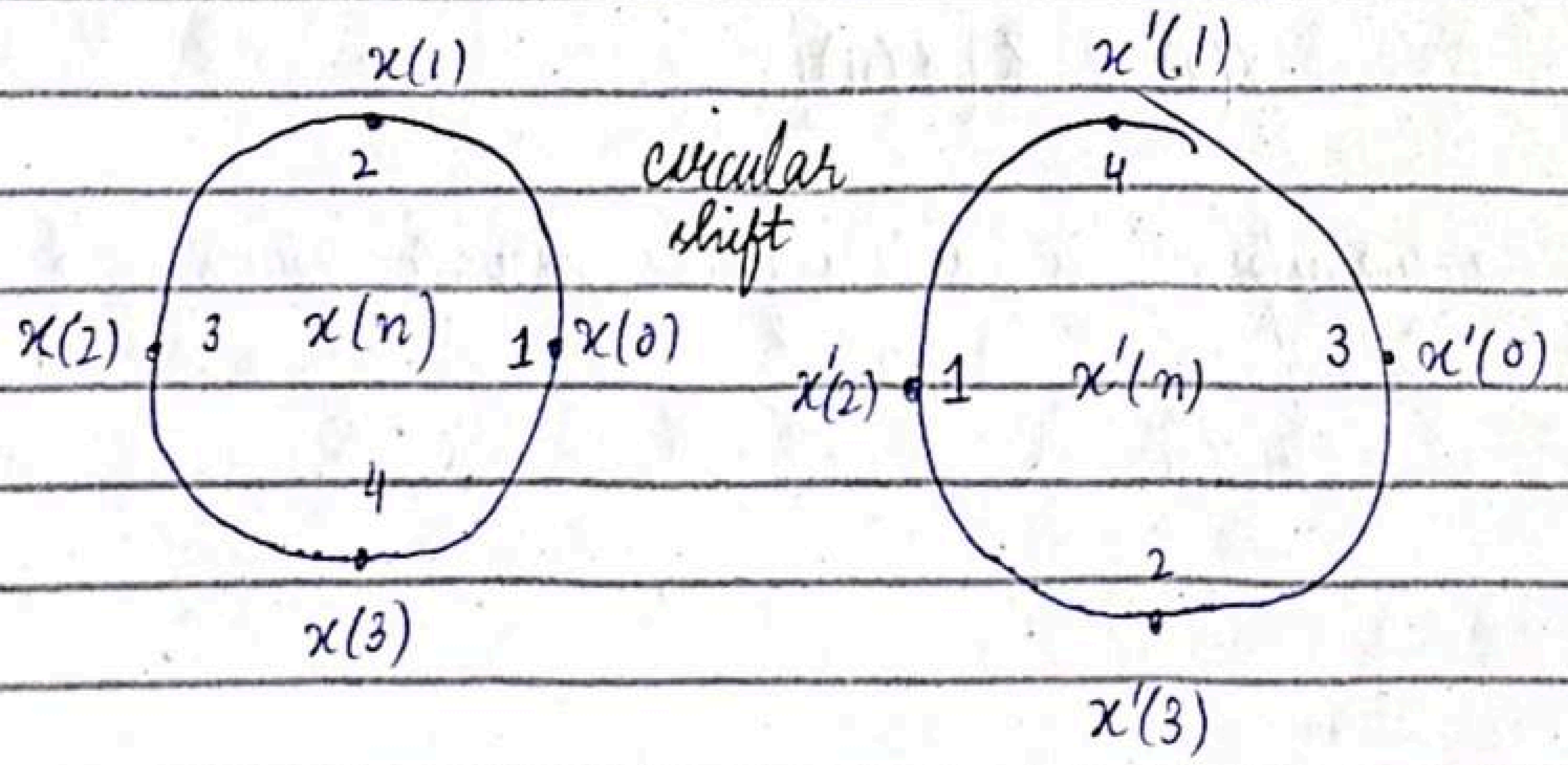

-4 -3 -2 -1 0 1 2 3 4 5 6 7 n →

$x_p(n-2)$


4
3
2
1
-6 -5 -4 -3 -2 -1 0 1 2 3 4 5 → n

$x'(n)$


3
4
1
2
0 1 2 3


x(1)
2
x(2) 3
x(n)
1 x(0)
4
x(3)
circular shift
x'(1)
4
x'(2) 1
x'(n)
3 x'(0)
2
x'(3)

25/03/25

Transform techniques are an important tool in the analysis of signals and linear time-invariant (LTI) systems. In this chapter we introduce the z-transform, develop its properties, and demonstrate its importance in the analysis and characterization $\rceil$ slides

The z-Transform

The direct z-Transform:-

The z-transform of a discrete time signal $x(n)$ is defined as power series.

$$X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$$

where 'z' $\rceil$

slides.

z-transform is denoted by

$$X(z) \cong Z[x(n)]$$

z-Transform is an infinite power series, it exists $\rceil$

slides.

Example 3.1.1

Determine z-transforms of the following finite duration signals

(a) $x_1(n) = \{1, 2, 5, 7, 0, 1\}$

(b) $x_2(n) = \{1, 2, \underset{\uparrow}{5}, 7, 0, 1\}$

solution:-

(a) $X_1(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$

$= 1 + 2z^{-1} + 5z^{-2} + 7z^{-3} + z^{-5} \quad z \neq 0$

ROC entire z-plane except $z=0$

$z \neq \infty$ (It will not exist)

(b) $X_2(z) = z^2 + 2z + 5 + 7z^{-1} + z^{-3}$

ROC entire plane except $z=0$ & $z=\infty$

z^k $(k>0)$ becomes unbounded for $z=\infty$

z^{-k} $(k>0)$ unbounded for $z=0$

Example 3.1.2

Determine the z-transform of the signal

$x(n) = \left(\frac{1}{2}\right)^n u(n)$

solution :-

$x(n)$ consists of an infinite number of non-zero values.

$$x(n) = \{1, (\tfrac{1}{2}), (\tfrac{1}{2})^2, (\tfrac{1}{2})^3, \cdots (\tfrac{1}{2})^n, \cdots\}$$

The z-transform of $x(n)$ is the finite power series.

$$X(z) = 1 + \frac{1}{2} z^{-1} + (\tfrac{1}{2})^2 z^{-2} + (\tfrac{1}{2})^n z^{-n} + \cdots$$

$$= \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^n z^{-n} = \sum_{n=0}^{\infty} \left(\frac{1}{2} z^{-1}\right)^n$$

This is infinite GM series

$$1 + A + A^2 + A^3 + \cdots = \frac{1}{1-A}, \quad |A| < 1$$

Consequently, for $|\frac{1}{2} z^{-1}| < 1$, or equivalently, for $|z| > \frac{1}{2}$

$$X(z) = \frac{1}{1 - \frac{1}{2} z^{-1}} \qquad \text{ROC } |z| > \frac{1}{2}$$

Example 3.1.3

Determine the z-Transform of signal

$$x(n) = \alpha^n u(n) = \begin{cases} \alpha^n & , n \geq 0 \\ 0 & , n < 0 \end{cases}$$

$\cancel{= 1 + a + a^2 + a^3 + a^4 + \dots\dots}$

$$X(z) = \sum_{n=0}^{\infty} \alpha^n z^{-n} = \sum_{n=0}^{\infty} (\alpha z^{-1})^n$$

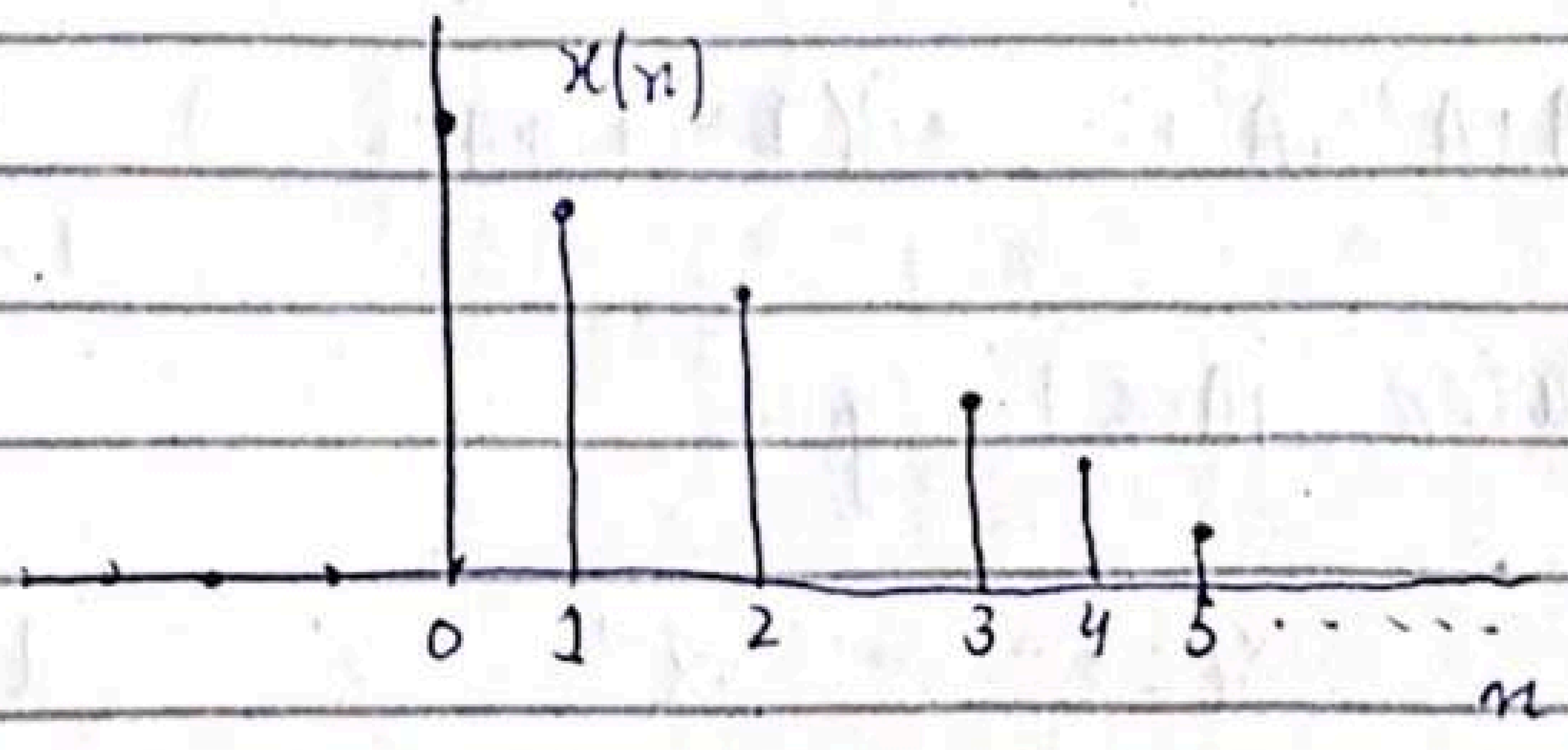
if $|\alpha z^{-1}| < 1$ or equivalently, $|z| > |\alpha|$, this power series converges to $1/(1 - \alpha z^{-1})$

Thuse we have z-Transform pair

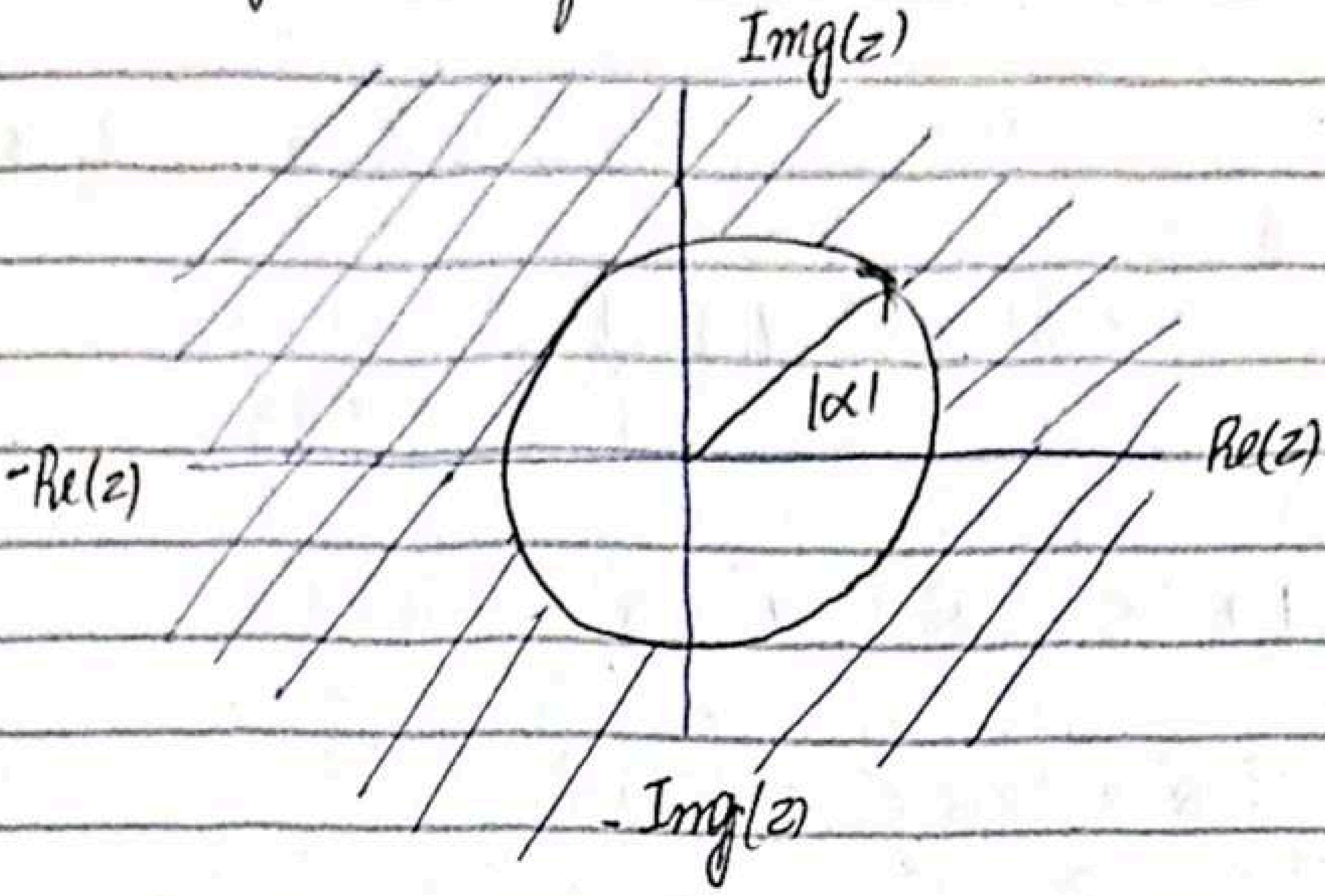
$$x(n) = \alpha^n u(n) \xleftrightarrow{z} X(z) = \frac{1}{1 - \alpha z^{-1}} \quad \text{ROC } |z| > |\alpha|$$

The ROC ↓
slides

The exponential signal $x(n) = \alpha^n u(n)$



ROC of z- Transform:



Example 3.1.4

Determine the z- transform of the signal

$$x(n) = -\alpha^n u(-n-1) = \begin{cases} 0 & n \geq 0 \\ -\alpha^n & n \leq -1 \end{cases}$$

we have

$$X(z) = \sum_{n=-\infty}^{-1} (-\alpha^n) z^{-n} = -\sum_{l=1}^{\infty} (\alpha^{-1} z)^l$$

where $l = -n$, using formula

$$A + A^2 + A^3 + \cdots = A(1 + A + A^2 + \cdots) = \frac{A}{1-A} \rightarrow |A| < 1 \text{ exists at}$$

when $|A| < 1$ gives

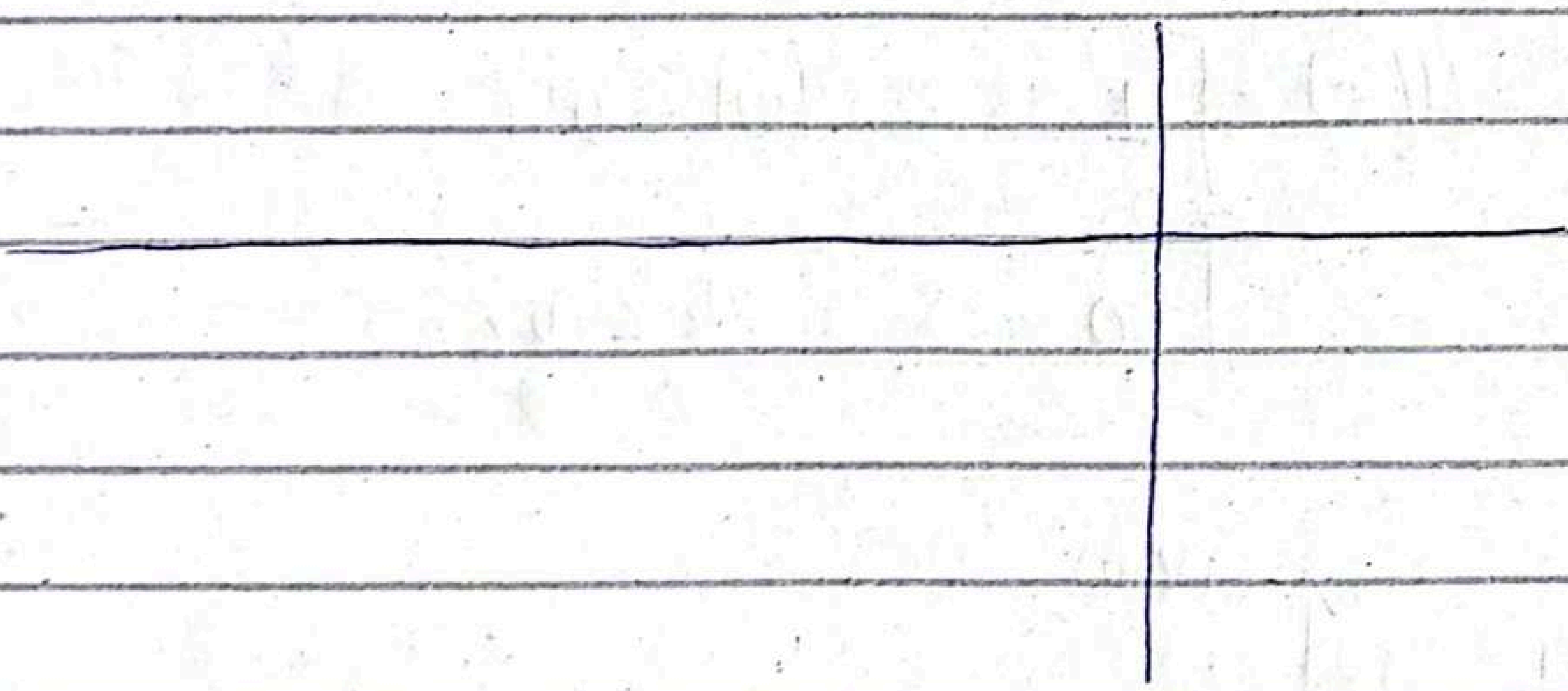
$$X(z) = -\frac{\alpha^{-1} z}{1 - \alpha^{-1} z} = \frac{1}{1 - \alpha z^{-1}}$$

provided that $|\alpha^{-1}z| < 1$ or equivalently $|z| < |\alpha|$

Hence $x(n) = -\alpha^{n} u(-n-1) \xleftrightarrow{z} X(z) = \dfrac{1}{1-\alpha z^{-1}}$

ROC $|z| < |\alpha|$

Anticausal signal $x(n)$ is



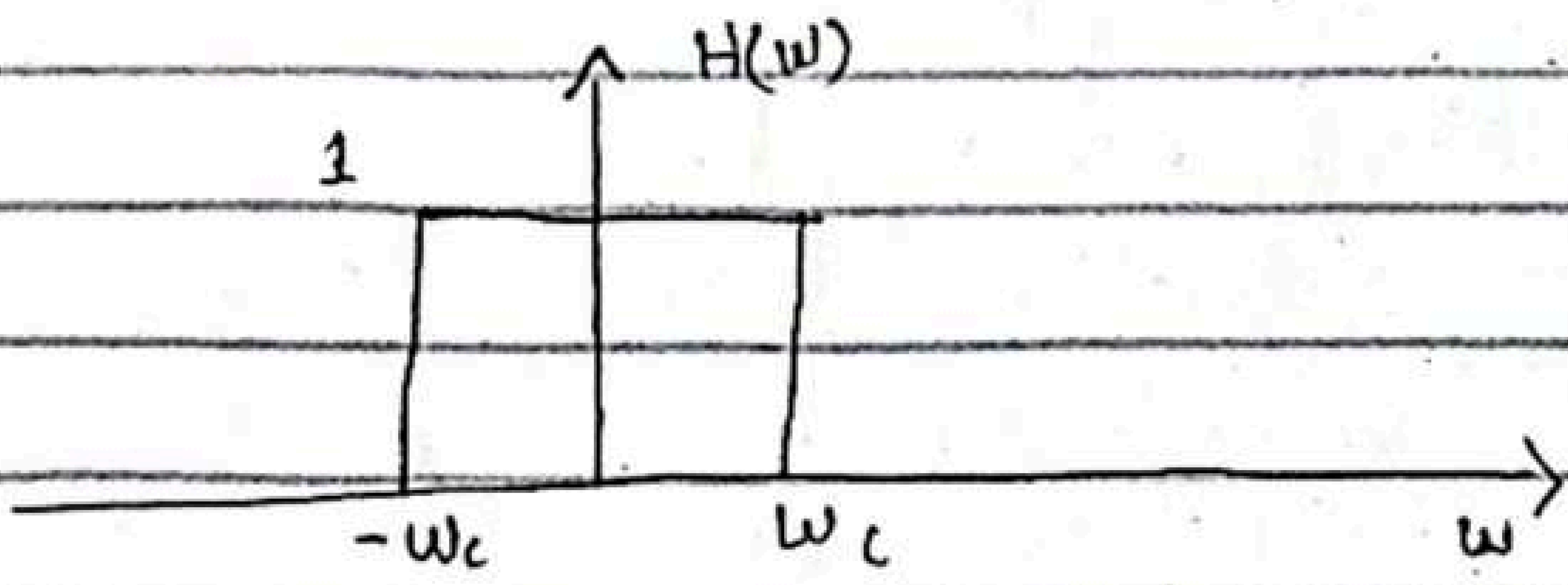
15/04/25

Digital Filters

*** Ideal Low Pass Filter :-**

The impulse response $h(n)$ of an ideal Low Pass filter with frequency characteristics is

$$H(w) = \begin{cases} 1 & |w| \le w_c \\ 0 & w_c < w < \pi \end{cases}$$



What would be its impulse response $h(n)$?

$$h(n) = ?$$

$$h(n) = \frac{1}{2\pi}\int_{-w_c}^{w_c} H(w)\, e^{jwn}\, dw$$

$$= \frac{1}{2\pi}\int_{-w_c}^{w_c} 1\, e^{jwn}\, dw$$

$$= \frac{1}{2\pi} \left. \frac{e^{jwn}}{jn} \right|_{-w_c}^{w_c} = \frac{1}{2\pi j n}\left[e^{jw_c n} - e^{-jw_c n}\right]$$

$$= \frac{1}{2\pi j n}\left(e^{jw_c n} - e^{-jw_c n}\right) = \left[\frac{e^{jw_c n} - e^{-jw_c n}}{2j}\right]\frac{1}{\pi n}$$

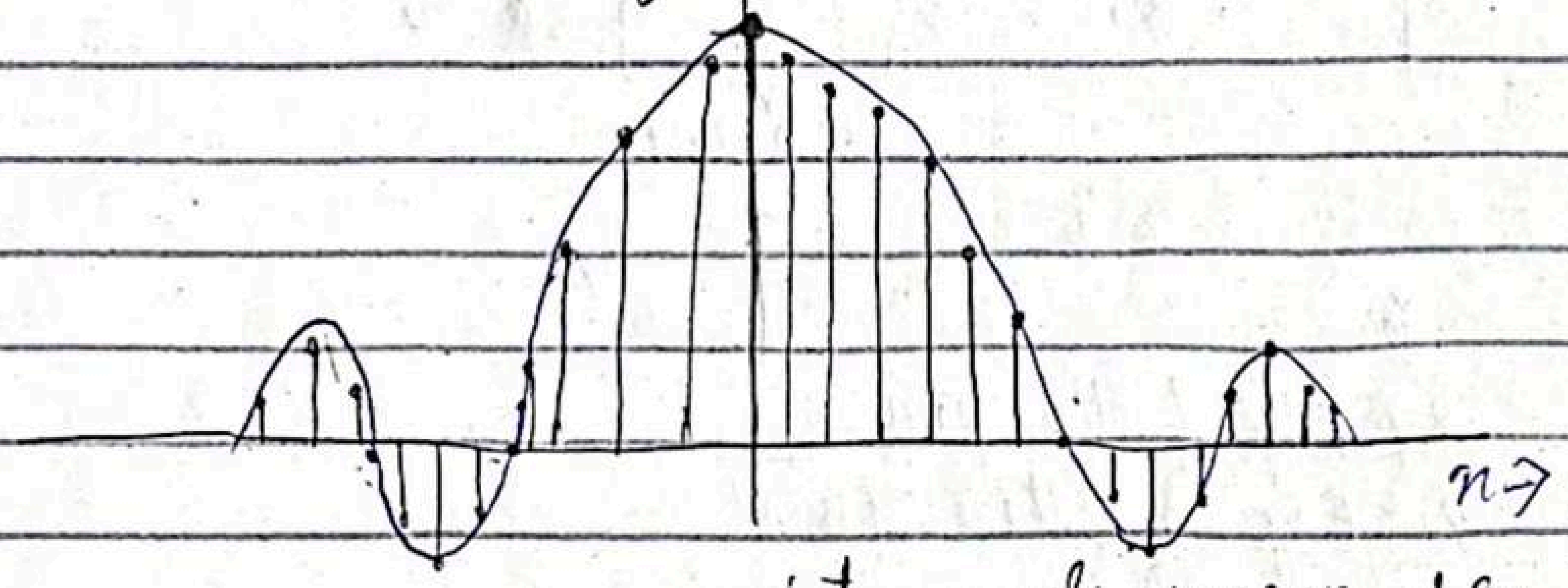
$$= \frac{\sin w_c n}{\pi n} \qquad n \neq 0$$

For $n=0$

$$h(0) = \frac{1}{2\pi}\int_{-w_c}^{w_c} e^{jw(0)}\,dw = \frac{1}{2\pi}\int_{-w_c}^{w_c}(1)\,dw = \frac{1}{2\pi}(w_c + w_c) = \frac{2w_c}{2\pi}$$

$$= \frac{w_c}{2\pi}$$

$$h(n) = \begin{cases} w_c/\pi & n=0 \\ \dfrac{\sin(nw_c)}{\pi n} & , n \neq 0 \end{cases}$$



unit sample response of an ideal LPF

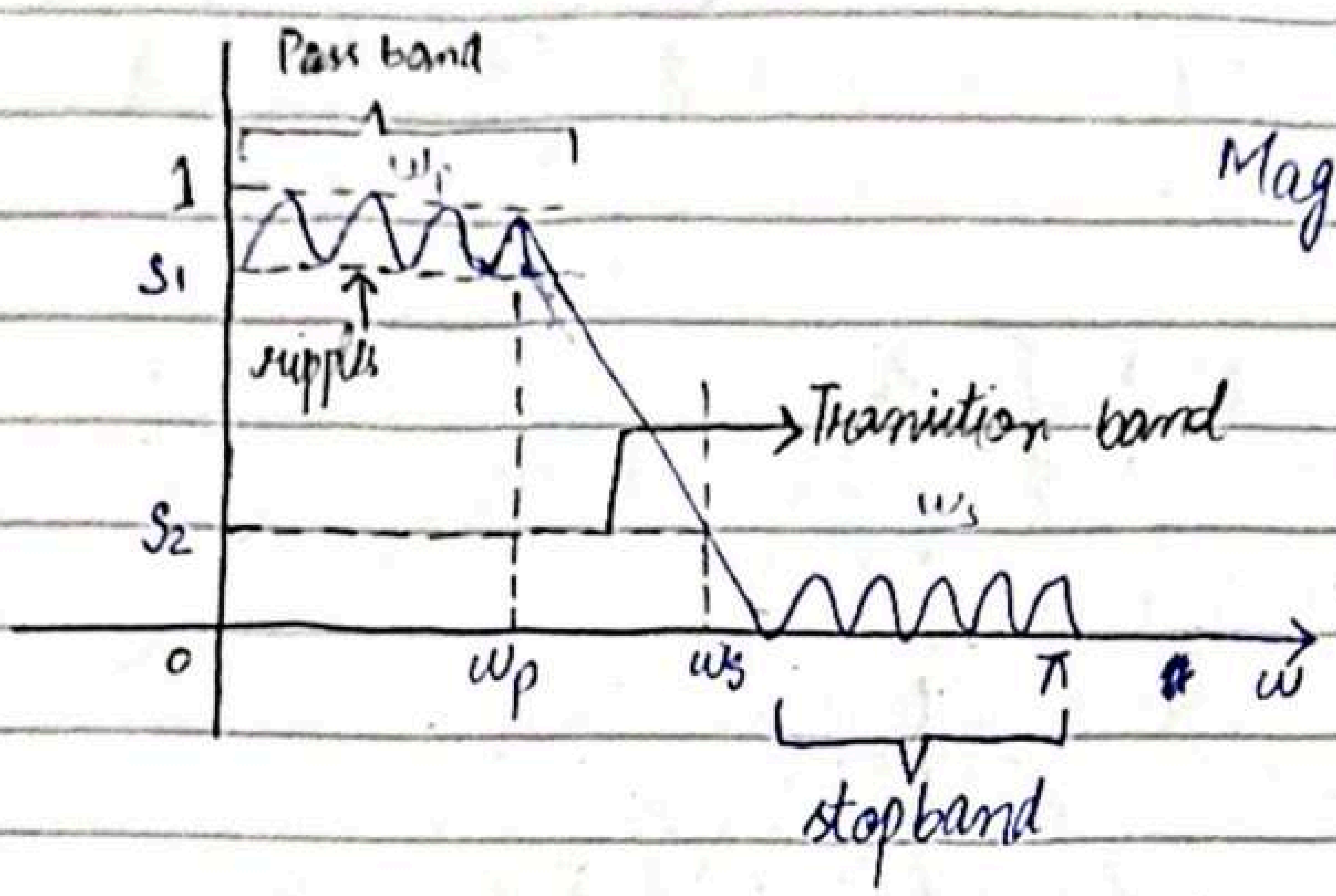
Ideal LPF is not phsically realizable because its impulse response in infinite

* Characteristics of Practical Frequency-selective Filters :-

* As we observed that ideal filters are unrealizable for real-time signal Processing applications. They are not absolutely necessary in most practical applications. A small amount of ripple is usually tolerable in passband. Similarly, a small non-zero value or small amount of ripple is also

tolerable in stopband

* The Transition of the frequency response from passband to stop band defines the transition band.



Magnitude of the characteristics of physically realizable filters

ω_p = Edge of the passband
ω_s = Edge of the stopband.

* Width of the passband is called bandwidth of the filter.

· Design of Linear-Phase FIR filters using Windows

Finite impulse response.

* In this method we begin with the desired frequency response specification $H_d(\omega)$ and determine the corresponding unit sample response $h_d(n)$. Indeed, $h_d(n)$ is related to $H_d(\omega)$ by Fourier Transform relation

$$H_d(\omega) = \sum_{n=0}^{\infty} h_d(n) e^{-j\omega n}$$

truncating
cut a channel,
limiting

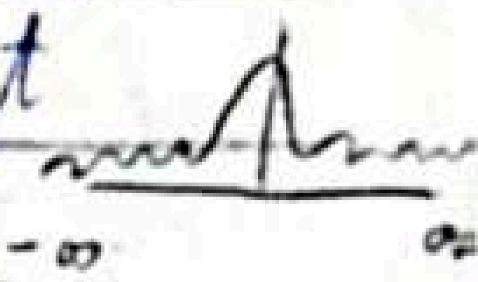
where

$$h_d(n) = \frac{1}{2\pi}\int_{-\pi}^{\pi} H_d(\omega)\, e^{j\omega n}\, d\omega \quad \text{—(i)}$$

* Thus, given $H_d(\omega)$, we can determine $h_d(n)$ by evaluating the integral in eq^n i.

(infinite IIR impulse response)

* In general, $h_d(n)$ is infinite in duration and must be ~~(truncation~~ truncated ~~yield~~ at some point, say at $n = M-1$, to yield an FIR filter of length M.



* Truncation of $h_d(n)$ to a length $M-1$ in equivalent to multiplying by $h_d(n)$ by a rectangular window, defined as:-

$$w(n) = \begin{cases} 1, & n = 0, 1, \ldots M-1 \\ 0 & \text{otherwise} \end{cases}$$

Thus, the unit sample response of the FIR filter becomes:-

$$h(n) = h_d(n)\, w(n)$$

$$h(n) = \begin{cases} h_d(n), & n = 0, 1, \ldots M-1 \\ 0 & \text{otherwise} \end{cases}$$

Characteristics Of Commonly Used Analog filters

→ IIR digital filters can easily be obtained by beginning with an analog filter and then using a mapping to transform the s-plane into the z-plane. Thus, the design of a digital filter is reduced to designing an appropriate analog filter and then performing the conversion from H(s) to H(z) plane.

Butter Worth Filter :-

Lowpass butterworth filters are all-pole filters characterized by the following magnitude-squared frequency response.

$$|H\Omega|^2 = \frac{1}{H\left(\frac{\Omega}{\Omega_c}\right)^{2N}}$$

where N is the order of filter, Ω_c is the -3dB cut-off frequency.

since, $H(s)\,H(-s)$ evaluated at $s=j\Omega$ is simply equal to $|H(\Omega)|^2$ it follows that

$$|H(\Omega)|^2 = H(s)H(-s) = \frac{1}{1+\left(\frac{-s^2}{\Omega_c^2}\right)^N} \quad \text{—— 8.3.48}$$

From 8.3.48, we find that:

$$\frac{-s^2}{\Omega_c^2} = (-1)^{1/N} = e^{j(2k+1)\pi/N}$$

$k = 0, 1, 2, \dots N-1$

and hence,

poles $\rightarrow$ $S_k = \Omega_c e^{j\pi/2} e^{j(2k+1)\pi/2N}$ $k = 0, 1, 2, \dots N-1$

Now,

$$|H(\Omega)|^2 = H(s)\, H(-s) = \frac{1}{1 + \left(\frac{\Omega}{\Omega_c}\right)^{2N}}$$

At $\Omega = 0 \Rightarrow H(0) = 1$
At $\Omega = \Omega_c \Rightarrow H(\Omega_c) = \frac{1}{2}$
At $\Omega \gg \Omega_c \Rightarrow \left(\frac{\Omega_c}{\Omega}\right)^{2N} \downarrow$
$|H(\Omega)|^2$ approach 0.

$$\frac{1}{1 + \left(\frac{\sqrt{\Omega^2}}{\sqrt{\Omega_c}}\right)^{2N}} = \frac{1}{1 + \left(\frac{\sqrt{-s^2}}{\Omega_c}\right)^{2N}}$$

$\Omega = \sqrt{\Omega^2}$

At $s = j\Omega \Rightarrow s^2 = j^2 \Omega^2 = -\Omega^2 \Rightarrow -s^2 = \Omega^2$

$$= \frac{1}{1 + \left(\frac{\sqrt{-s^2}}{\Omega_c}\right)^{2N}} = \frac{1}{1 + \left(\frac{-s^2}{\Omega_c^2}\right)^{N}}$$

$$|H(\Omega)|^2 = H(s)\, H(s) = \frac{1}{1 + \left(\frac{-s^2}{\Omega_c^2}\right)^{N}}$$

Find equation for poles

$$1 + \left(\frac{-s^2}{\Omega_c^2}\right)^N = 0$$

Multiply by power $1/N$

$$\left(\frac{-s^2}{\Omega_c^2}\right)^N = -1 \Rightarrow \frac{-s^2}{\Omega_c^2} = (-1)^{1/N}$$

we know
$-1 = e^{j\pi}$
since $e^{j\pi} = \cos\pi + j\sin\pi = -1$

$e^{j\pi} = e^{j(\pi + 2\pi k)}$
$e^{j\pi} = e^{j(2k+1)\pi}$
adding $2\pi k$ does not change any value.

-3dB cutoff freq.

$$-\frac{s^2}{\Omega_c^2} = e^{j\pi/N} = e^{j(2k+1)\pi/N} \quad \text{for } k = 0, 1, 2, \ldots N-1$$

$$-s^2 = \Omega_c^2 e^{j(2k+1)\pi/N} \Rightarrow s^2 = \Omega_c^2 (-1) e^{j(2k+1)\pi/N}$$

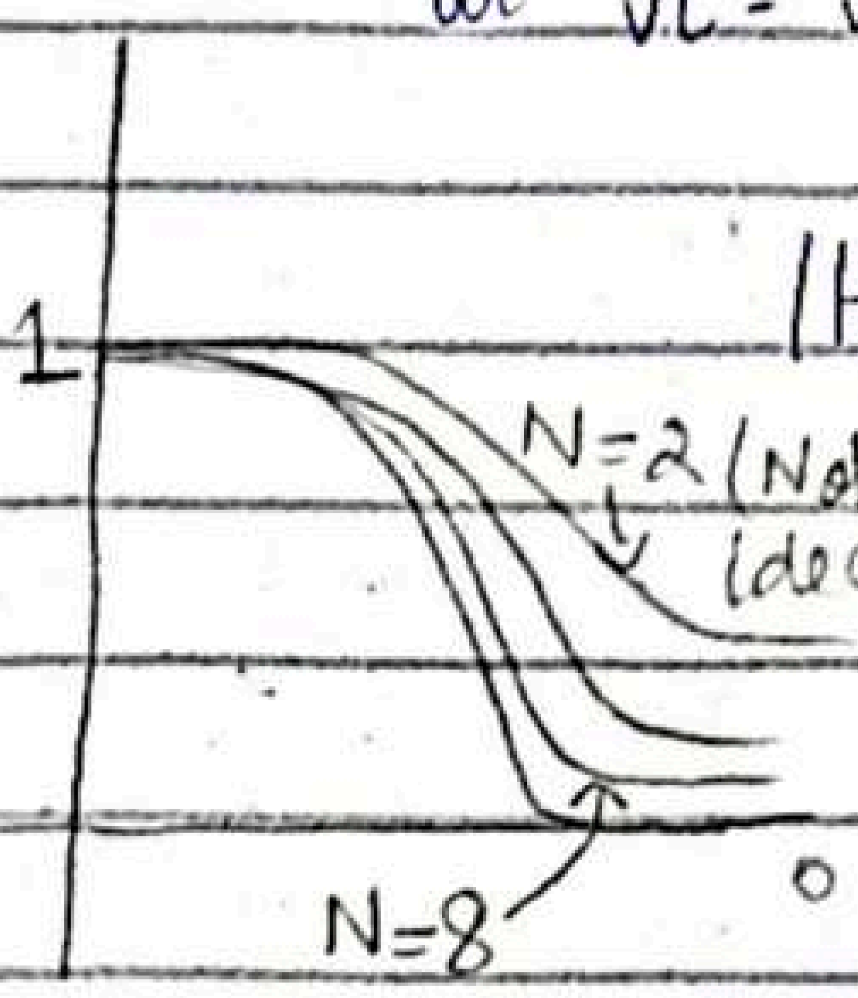
$$\sqrt{s^2} = \sqrt{\Omega_c^2}\sqrt{e^{j(2k+1)\pi/N}} \qquad \sqrt{e^{j\pi}} = \sqrt{\Omega_c^2}\left(e^{j(2k+1)\pi/N}\right)^{1/2}$$

$(e^{j\pi})^{1/2}$

$$s_k = \Omega_c e^{j\pi/2} e^{j(2k+1)\pi/2N} \rightarrow \text{poles. (eq to find poles)}$$

$$|H(\Omega)|^2 = H(s)H(-s) = \frac{1}{1+\left(\frac{-s^2}{\Omega_c^2}\right)^N} \quad \text{— } 8.3.47$$

The order of the filter can be obtained from eq (8.3.47). We want "N" to be have attenuation δ_2^2 at a specified frequency Ω_s. Thus, at $\Omega = \Omega_s$,



closer to ideal

Cross multiply

$$|H(\Omega_s)|^2 = \frac{1}{1+\left(\frac{\Omega_s}{\Omega_c}\right)^{2N}} = \delta^2 \Rightarrow \frac{1}{\delta^2} = 1 + \left(\frac{\Omega_s}{\Omega_c}\right)^{2N}$$

$$\left(\frac{\Omega_s}{\Omega_c}\right)^{2N} = \frac{1}{\delta^2} - 1 \Rightarrow \text{Take Natural log}$$

$$2N\log\left(\frac{\Omega_s}{\Omega_c}\right) = \log\left(\frac{1}{\delta^2} - 1\right) \Rightarrow N = \frac{\log(1/\delta^2 - 1)}{2\log\left(\frac{\Omega_s}{\Omega_c}\right)}$$

N → order of filter; Ω_s → stop band freq; Ω_c → cut-off freq

Example 8.3.6

Determine the order and poles of a low pass Butterworth filter that has a -3dB bandwidth of 500Hz and an attenuation of 40dB at 1000 Hz.

solution:-

$N = ?$ $\qquad \Omega = 2\pi f = 2\pi \Omega$

$S_k = ?$

$\Omega_c = 500\,Hz = 1000 = 500 \times 2\pi = 1000\pi$

$\Omega_s = 1000\,Hz = 2000\pi$

$$N = \frac{\log\left(\frac{1}{\delta^2} - 1\right)}{2\log\left(\frac{\Omega_s}{\Omega_c}\right)}$$

$$S_k = \Omega_c\, e^{j\pi/2}\, e^{j(2k+1)\pi/2N}$$

$$= (500)\, e^{j\pi/2}\, e^{j\pi}$$

$S_Q = 1$

$$= \frac{\log\left(\frac{1}{40^2} - 1\right)}{2\log\left(\frac{1000}{500}\right)}$$

$$\boxed{20\log\left(\frac{1}{\delta}\right) = dB}\ 40$$

$$20\log\left(\frac{1}{\delta}\right) = 40$$

$$N = \frac{\log\left(\frac{1}{(0.01)^2} - 1\right)}{2\log\left(\frac{2000\pi}{1000\pi}\right)}$$

$$\log\left(\frac{1}{\delta}\right) = 2$$

$$\delta = \frac{1}{10^2} = 0.01$$

$N = 6.643 \approx 7$

$\delta = 0.01$

Poles:-

$$S_k = \Omega_c\, e^{j\pi/2}\, e^{j(2k+1)\pi/2N} \qquad k = 0, 1, \cdots N-1$$

$k=0$ $\quad = (1500)\, e^{j\pi/2}\, e^{j(0+1)\pi/2\times 7}$ $\qquad k = 0 \to 6$

7 poles

24/04/25

Butter Worth filters:-

Magnitude of frequency response $= |H(\Omega)|^2 = \frac{1}{1+\left(\frac{\Omega}{\Omega_c}\right)^2}$

$$= \frac{1}{1+\left(\frac{-s^2}{\Omega_c^2}\right)^N}$$

Ω_c = -3dB / cut off frequency

$S_k = \Omega_c e^{j\pi/2} e^{j(2k+1)\pi/2N}$ for $k = 0, 1, \dots N-1$

$$N = \frac{\log\left(\frac{1}{\delta^2} - 1\right)}{2\log\left(\frac{\Omega_s}{\Omega_c}\right)}$$

$$|H(\Omega)|^2 = \frac{1}{1+\left(\frac{\Omega}{\Omega_c}\right)^{2N}}$$

* At $\Omega = 0$

$$|H(0)|^2 = \frac{1}{1+0} = 1$$

* At $\Omega = \Omega_c$

$$|H(\Omega_c)|^2 = \frac{1}{1+\left(\frac{\Omega_c}{\Omega_c}\right)^{2N}} = \frac{1}{2}$$

* At $\Omega \gg \Omega_c$ let $\Omega = 1000$ and $\Omega_c = 1$

$$|H(\Omega_c)|^2 = \frac{1}{1+\left(\frac{1000}{1}\right)^{2N}} \simeq \frac{1}{\left(\frac{1000}{1}\right)^{2N}} = \left(\frac{\overset{\Omega_c = 1}{1}}{\underset{\Omega = 1000}{1000}}\right)^{2N}$$

$$= \left(\frac{\Omega_c}{\Omega}\right)^{2N}$$

Hence, $|H(\Omega)|^2$ approaches zero as $\Omega \gg \Omega_c$ so for higher frequencies than Ω_c, $|H(\Omega)|^2$ behaves like stop-band.

$\therefore e^{-j\omega n_0} \longrightarrow x(n-n_0)$

06/05/25

Design of FIR filters with windows

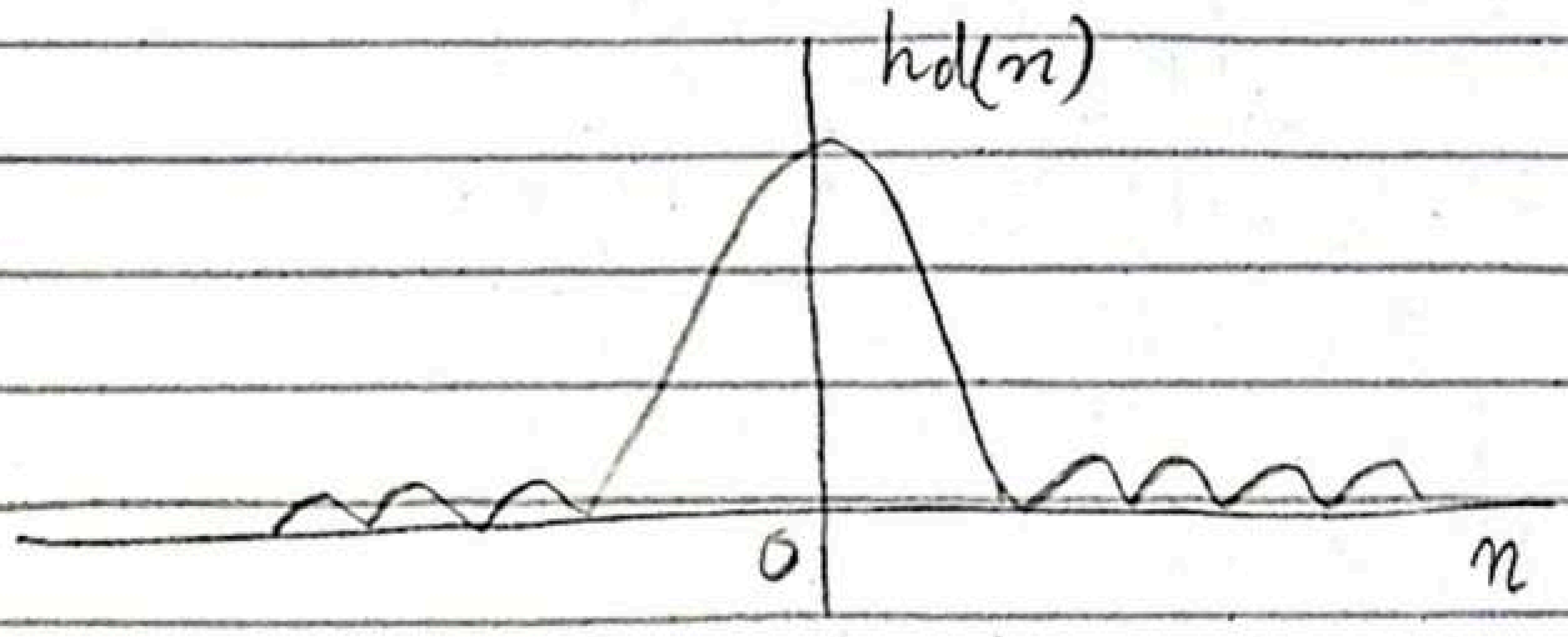
→ In real life applications, we initiate designing filter in frequency domain. For example, we initiate from ω_c/Ω_c, Ω_{ll} etc.

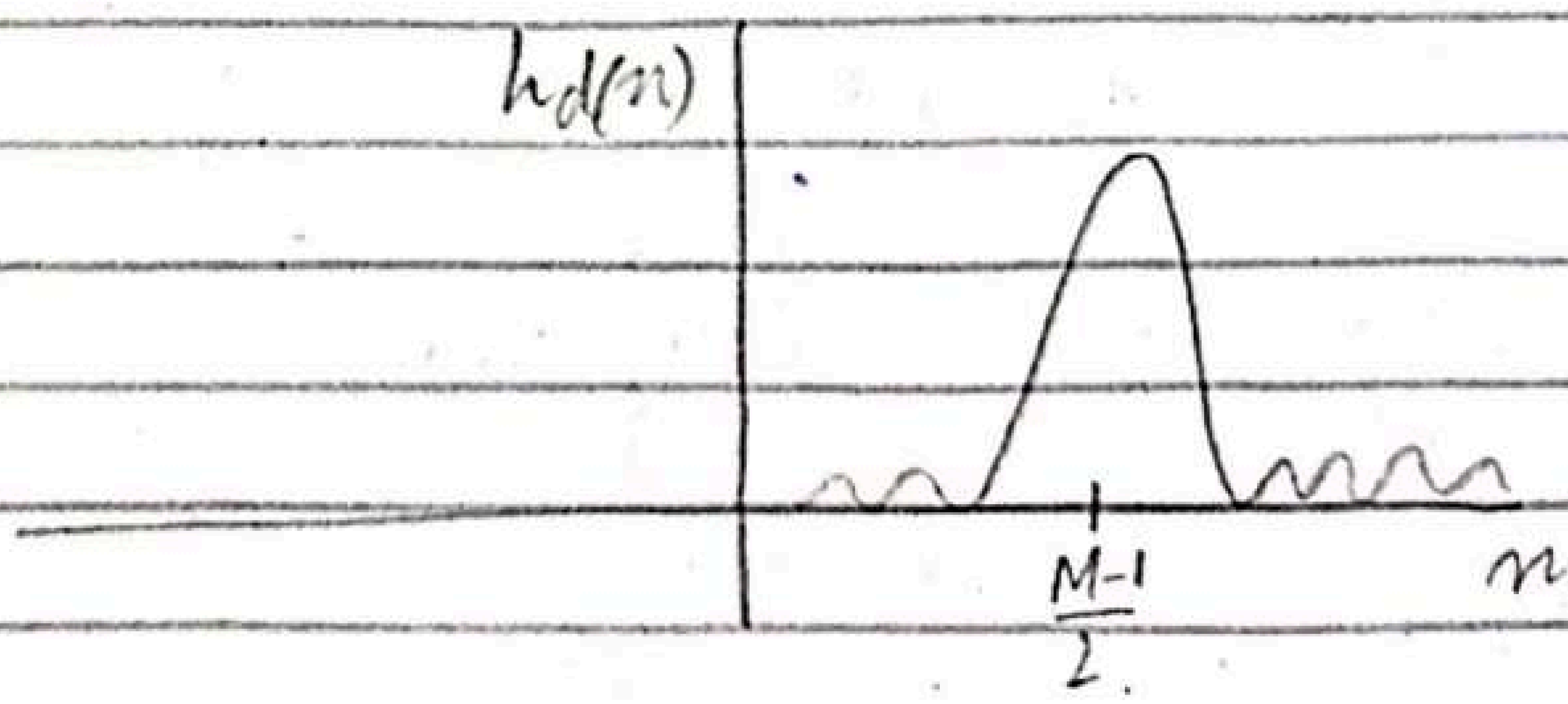
→ second, a designed filter should be causal and its length should be 'M' from $n=0$ to $n=M-1$

→ To hold the symmetry within desired impulse response $hd(n)$ Hence, we shift the center to $\left(\frac{M-1}{2}\right)$ by multiplying $Hd(\omega)$ with $e^{-j\omega\left(\frac{M-1}{2}\right)}$ Hence,

if $M=5$, then the center will be at $\frac{5-1}{2}=2$ i.e.

$$hd(n) = [0 \quad 1 \quad \underset{\uparrow}{2} \quad 3 \quad 4]$$





Design a LPF of length $M=5$, $\omega_c = \frac{\pi}{2}$

Case 1:-

Without multiplying $\exp(-j\omega(\frac{M-1}{2}))$

$$H(\omega) = \begin{cases} 1 , & |\omega| \leq \omega_c \\ 0 & \text{otherwise} \end{cases}$$

$$h(n) = \begin{cases} \frac{1}{2} = 0.5 & n=0 \\ \frac{0.5\sin(\frac{\pi n}{2})}{\pi n} & n \neq 0 \end{cases}$$

use this

$h(0) = 0.5$

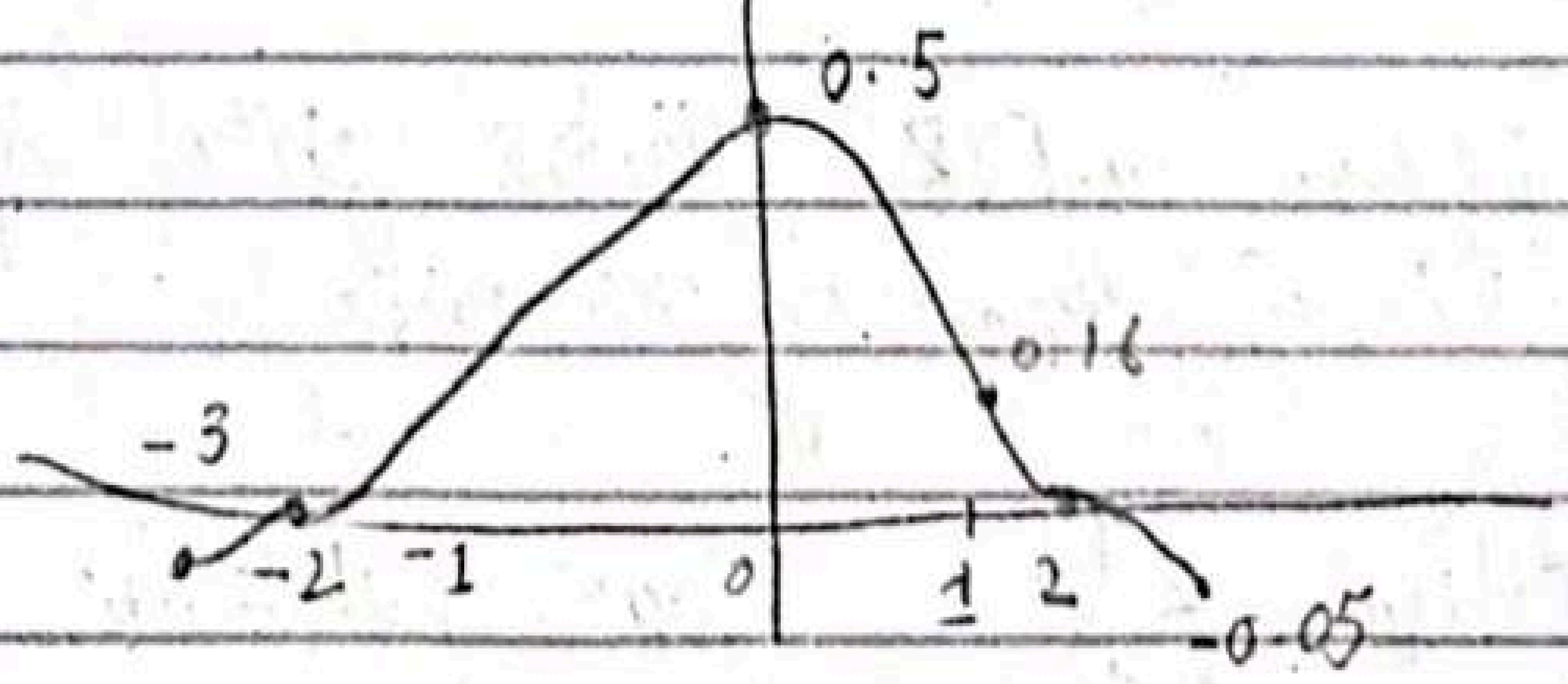
$h(-1) = 0.16$

$h(1) = 0.16$

$h(2) = 0$

$h(-2) = 0$

$h(3) = h(-3) = -0.05$



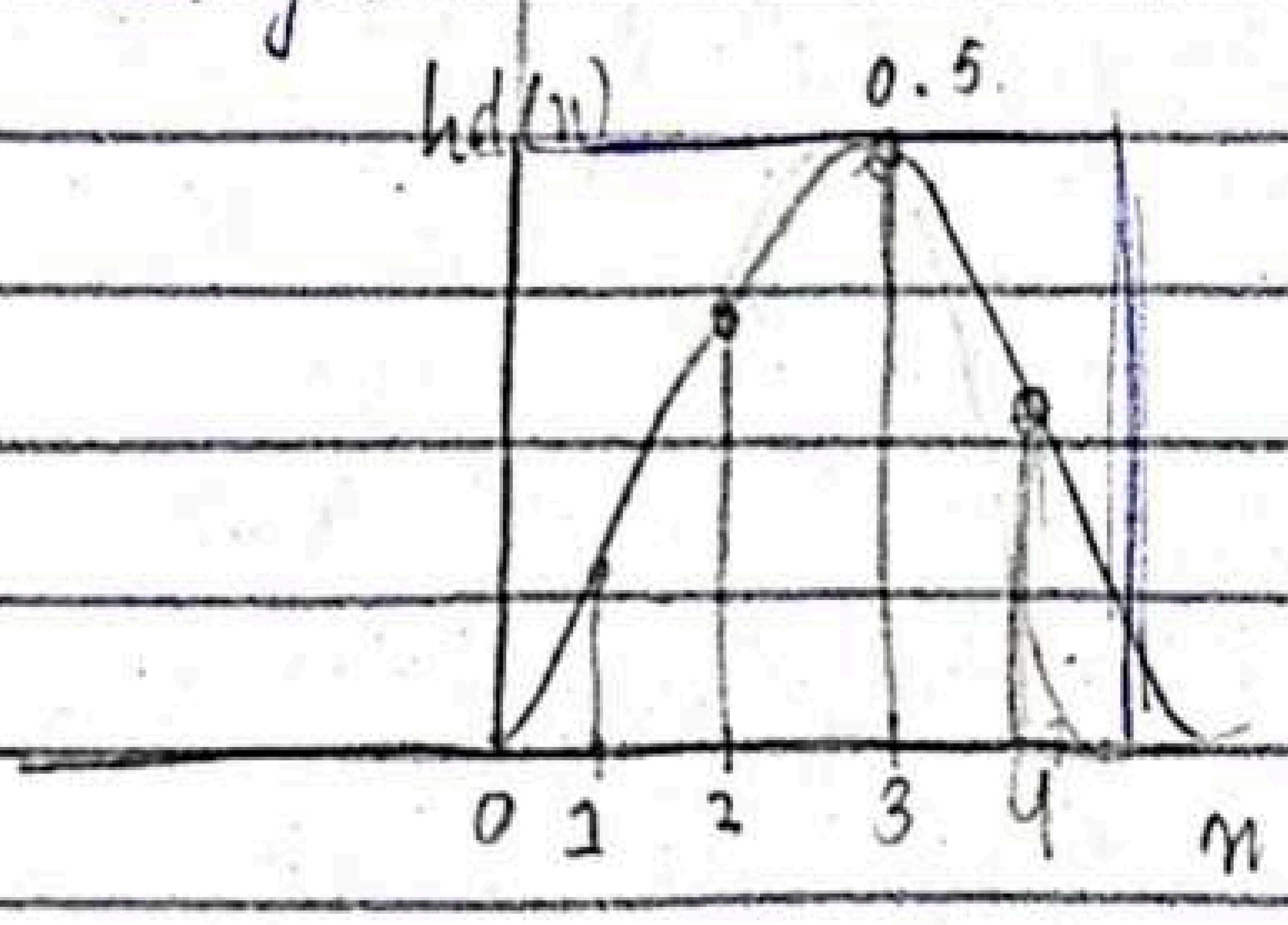
Case 2:-

With Multiplying $\exp\left(-j\omega\left(\frac{M-1}{2}\right)\right)$

As filter with center at 2, so compute for $n = 0, 1, 2, 3, 4$

$hd(0) = h(n-2) = 0$
$hd(1) = h(-1) = 0.16$
$hd(2) = 0.5$
$hd(3) = 0.16$
$hd(4) = 0$

Now $hd(n)$ is from 0 to 4, which is practically realizable.



$$\begin{cases} 2 \cdot e^{-j\omega 2} & |\omega| < \omega_c \\ 0 & \text{otherwise} \end{cases}$$

H.W

Design of FIR filters with windows

Q 8.1 Design a FIR digital filter approximating the ideal frequency response

$$H_d(\omega) = \begin{cases} 1 & \text{for } |\omega| \le \pi/6 \\ 0 & \text{for } \pi/6 < \omega \le \pi \end{cases}$$

(a) Determine the coefficients of 25-tap filter based on the window method with a rectangular window.

8/05/25

Question :-

Design a FIR filter approximately the ideal frequency response

$$H_d(\omega) = \begin{cases} 1, & |\omega| \le \pi/6 \\ 0, & \frac{\pi}{6} \le \omega \le \pi \end{cases}$$

Determine the co-efficients of 25-tap filter by using hamming window.

impulse response

$$h_d(n) = h(n) \cdot w(n)$$

$M-1$

$25-1$

$\frac{24}{2} = 12$

$$h(n) = \frac{1}{2\pi}\int_{-\pi/6}^{\pi/6} H_d(\omega)\, e^{j\omega n}\, d\omega$$

$$= \frac{1}{2\pi}\int_{-\pi/6}^{\pi/6} e^{j\omega(n-12)}\, d\omega$$

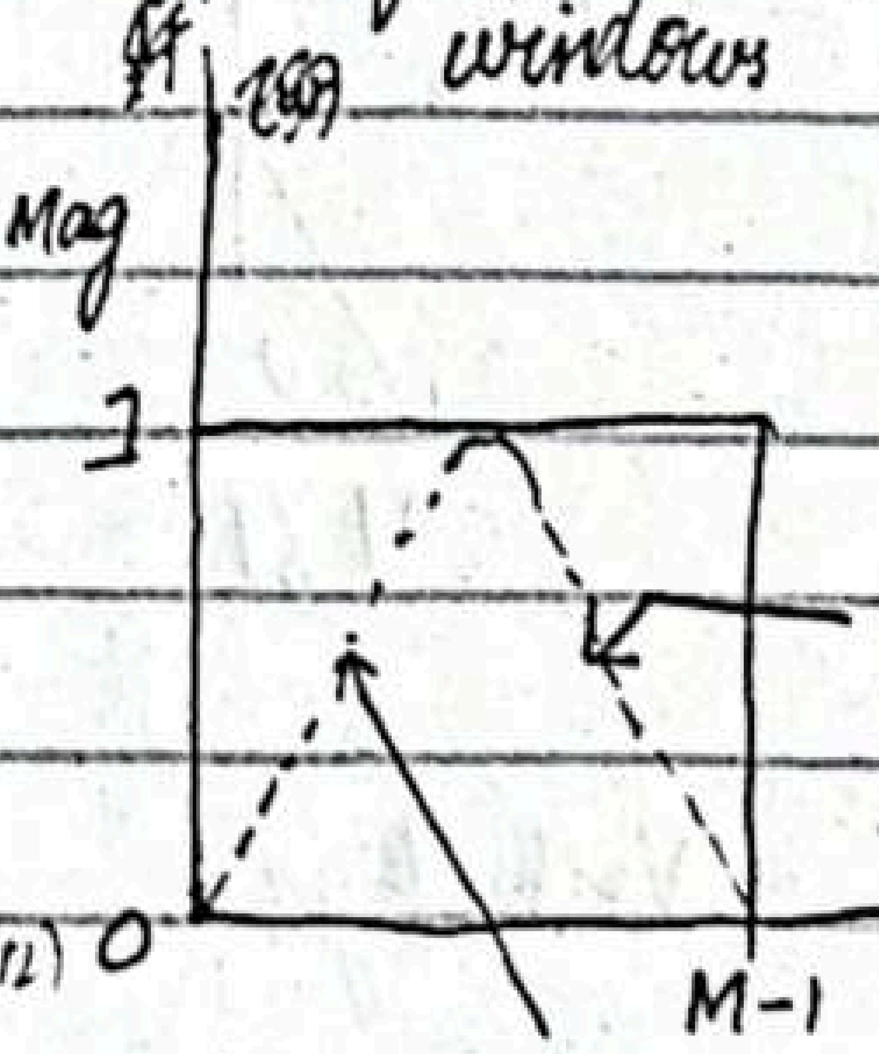
$$= \frac{1}{2\pi} \frac{e^{j\omega(n-12)}}{jn}$$

$$= \frac{1}{2j\pi(n-12)} \left[e^{j(\pi/6)(n-12)} - e^{-j(\pi/6)(n-12)} \right]$$

$$= \frac{1}{\pi(n-12)} \cdot \frac{e^{j\frac{\pi}{6}(n-12)} - e^{-j\frac{\pi}{6}(n-12)}}{2j}$$

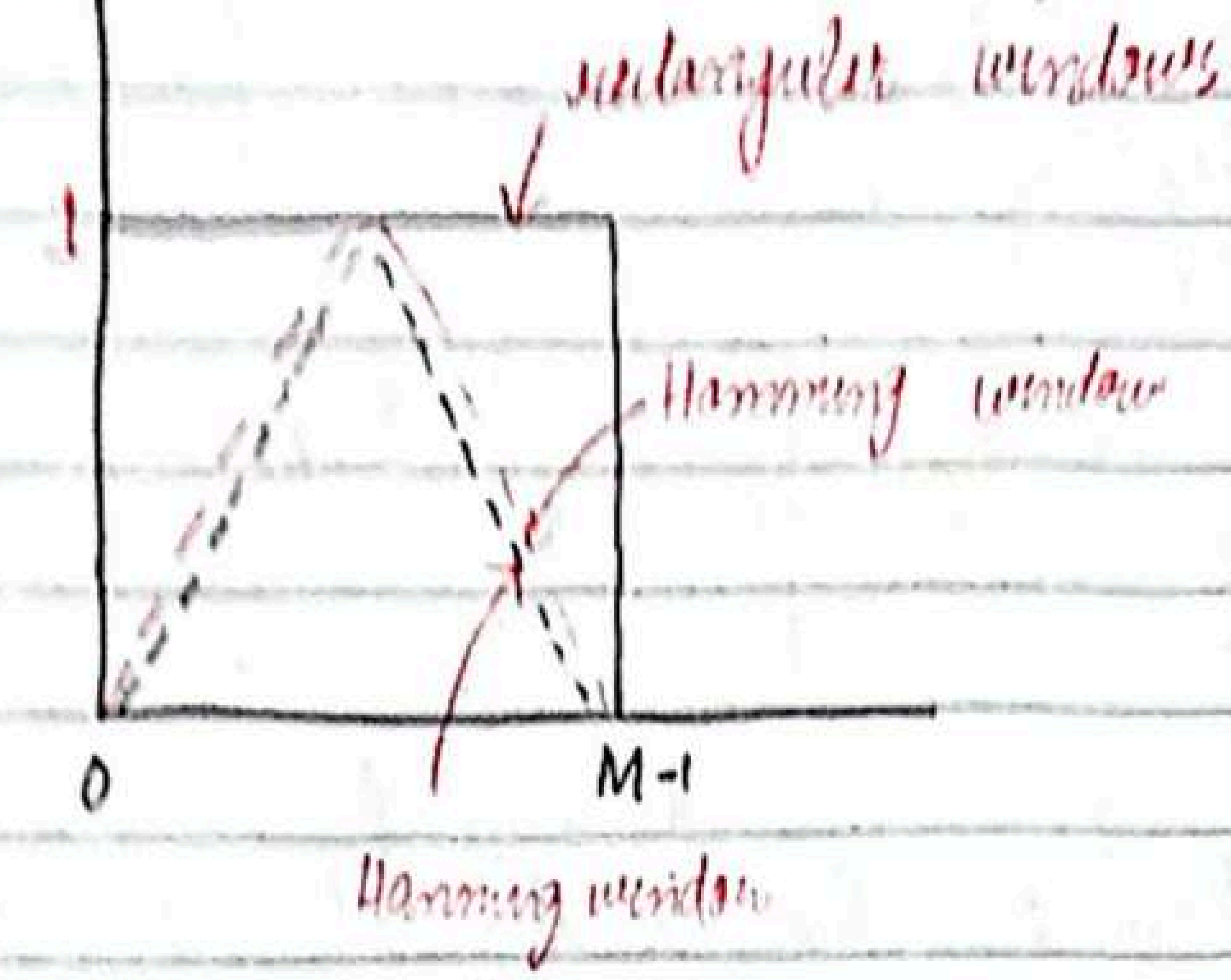
$$= \frac{1}{\pi(n-12)} \sin\left(\frac{\pi}{6}(n-12)\right)$$

shapes of diff windows

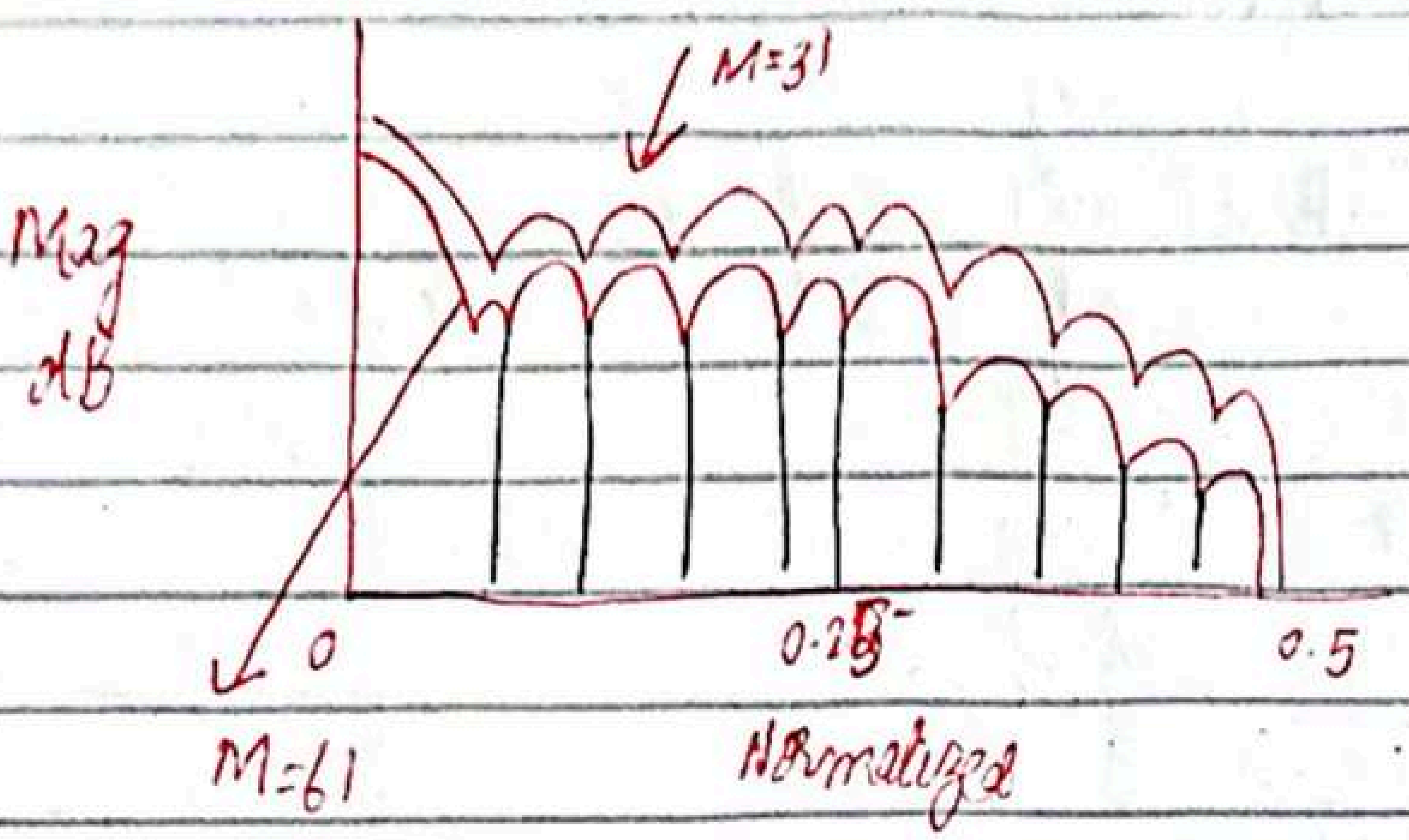


$$h(n) = \frac{1}{\pi(n-12)} \sin\left(\frac{\pi}{6}(n-12)\right)$$

Shapes of diff windows:-

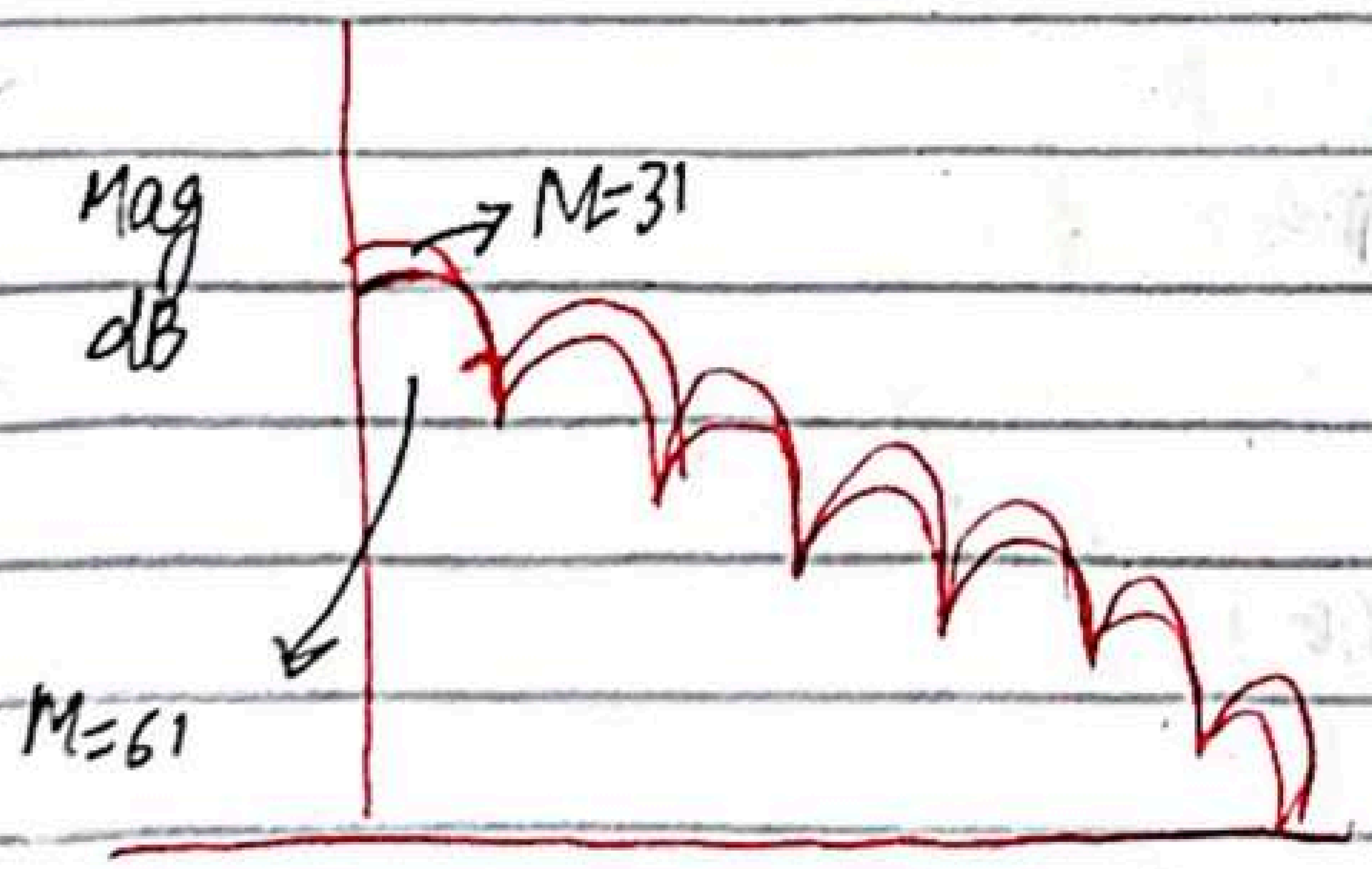


Frequency response of Hanning window.



Normalized

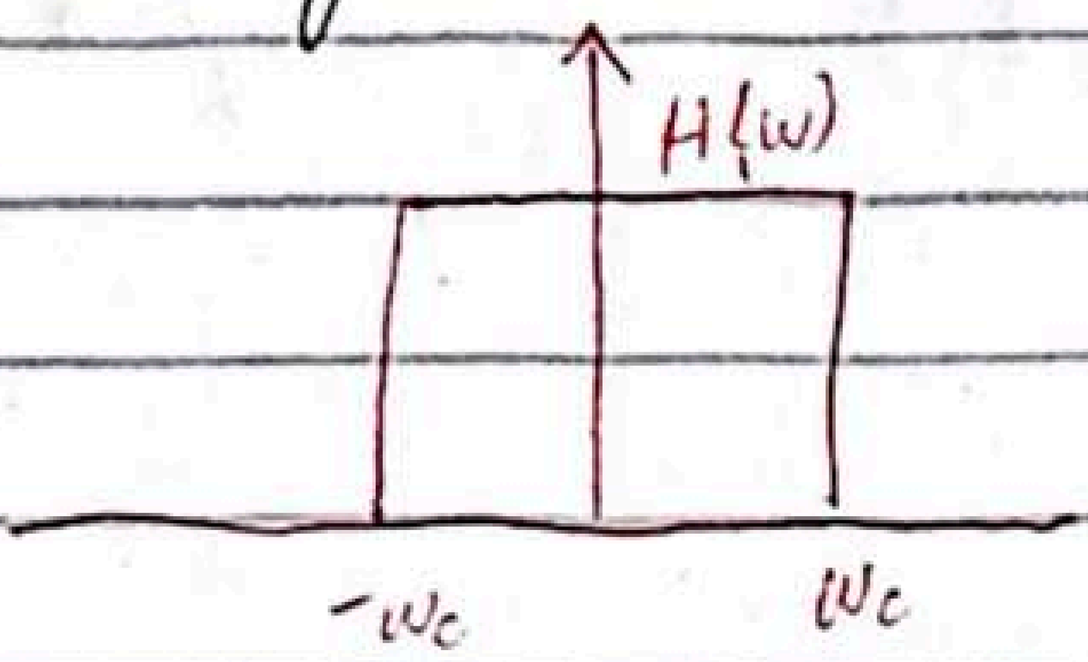
Frequency response of Red window

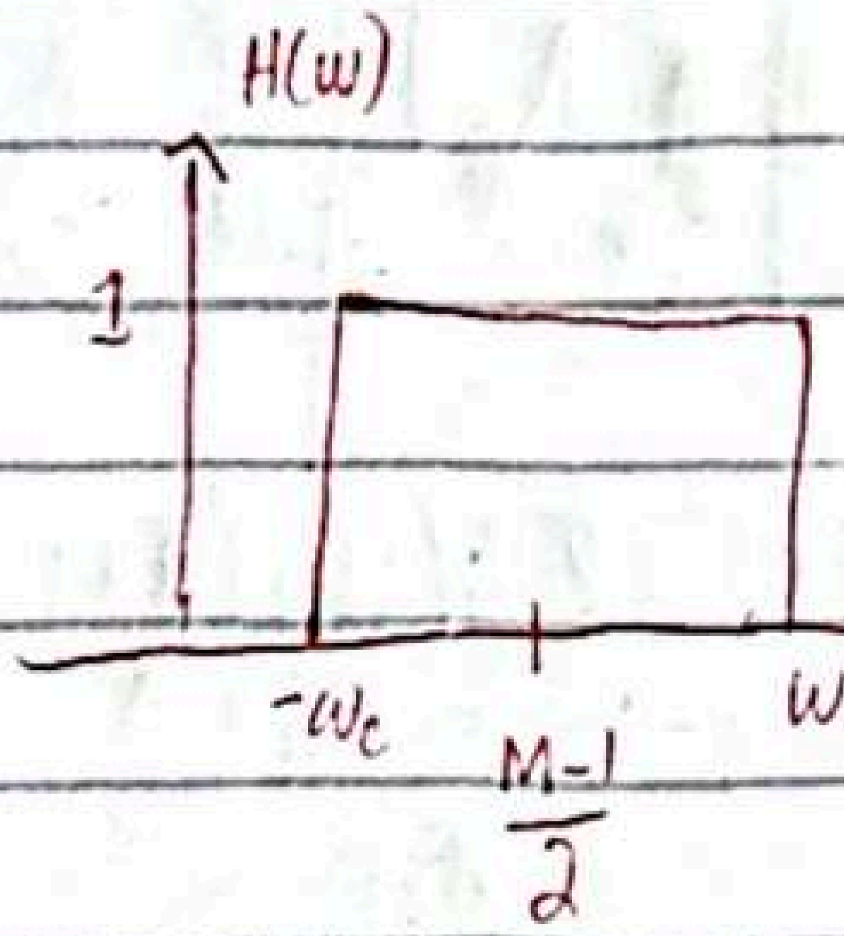


$$h_d(n) = h(n) \cdot w(n)$$

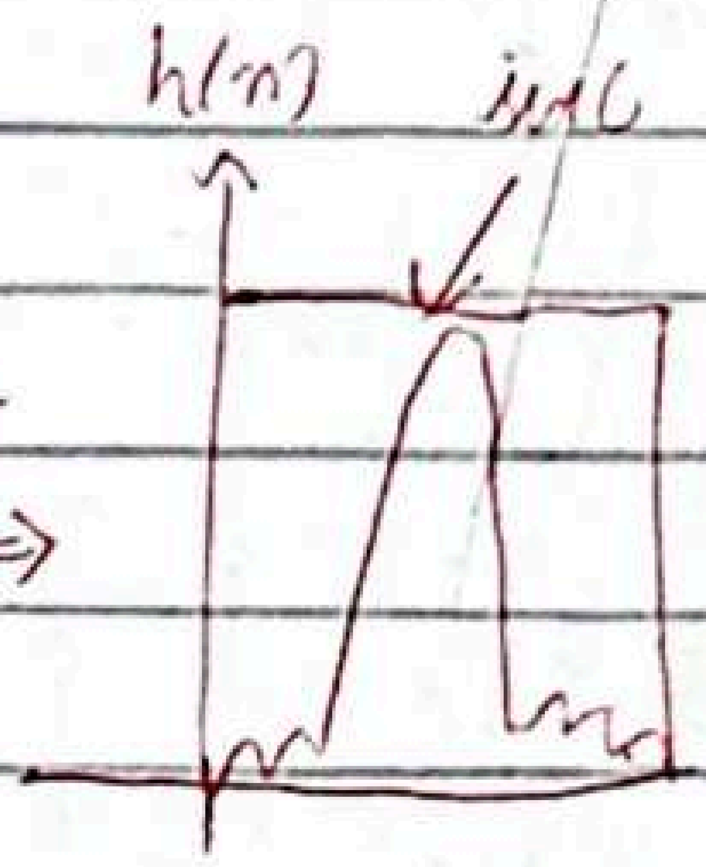
$$= \left[\frac{\sin\left(\frac{\pi}{6}(n-12)\right)}{\pi(n-12)}\right] \cdot \left[0.54 - 0.46\left(\cos\frac{2\pi n}{M-1}\right)\right]$$

rectangular:-





IFFT $\Rightarrow$



$w(n) \rightarrow$ Hamming window

$$w(n)_{\text{hanning}} = \frac{1}{2}\left(1 - \cos\frac{2\pi n}{M-1}\right)$$

$$w(0) = \frac{1}{2}\left(1 - \cos\frac{2\pi(0)}{M-1}\right)$$

$$= \frac{1}{2}\left(1 - \frac{\cos(0)}{24}\right)$$

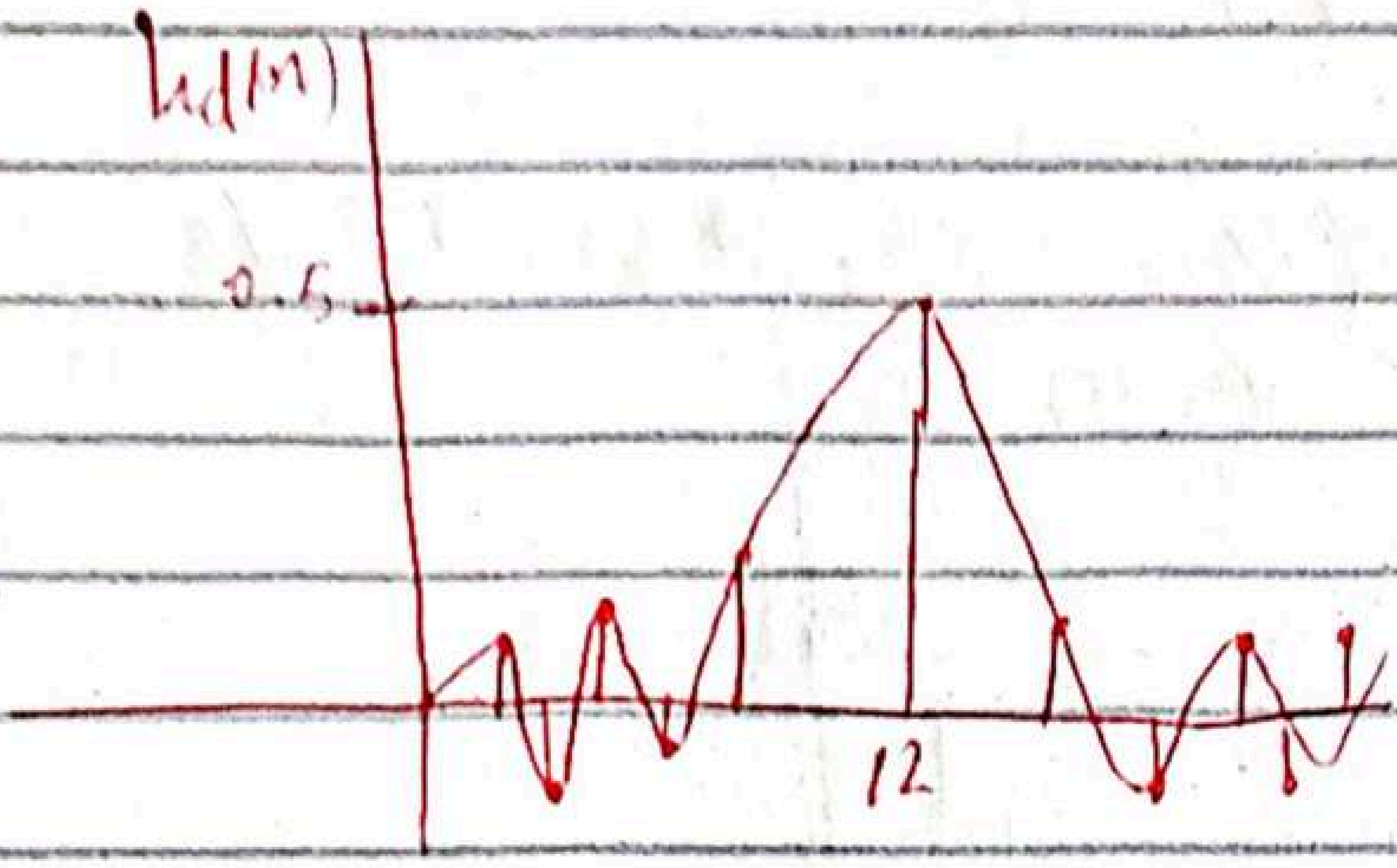
$$\cancel{= \frac{1}{2}\left(1 - \frac{1}{24}\right) = \frac{1}{2}\left(\frac{23}{24}\right) = \frac{23}{48}}$$

$$\frac{24}{2}$$

$$hd(0) = \left[\frac{\sin\frac{\pi}{6}(n-12)}{\pi(n-12)}\right] \cdot \left[0.54 - 0.4\left(\cos\frac{2\pi(0)}{24}\right)\right]$$

$= 0$

$hd(1) = 1.38m$	$hd(7) =$
$hd(2) = -3.9m$	$hd(8) =$
$hd(3) = -7.59m$	$hd(9) =$
$hd(4) = -10.01$	$hd(10) =$
$hd(5) = -9.57m$	$hd(11) =$
$hd(6) = 0$	$hd(12) =$



There will be symmetry

$$h_d(n) = \frac{0.5(\sin(\frac{\pi n}{2})}{\pi n} \quad \alpha$$